



Grupo de
Genética e
Genômica da
Conservação
APTA



Introdução à Modelagem de Nicho Ecológico

WEBINAR NEXT GENERATION



Maurício Vancine

16/07/2020



Webinar

Tópicos

1. Apresentações
2. Introdução aos Modelos de Nicho Ecológico - *Ecological Niche Models* (ENMs)
3. Nicho Ecológico e Distribuição de Espécies
4. Construção dos ENMs passo a passo
5. Dados de entrada: ocorrências e variáveis ambientais
6. Ajuste dos modelos
7. Avaliação dos modelos
8. Predição dos modelos
9. Aplicações e mais informações
10. Prática no R

1. Apresentações

Maurício Vancine

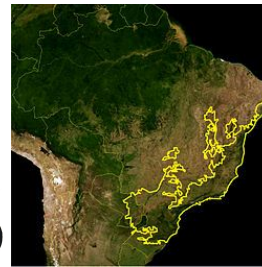
Ecólogo (2015) | Mestre em Zoologia (2018) |
Doutorando em Zoologia (2020-?)

Pesquisa

Ecologia Espacial (Ecologia da Paisagem)

Ecologia Quantitativa (SDM e JSMD)

Ecologia e Conservação de Anfíbios



unesp 

UNIVERSIDADE ESTADUAL PAULISTA
"JÚLIO DE MESQUITA FILHO"



Prof. Milton Ribeiro



Prof. Célio Haddad

Maurício Vancine

Ecólogo (2015) | Mestre em Zoologia (2018) |
Doutorando em Zoologia (2020-?)

Pesquisa

Ecologia Espacial (Ecologia da Paisagem)

Ecologia Quantitativa (SDM e JSMD)

Ecologia e Conservação de Anfíbios

Especialidades

Modelos de Nicho Ecológico (ENMs)

Análise de Dados Ecológicos e Geoespaciais

Open Source [R, QGIS, GRASS GIS, Linux, Libreoffice, ...]

Contato e informações

✉ mauricio.vancine@gmail.com

🐦 [@mauriciovancine](https://twitter.com/mauriciovancine)

🔗 mauriciovancine.netlify.com



2. Introdução aos Modelos de Nicho Ecológico (ENMs)

Uma abordagem, muitos nomes...

Ecology, 93(7), 2012, pp. 1527–1539
© 2012 by the Ecological Society of America

Uses and misuses of bioclimatic envelope modeling

MIGUEL B. ARAÚJO^{1,2,3,5} AND A. TOWNSEND PETERSON⁴

1. **Modelos de Envelopes Climáticos** (*Bioclimatic Envelope Models*)

Estimado um espaço multivariado de variáveis climáticas (envelope)

2. **Modelos de Nicho Ecológico** (*Ecological Niche Models*)

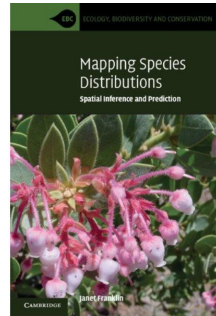
Vincula o envelope à teoria de nicho ecológico (Grinnell e Hutchinson)

3. **Modelos de Adequabilidade de Habitat** (*Habitat Suitability Models*)

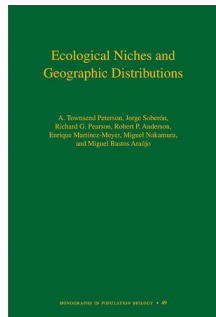
Envelope relacionado ao “habitat”, como espaço físico e recursos

4. **Modelos de Nicho Ecológico** (*Species Distribution Models*)

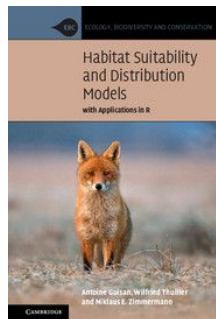
Modelar a distribuição geográfica das espécies



Franklin (2009)



Peterson et al. (2011)



Guisan et al. (2017)

Uma abordagem, muitos nomes...

Ecology, 93(7), 2012, pp. 1527–1539
© 2012 by the Ecological Society of America

Uses and misuses of bioclimatic envelope modeling

MIGUEL B. ARAÚJO^{1,2,3,5} AND A. TOWNSEND PETERSON⁴

1. **Modelos de Envelopes Climáticos** (*Bioclimatic Envelope Models*)

Estimado um espaço multivariado de variáveis climáticas (envelope)

2. **Modelos de Nicho Ecológico** (*Ecological Niche Models*)

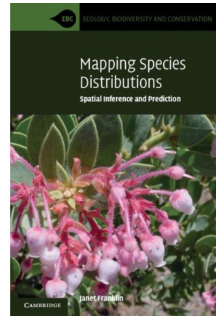
Vincula o envelope à teoria de nicho ecológico (Grinnell e Hutchinson)

3. **Modelos de Adequabilidade de Habitat** (*Habitat Suitability Models*)

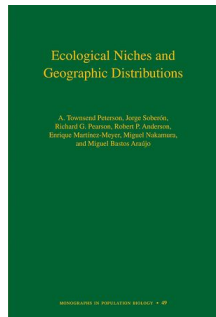
Envelope relacionado ao “habitat”, como espaço físico e recursos

4. **Modelos de Nicho Ecológico** (*Species Distribution Models*)

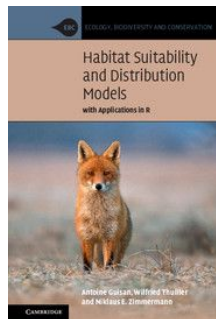
Modelar a distribuição geográfica das espécies



Franklin (2009)



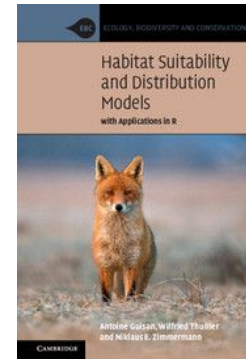
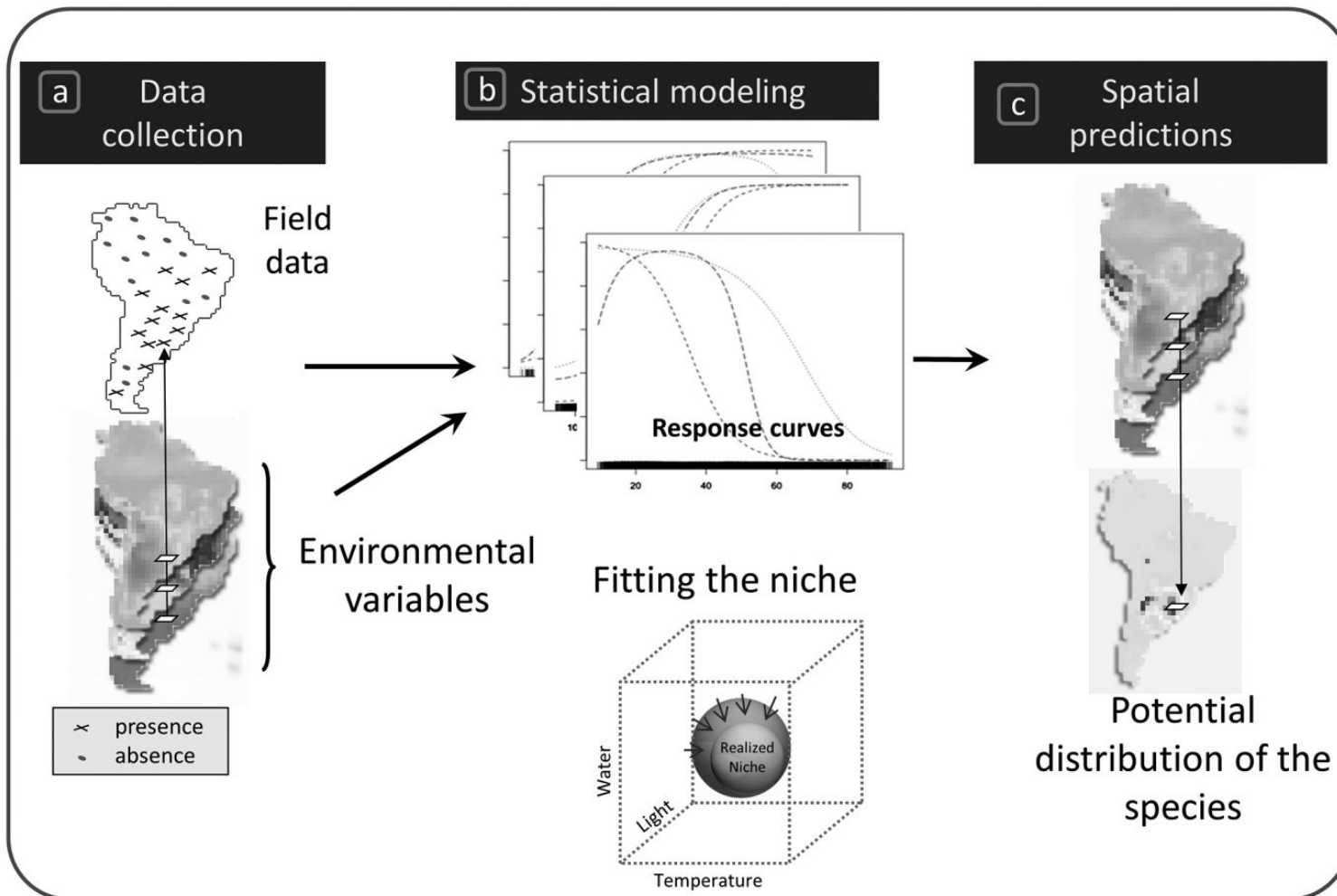
Peterson et al. (2011)



Guisan et al. (2017)

Modelos de Nicho Ecológico (ENMs)

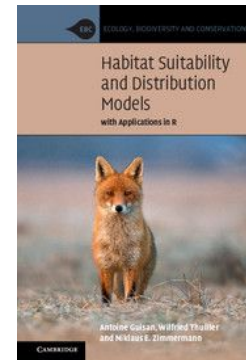
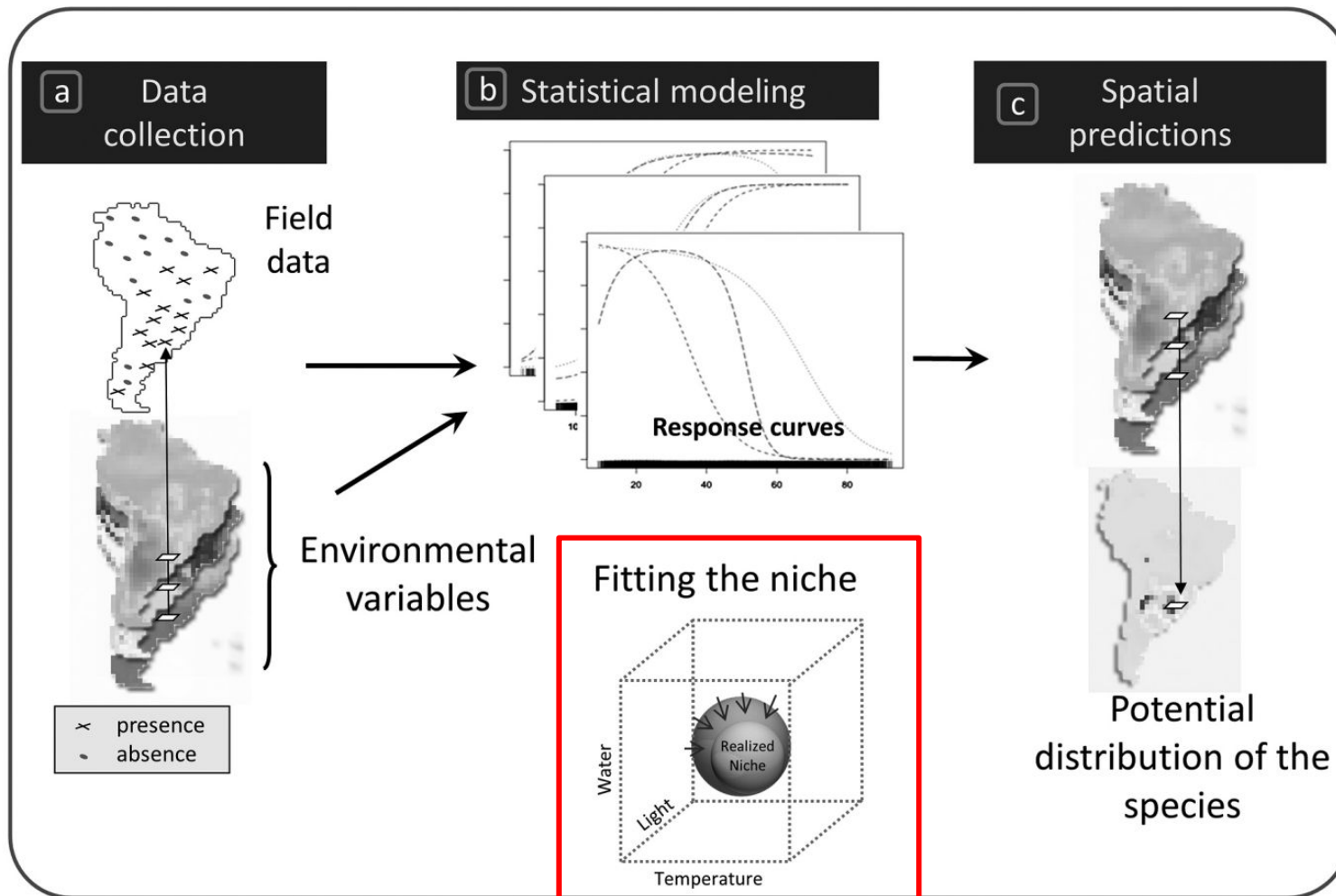
Visão geral



Guisan et al. (2017)

Modelos de Nicho Ecológico (ENMs)

Visão geral



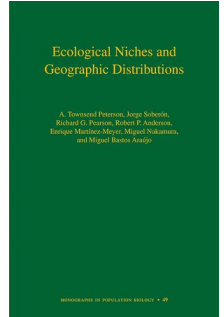
Guisan et al. (2017)

3. Nicho ecológico e distribuição das espécies

O que determina a distribuição de espécies?

Espaço Geográfico (G)

G

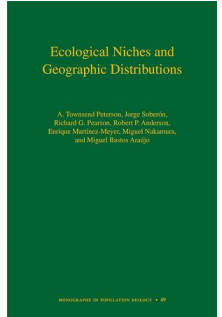


Peterson et al. (2011)

O que determina a distribuição de espécies?

Espaço Geográfico (G)

G



Peterson et al. (2011)

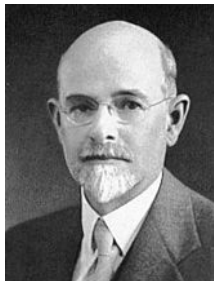
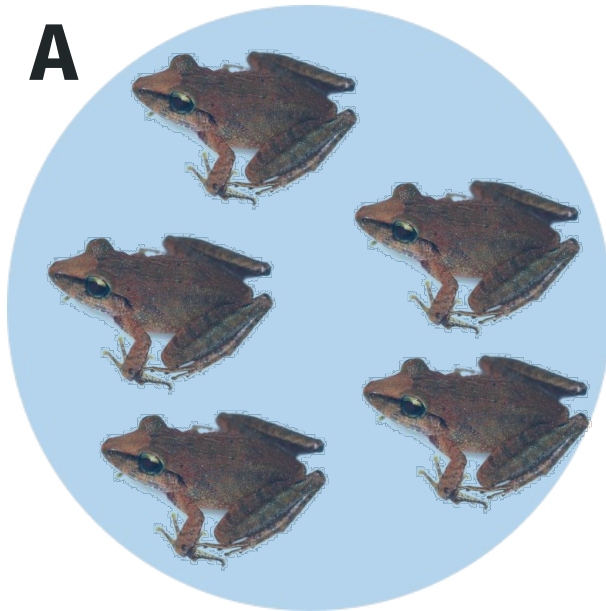
O que determina a distribuição de espécies?

Condições Abióticas (A)

G

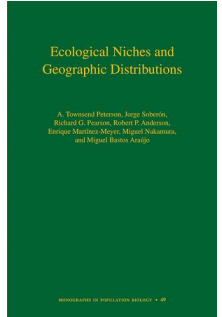


A



Joseph Grinnell (1917)

Requerimentos ambientais “condições climáticas”

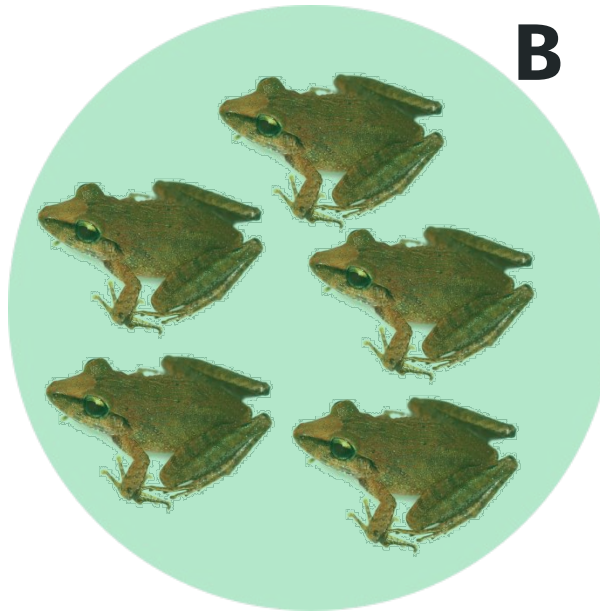


Peterson et al. (2011)

O que determina a distribuição de espécies?

Condições Bióticas (B)

G

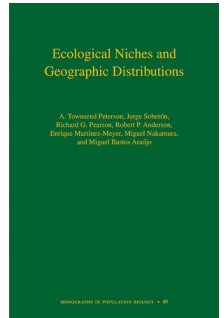
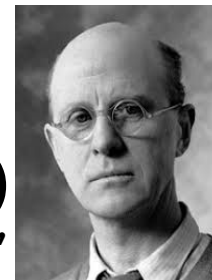


B



Charles Elton (1927)

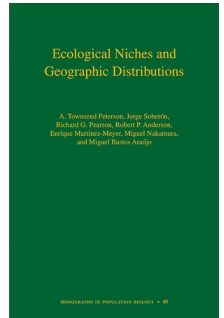
Papel funcional dos organismos “impacto”



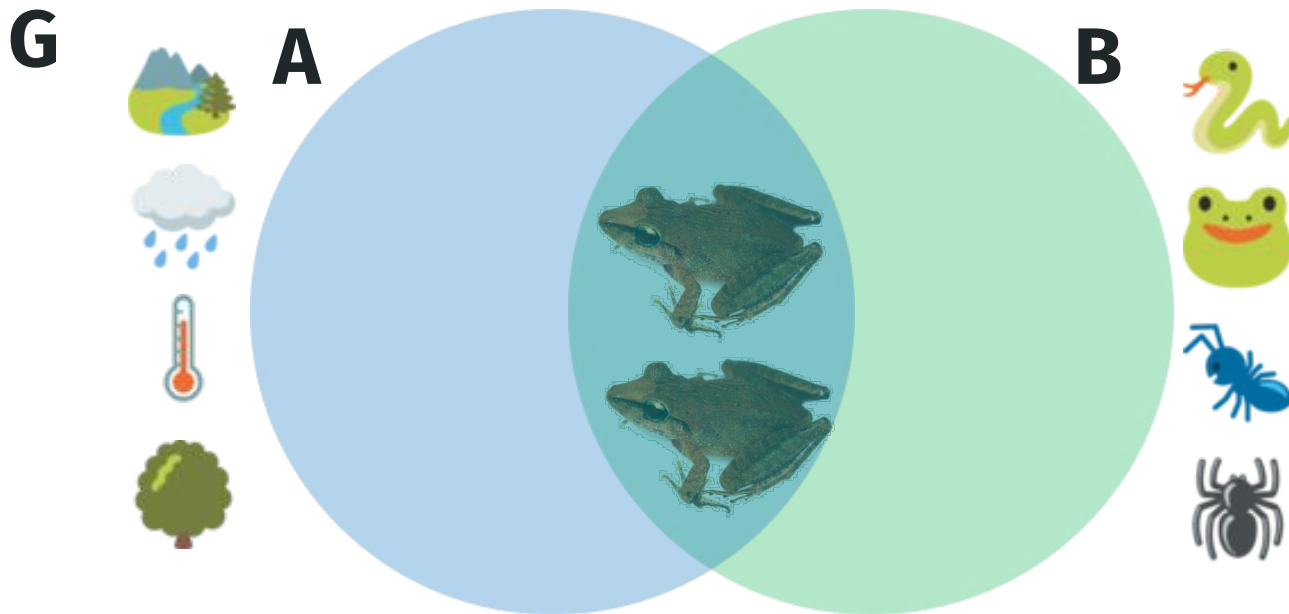
Peterson et al. (2011)

O que determina a distribuição de espécies?

Relação entre condições abióticas e bióticas



Peterson et al. (2011)



George E. Hutchinson (1957)

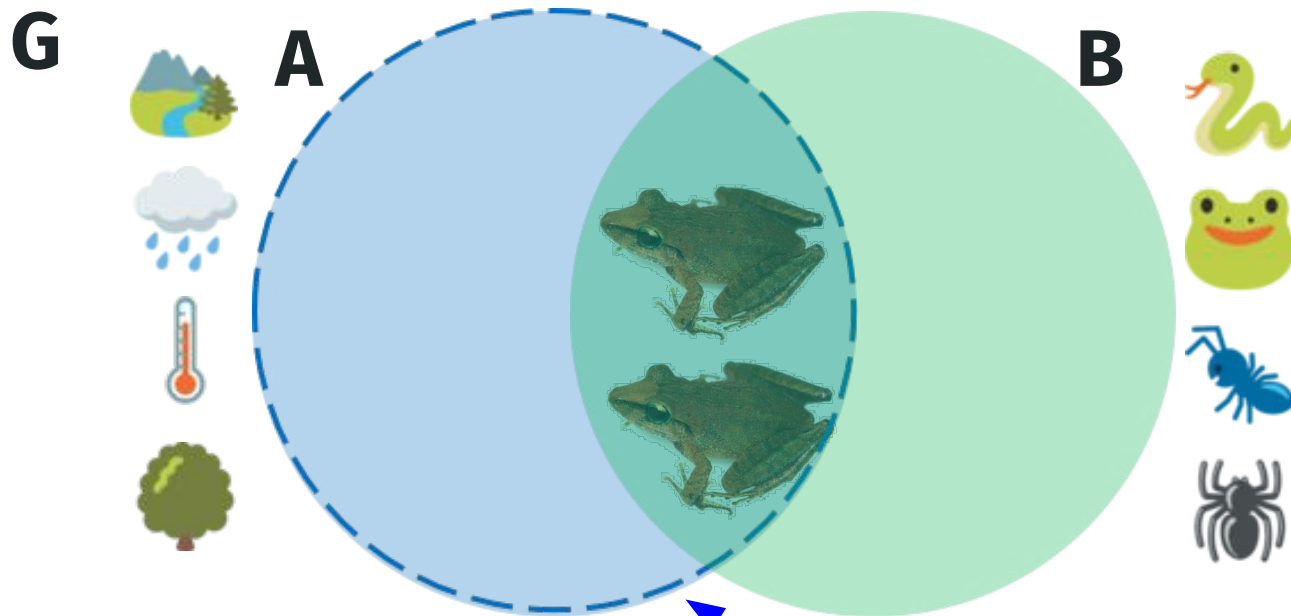
Requerimentos ambientais (**Nicho Fundamental**)

+

Requerimentos biológicos (**Nicho Realizado**)

O que determina a distribuição de espécies?

Nicho Fundamental

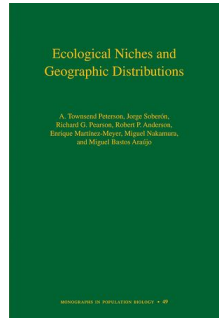


George E. Hutchinson (1957)

Requerimentos ambientais (Nicho Fundamental)

+

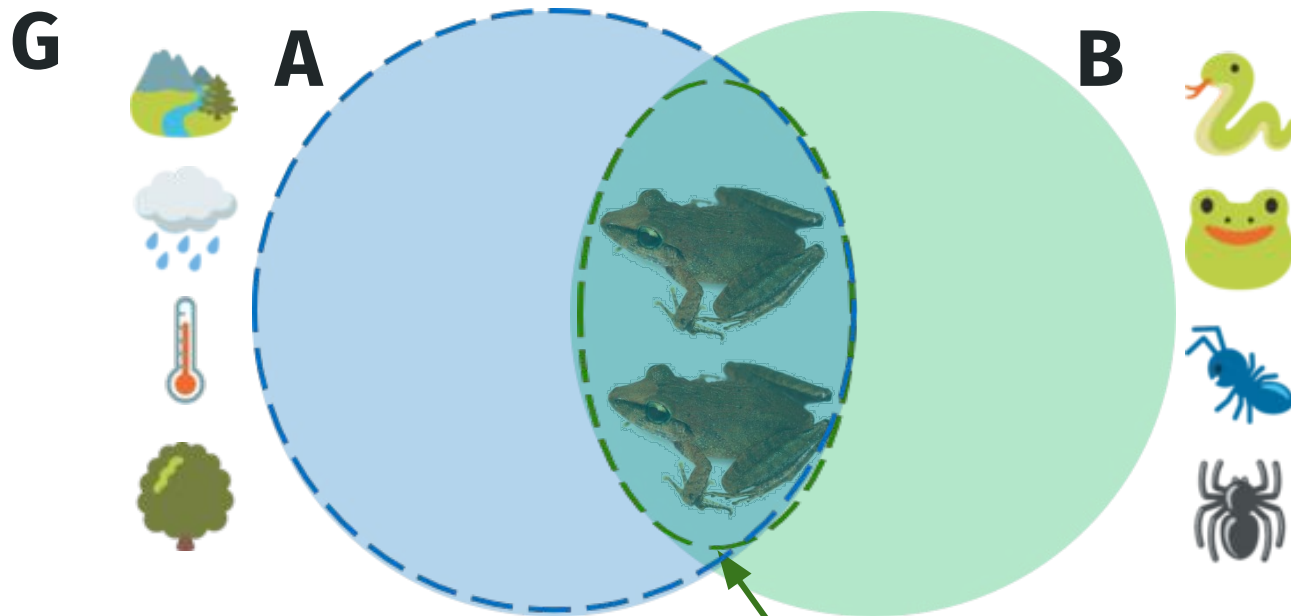
Requerimentos biológicos (Nicho Realizado)



Peterson et al. (2011)

O que determina a distribuição de espécies?

Nicho Realizado

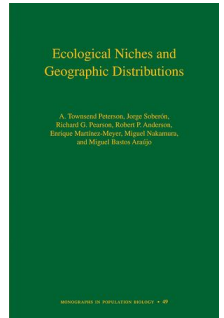


George E. Hutchinson (1957)

Requerimentos ambientais (Nicho Fundamental)

+

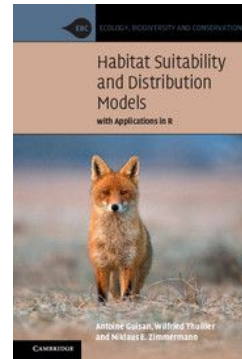
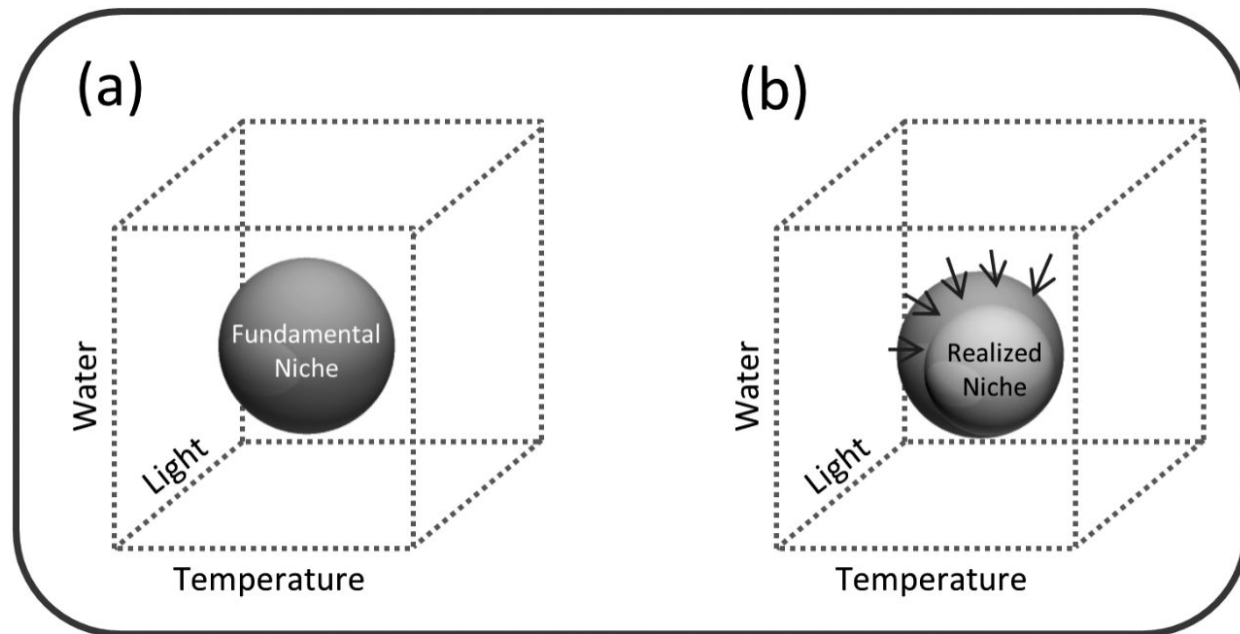
Requerimentos biológicos (Nicho Realizado)



Peterson et al. (2011)

O que determina a distribuição de espécies?

Hipervolume n-dimensional



Guisan et al. (2017)



George E. Hutchinson (1957)

Requerimentos ambientais (**Nicho Fundamental**)

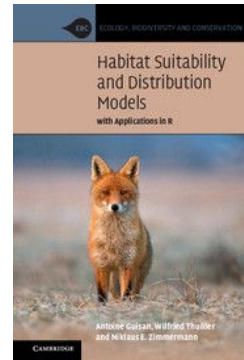
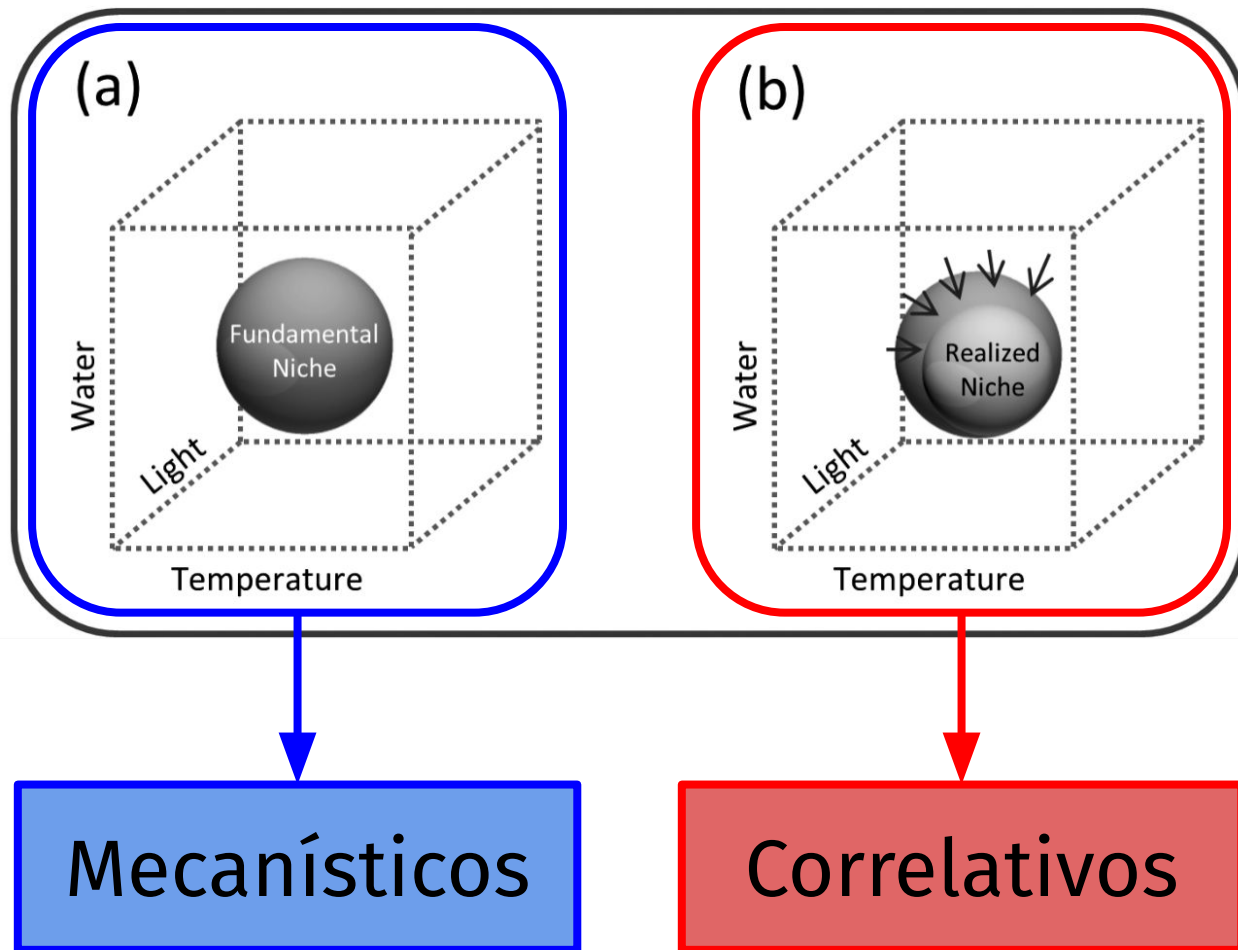
+

Requerimentos biológicos (**Nicho Realizado**)

Os ENMs estimam o nicho
fundamental ou realizado?

Nicho fundamental e realizado

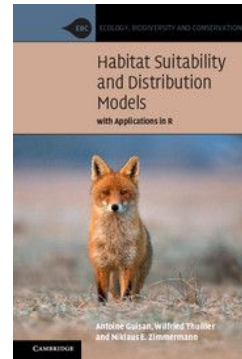
Modelos mecanísticos e correlativos



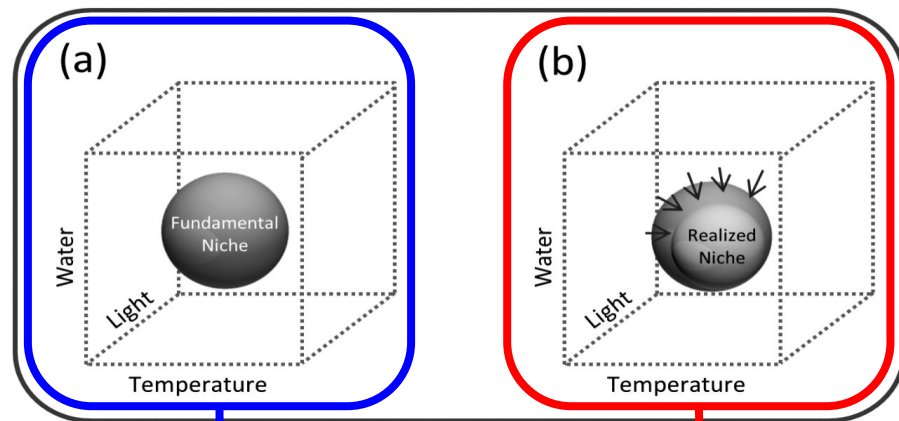
Guisan et al. (2017)

Nicho fundamental e realizado

Modelos mecanísticos e correlativos

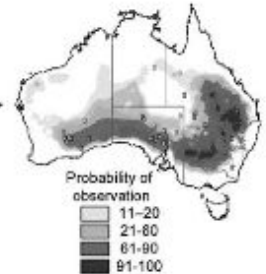
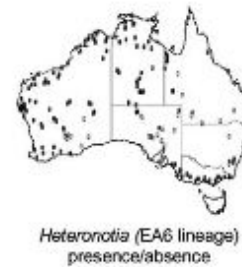
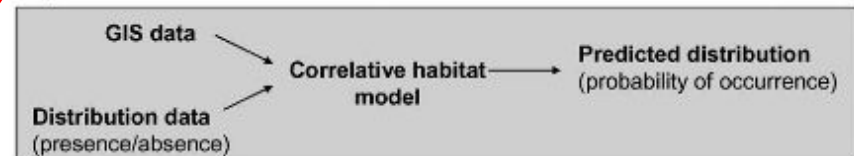
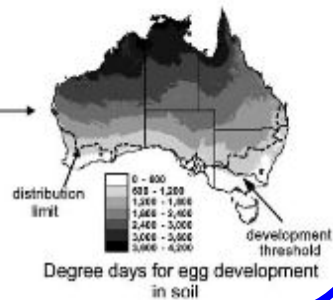
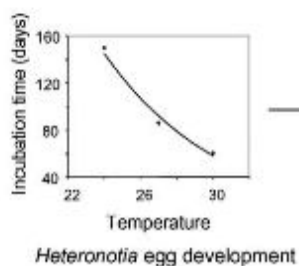
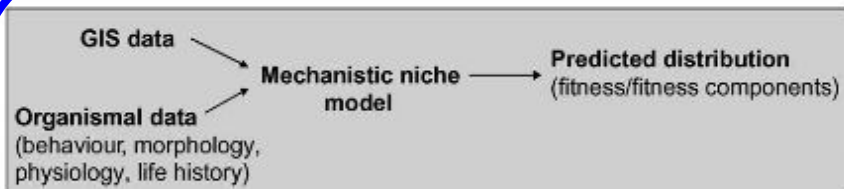


Guisan et al. (2017)



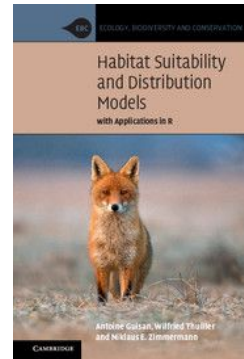
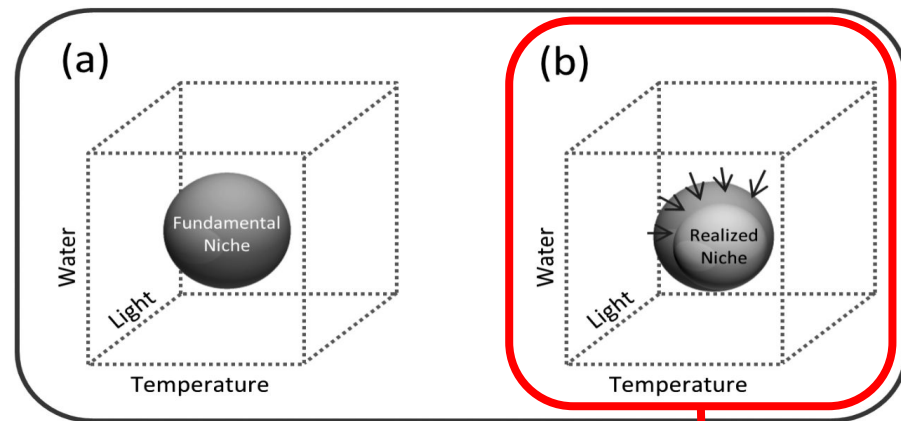
Mecanísticos

Correlativos



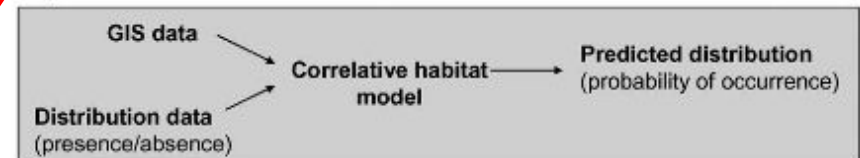
Nicho realizado

Modelos correlativos

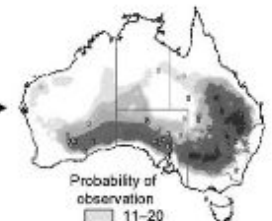


Guisan et al. (2017)

Correlativos



Heteronotia (EAG lineage)
presence/absence



Probability of
observation
11-20
21-60
61-90
91-100

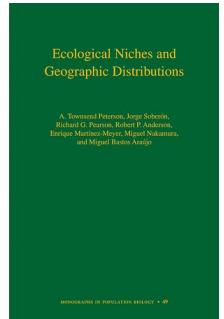
Modelos correlativos

Ocorrências

Espaço geográfico (G)



Jackson & Overpack (2000)

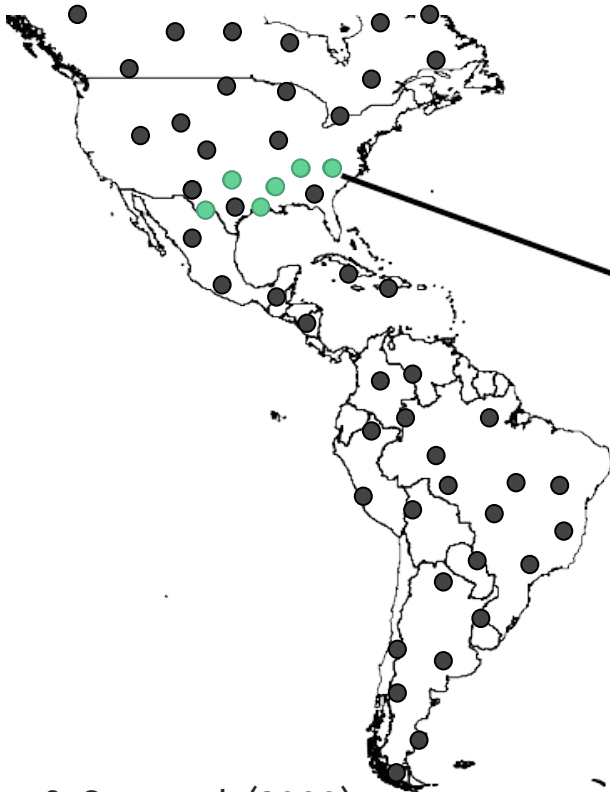


Peterson et al. (2011)

Modelos correlativos

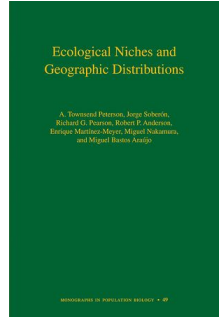
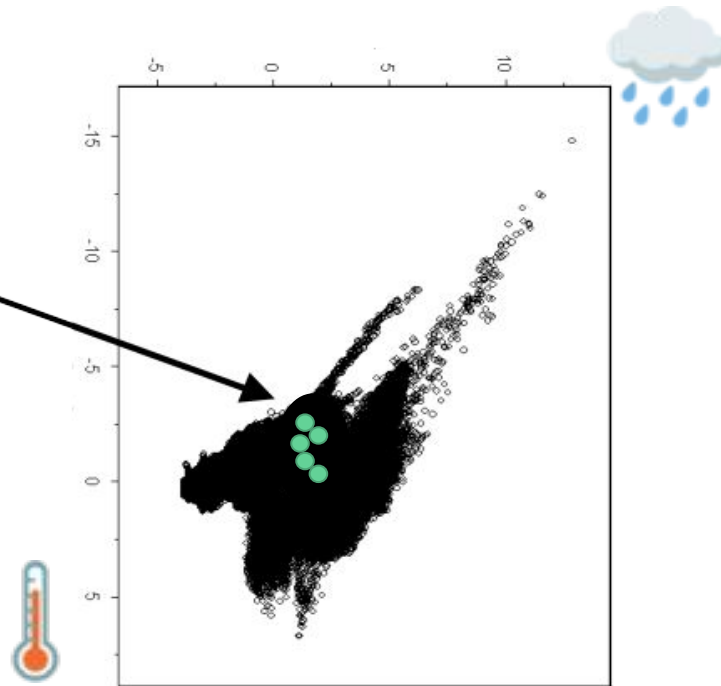
Condições ambientais

Espaço geográfico (G)



Jackson & Overpack (2000)

Espaço ambiental (E)

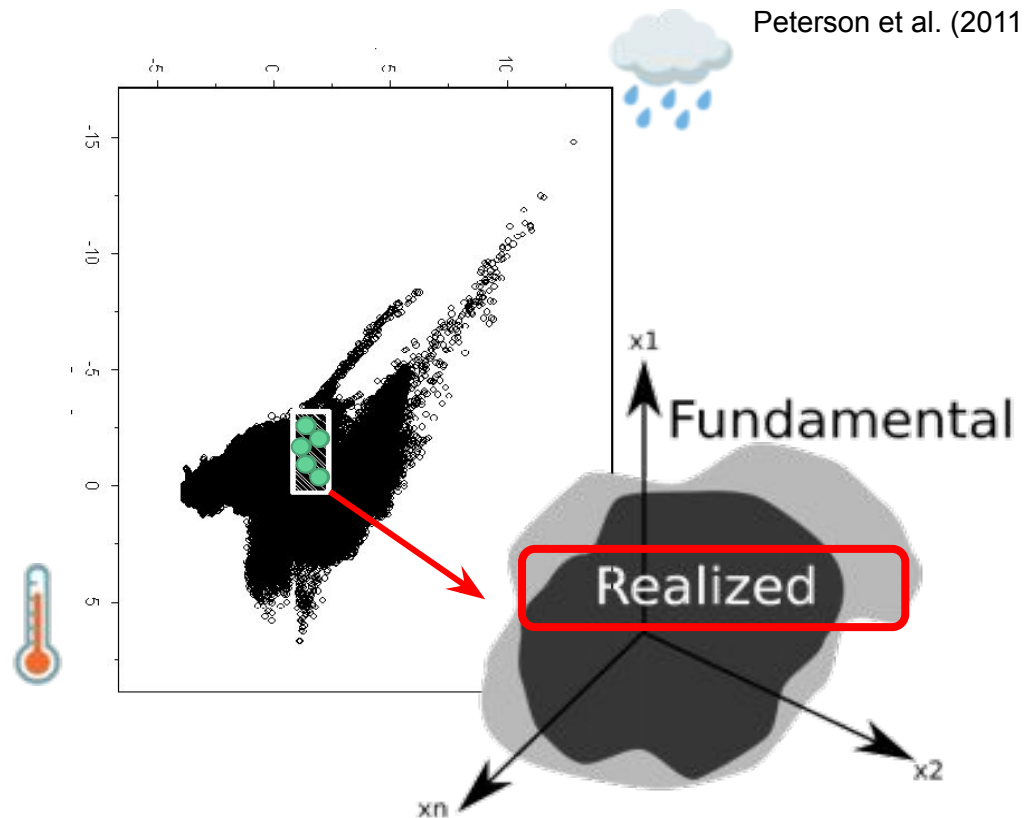


Peterson et al. (2011)

Modelos correlativos

Estimativa do nicho realizado

Espaço ambiental (E)



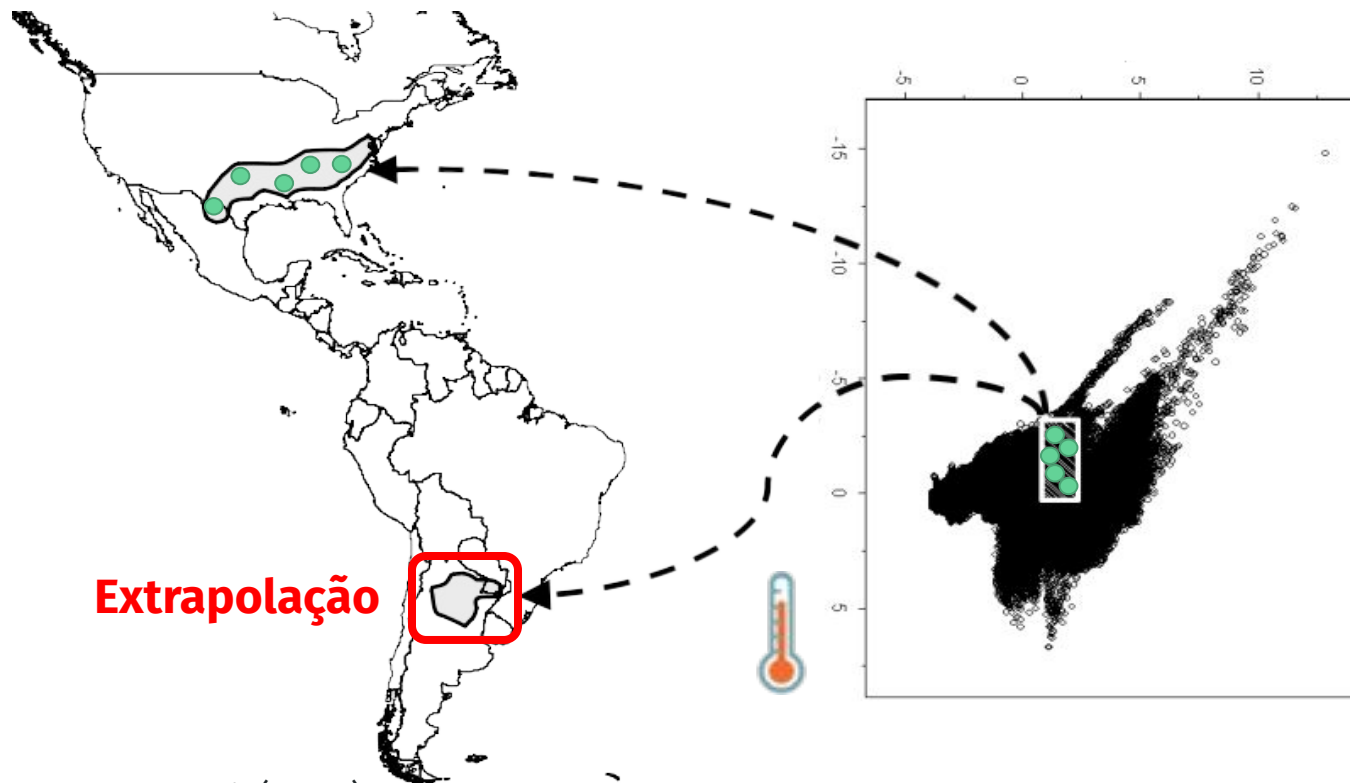
Jackson & Overpack (2000)

Modelos correlativos

Predição do nicho realizado estimado

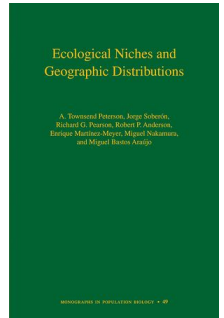
Espaço geográfico (G)

Espaço ambiental (E)



Extrapolação

Jackson & Overpack (2000)

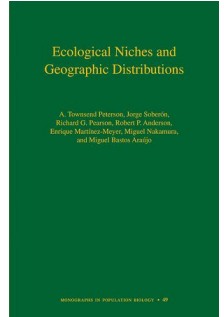
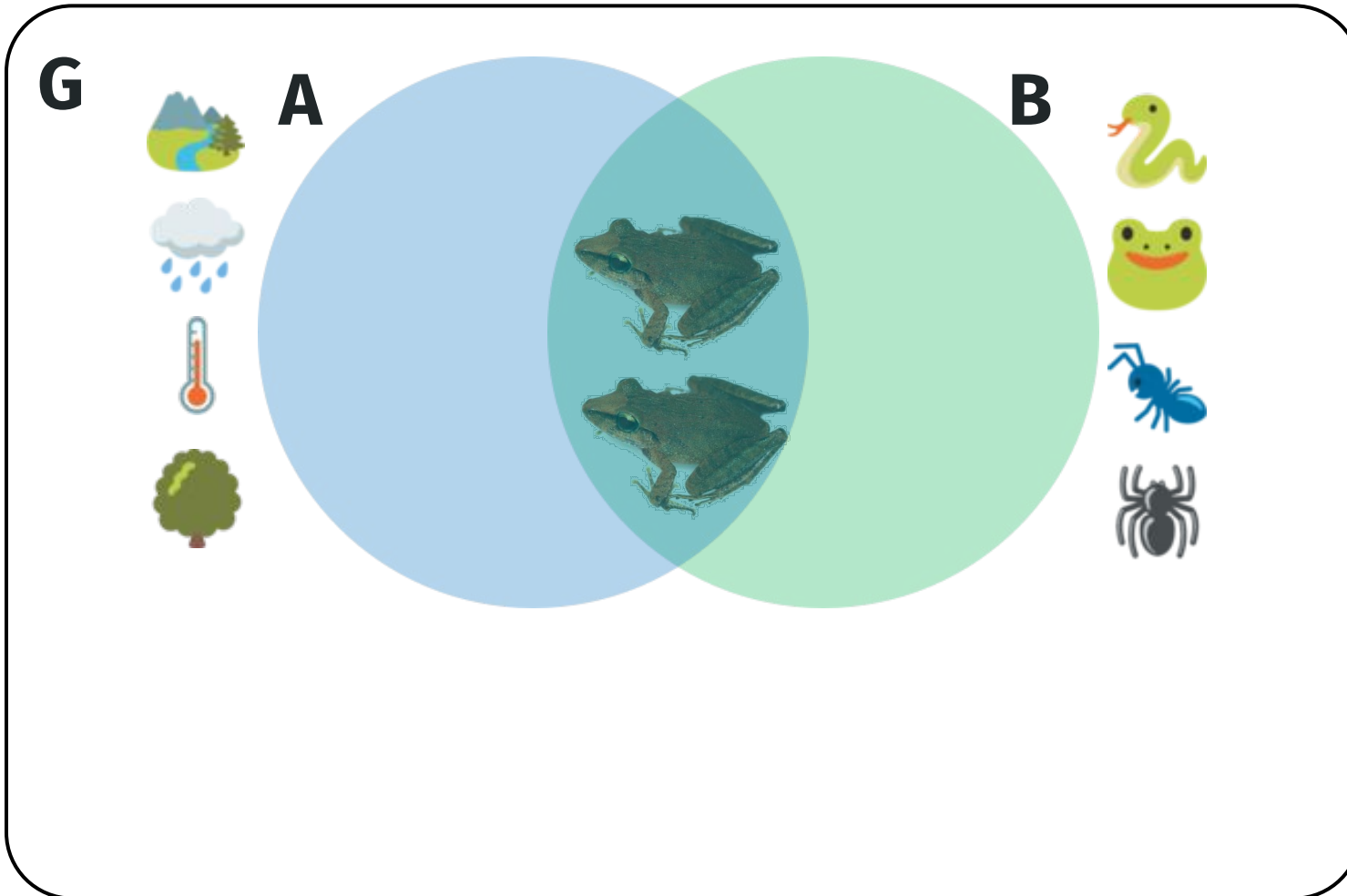


Peterson et al. (2011)

E como contornar essa
extrapolação?

O que determina a distribuição de espécies?

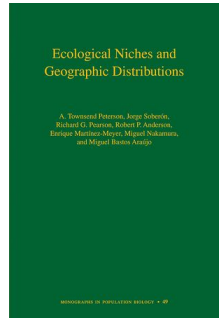
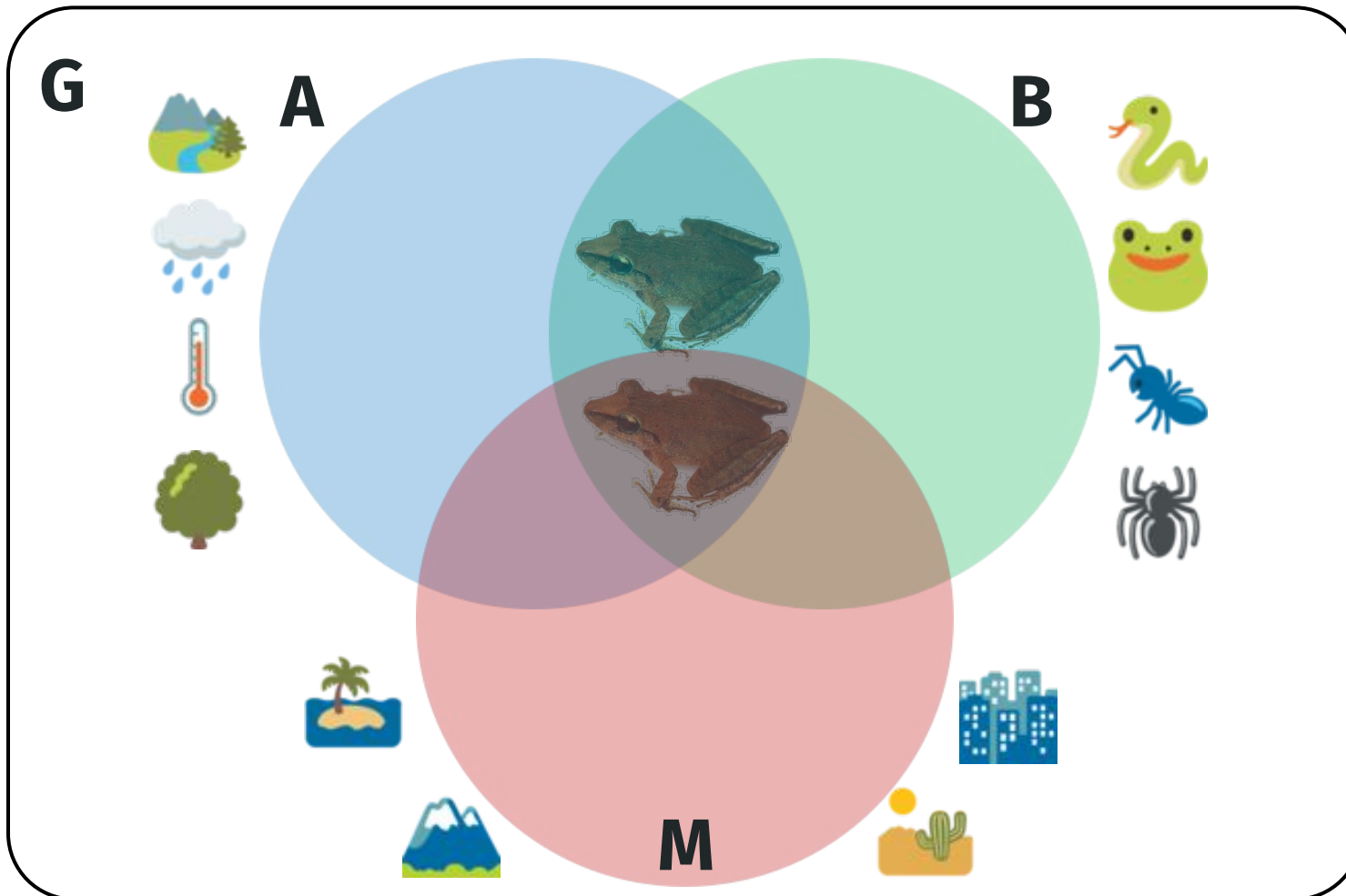
Nicho Ecológico



Peterson et al. (2011)

O que determina a distribuição de espécies?

Nicho Ecológico limitado pelo movimento



Peterson et al. (2011)

A teoria dos modelos...

O que determina a distribuição das espécies?

Teoria

$$\frac{1}{x_i^j} \frac{dx_i^j}{dt} = \underbrace{r_i(\bar{e}^j)}_{\text{Ambiente}} - \underbrace{\varphi_i^j(\bar{x}^j; \vec{R}_i^j)}_{\text{Interações Bióticas}} + \underbrace{\psi^j(\bar{x}; \mathbf{T})}_{\text{Movimento}}$$

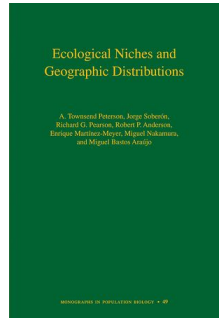
G

Ambiente
(fatores abióticos)

Interações
Bióticas

Movimento

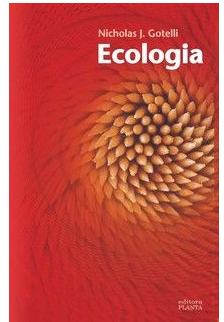
i = espécie
 j = **célula do raster**
 x = densidade
 t = tempo
 r = cresc. da pop.
 e = vetor de cond. amb.
 φ = termo de regulação
 R = interação com outras esp.
 T = matriz de transição (cap. de mov.)
 ψ = termo de movimento



Peterson et al. (2011)

O que determina a distribuição das espécies?

Ecologia de Populações



Gotelli (2007)

$$\frac{dN}{dt} = rN$$

Crescimento
exponencial

A taxa de crescimento per capita (r) não muda, mesmo se a população aumentar muito.

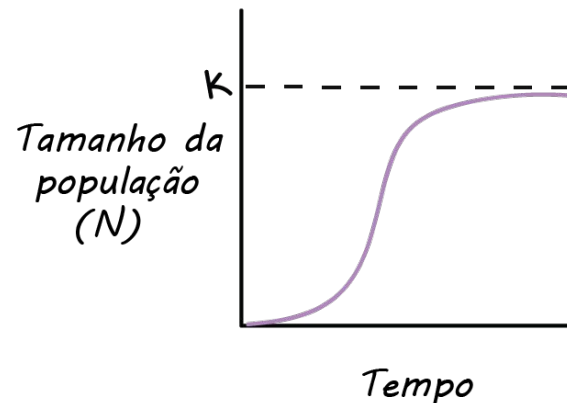
$$\frac{dN}{dt} = r_{\text{máx}} N$$



Crescimento
logístico

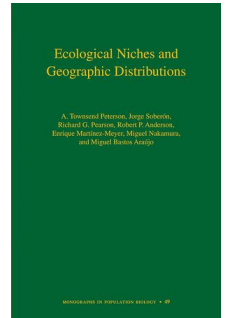
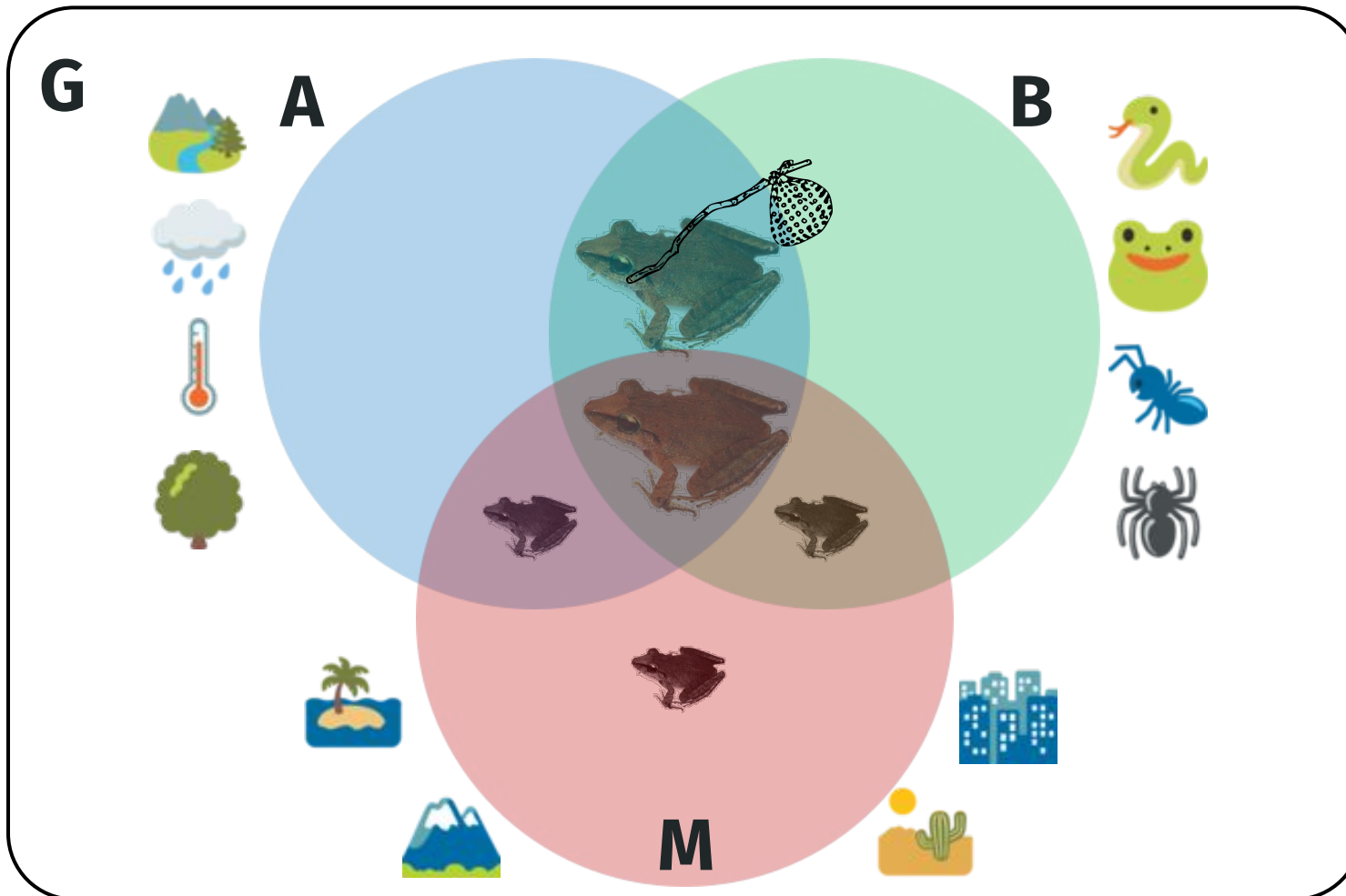
A taxa de crescimento per capita (r) diminui à medida que a população se aproxima de seu tamanho máximo.

$$\frac{dN}{dt} = r_{\text{máx}} \left(\frac{K-N}{K} \right) N$$



O que determina a distribuição de espécies?

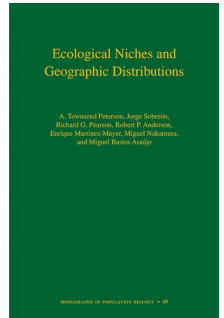
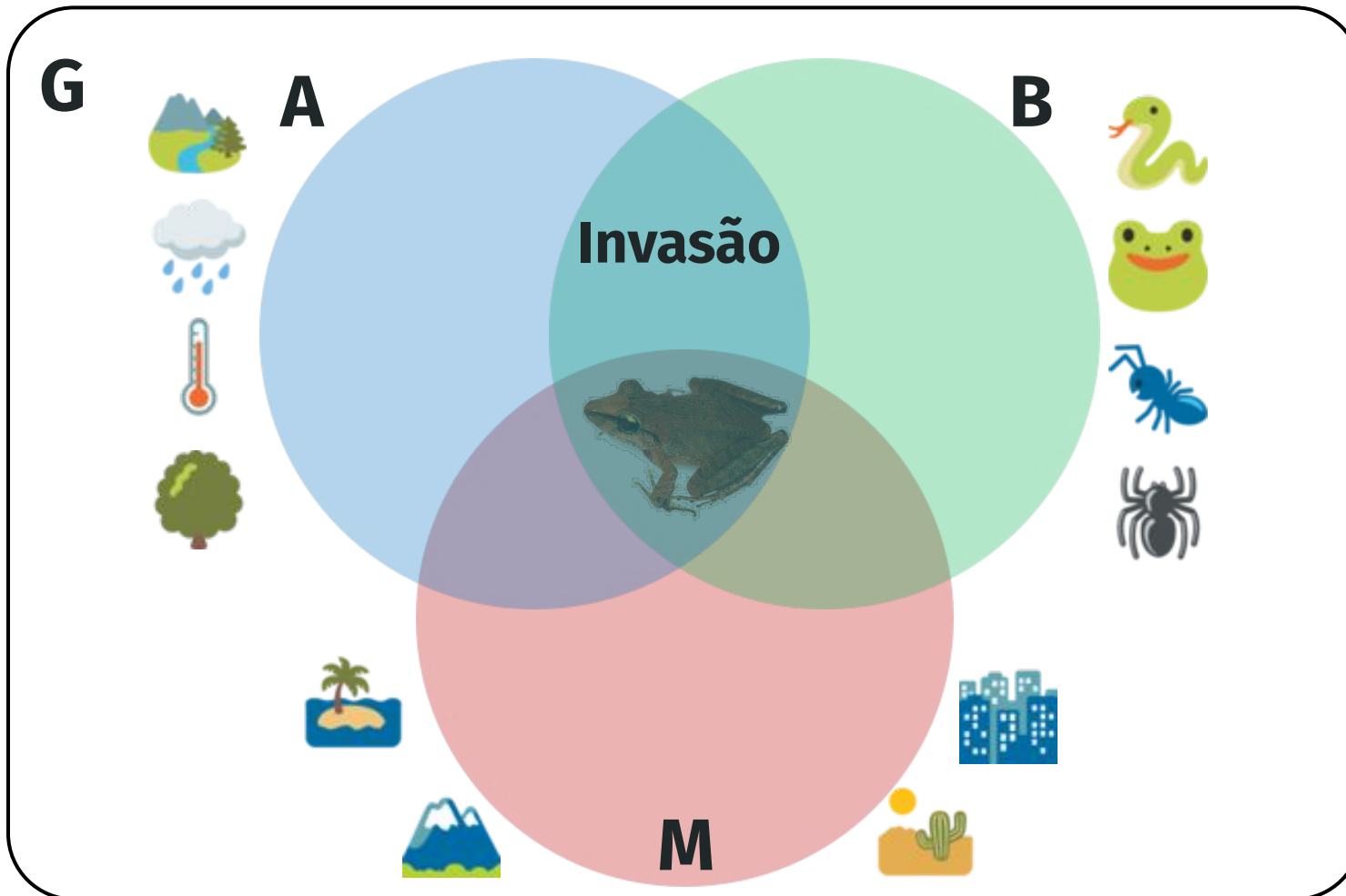
Populações fonte e ralo (*source-sink*)



Peterson et al. (2011)

O que determina a distribuição de espécies?

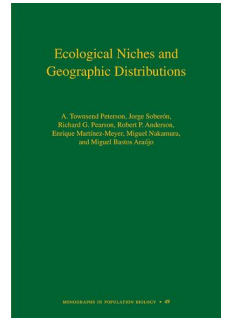
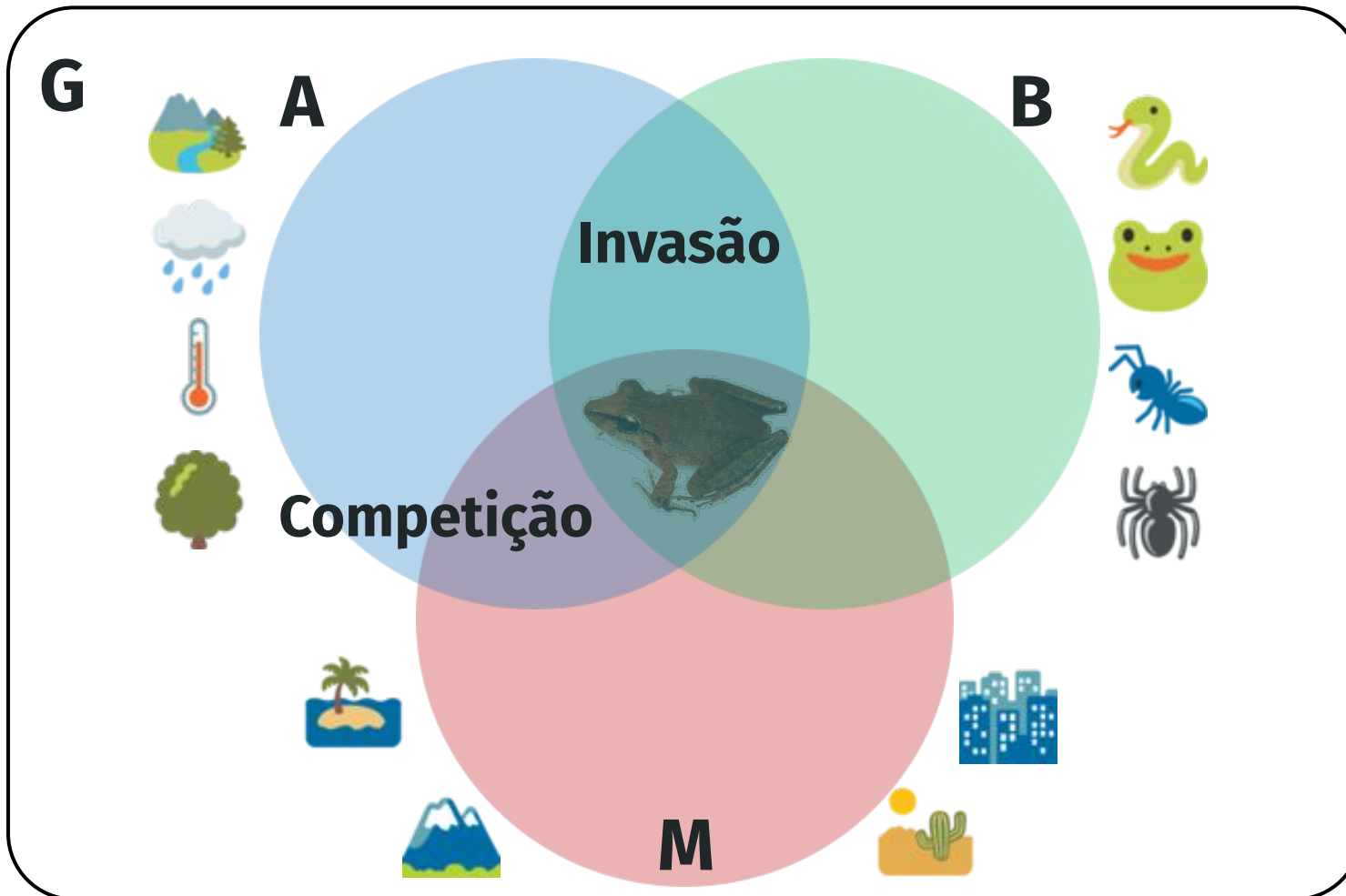
Populações fonte e ralo (*source-sink*)



Peterson et al. (2011)

O que determina a distribuição de espécies?

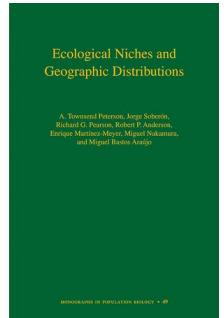
Populações fonte e ralo (*source-sink*)



Peterson et al. (2011)

O que determina a distribuição de espécies?

Populações fonte e ralo (*source-sink*)

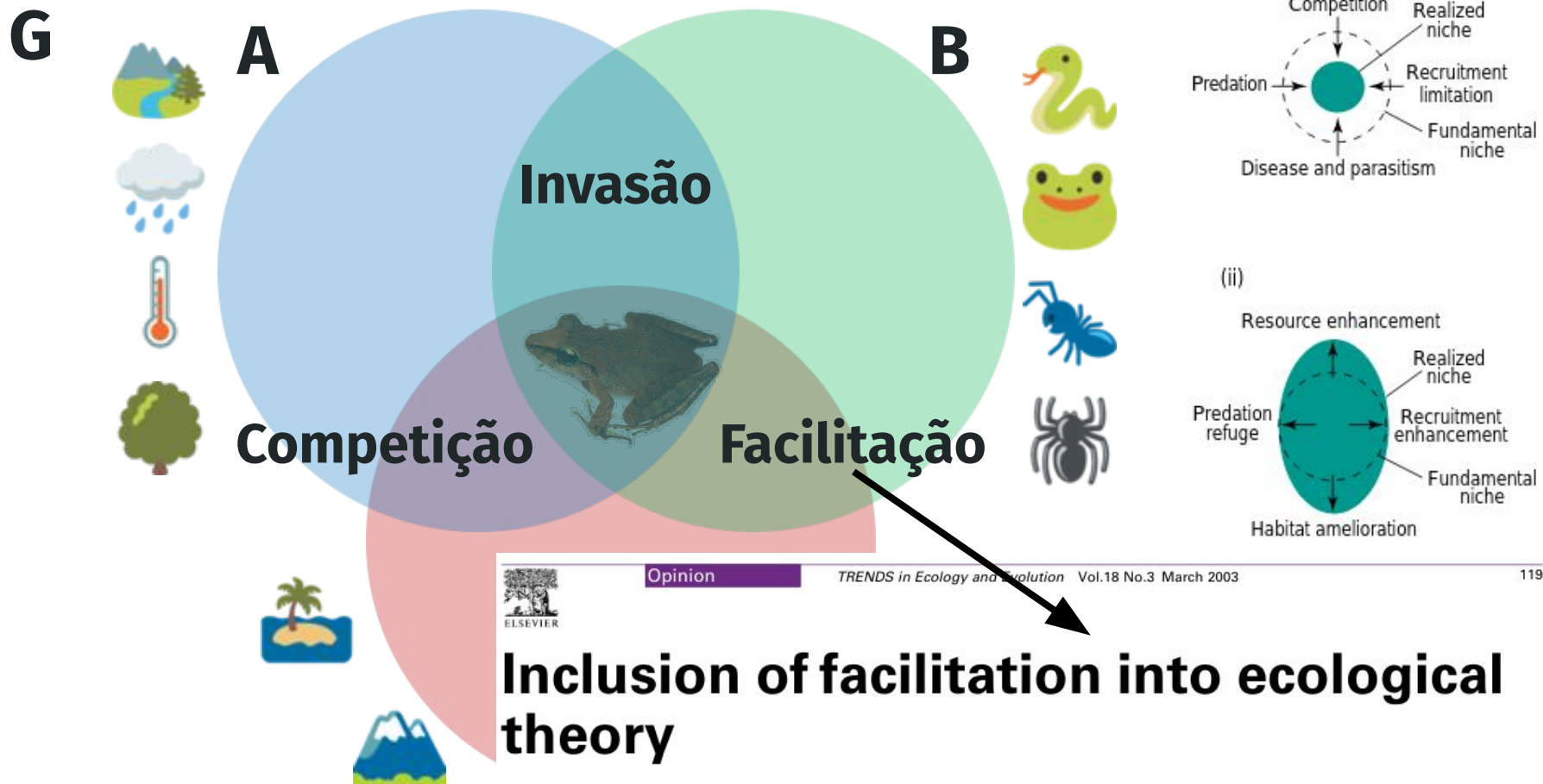


Peterson et al. (2011)



O que determina a distribuição de espécies?

Populações fonte e ralo (*source-sink*)



Opinion

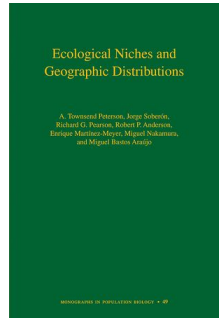
TRENDS in Ecology and Evolution Vol.18 No.3 March 2003

119

John F. Bruno¹, John J. Stachowicz² and Mark D. Bertness³

O que determina a distribuição de espécies?

Populações fonte e ralo (*source-sink*)

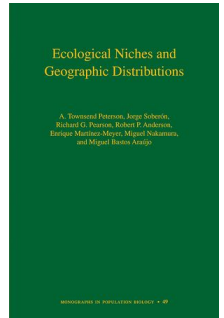


Peterson et al. (2011)

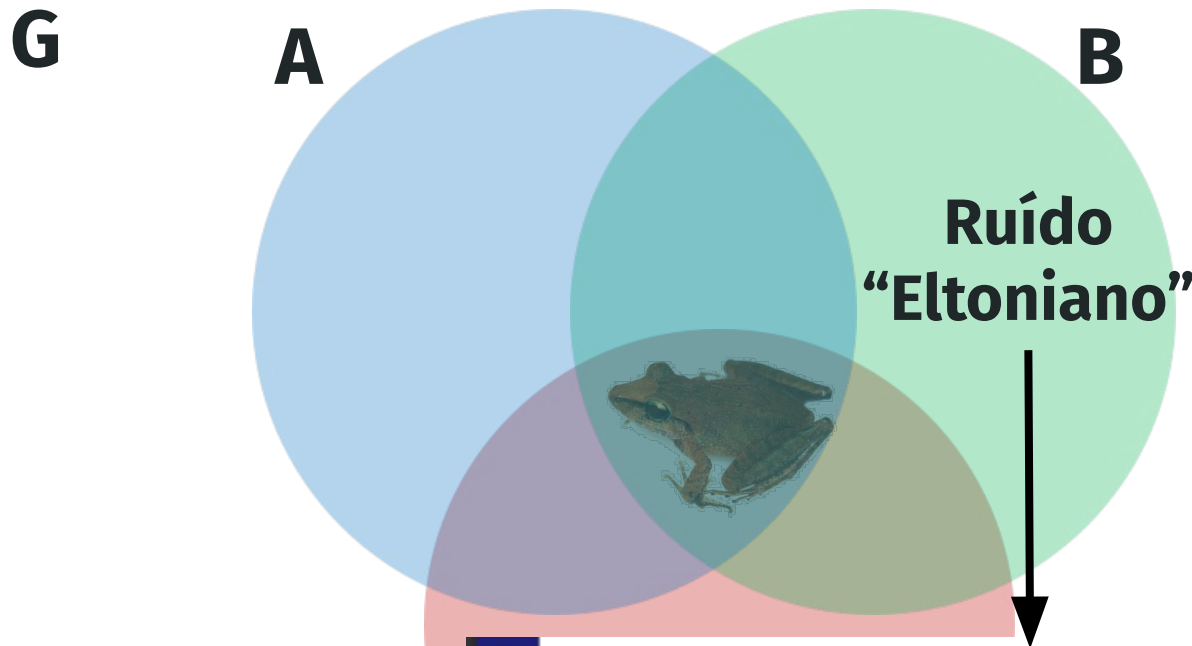
E as interações bióticas?

O que determina a distribuição de espécies?

Interações bióticas “ignoradas”



Peterson et al. (2011)



PNAS

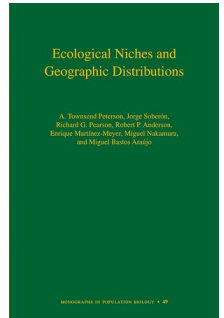
Niches and distributional areas: Concepts, methods, and assumptions

Jorge Soberón^{a,1} and Miguel Nakamura^b

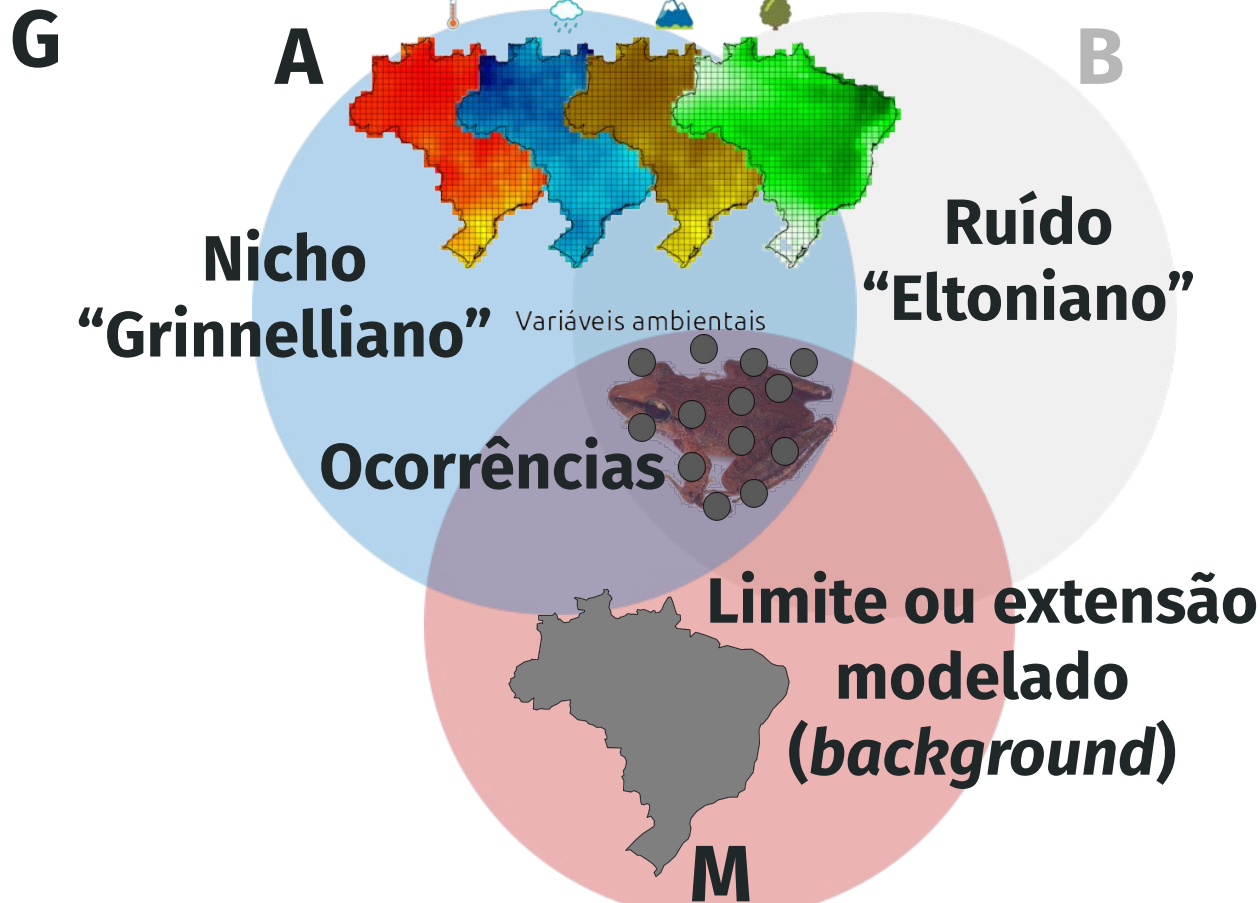
^aBiodiversity Institute, University of Kansas, Dyche Hall, 1345 Jayhawk Boulevard, Lawrence, KS 66045; and ^bCentro de Investigación en Matemáticas, A. C. Jalisco s/n, Col. Valenciana, Guanajuato, 36240, México

O que determina a distribuição de espécies?

Estimativa do nicho Grinnelliano realizado



Peterson et al. (2011)



Área em desenvolvimento

Como inserir as interações bióticas nos ENMs?

RESEARCH PAPER

WILEY *Journal of Biogeography*

Using biotic interactions in broad-scale estimates of species' distributions

Iulian Gherghel^{1,2,3} | François Brischoux⁴ | Monica Papeş⁵

BIOLOGICAL
REVIEWS

Cambridge
Philosophical Society

Open Access

The role of biotic interactions in shaping distributions and realised assemblages of species: implications for species distribution modelling

Mary Susanne Wisz, Julien Pottier, W. Daniel Kissling, Loïc Pellissier, Jonathan Lenoir, Christian F. Damgaard, Carsten F. Dormann, Mads C. Forchhammer, John-Arvid Grytnes ... [See all authors](#)

*Journal of
Biogeography*

Original Article | Full Access

The importance of biotic interactions in species distribution models: a test of the Eltonian noise hypothesis using parrots

Carlos B. de Araújo | Luiz Octavio Marcondes-Machado, Gabriel C. Costa

Ecology and Evolution

Open Access

ORIGINAL RESEARCH | Open Access | CC BY

Effects of biotic interactions on modeled species' distribution can be masked by environmental gradients

William Godsoe | Janet Franklin, F. Guillaume Blanchet

RESEARCH REVIEWS

WILEY *Global Ecology and Biogeography*

A Journal of
Macroecology

Biotic interactions in species distribution modelling: 10 questions to guide interpretation and avoid false conclusions

Carsten F. Dormann¹ | Maria Bobrowski² | D. Matthias Dehling³ |
David J. Harris⁴ | Florian Hartig^{1,5} | Heike Lischke⁶ | Marco D. Moretti⁷ |
Jörn Pagel⁸ | Stefan Pinkert⁹ | Matthias Schleuning¹⁰ |
Susanne I. Schmidt¹¹ | Christine S. Sheppard⁸ | Manuel J. Steinbauer^{12,13} |
Dirk Zeuss¹⁴ | Casper Kraan^{15,16}

Biotic interactions and climate in species distribution modelling

Daniel P. Bebber, Sarah J. Gurr

doi: <https://doi.org/10.1101/520320>

4. SDM passo a passo

SDM passo a passo

Estrutura dos ENMs

ECOGRAPHY

Review and synthesis

A standard protocol for reporting species distribution models

Damaris Zurell, Janet Franklin, Christian König, Phil J. Bouchet, Carsten F. Dormann, Jane Elith, Guillermo Fandos, Xiao Feng, Gurrutza Guillera-Aroita, Antoine Guisan, José J. Lahoz-Monfort, Pedro J. Leitão, Daniel S. Park, A. Townsend Peterson, Giovanni Rapacciuolo, Dirk R. Schmatz, Boris Schröder, Josep M. Serra-Díaz, Wilfried Thuiller, Katherine L. Yates, Niklaus E. Zimmermann and Cory Merow

Ecography

43: 1–17, 2020

doi: 10.1111/ecog.04960

Subject Editor: David Nogués-Bravo

Editor-in-Chief: Miguel Araújo

Accepted 30 April 2020

Conceptualisation

1

Data preparation

2

Model fitting

3

**Assessment of
model**

4

**Predictions in
space and time**

5

<https://doi.org/10.1111/ecog.04960>

Conceitualização

Perguntas associadas à distribuição das espécies

Teoria -> Perguntas -> Hipóteses ->
Estatística (modelos) -> Respostas

Conceitualização

Perguntas associadas à distribuição das espécies

Teoria -> Perguntas -> Hipóteses ->
Estatística (modelos) -> Respostas

1. Padrões de diversidade
2. Mudanças climáticas (futuro)
3. Mudanças climáticas (passado)
4. Invasão de espécies
5. Transmissão de doenças
6. Interações entre espécies
7. Processos de diversificação
8. Dispersão de espécies
9. Processos de extinção
10. Conservação-evolução do nicho
11. Testar hipóteses filogeográficas
12. Estabelecer refúgios climáticos
13. Estabelecer hotspots
14. Estabelecer áreas protegidas
15. Eficiência das áreas protegidas

SDM passo a passo

Estrutura dos ENMs

ECOGRAPHY

Review and synthesis

A standard protocol for reporting species distribution models

Damaris Zurell, Janet Franklin, Christian König, Phil J. Bouchet, Carsten F. Dormann, Jane Elith, Guillermo Fandos, Xiao Feng, Gurrutza Guillera-Aroita, Antoine Guisan, José J. Lahoz-Monfort, Pedro J. Leitão, Daniel S. Park, A. Townsend Peterson, Giovanni Rapacciuolo, Dirk R. Schmatz, Boris Schröder, Josep M. Serra-Díaz, Wilfried Thuiller, Katherine L. Yates, Niklaus E. Zimmermann and Cory Merow

Ecography

43: 1–17, 2020

doi: 10.1111/ecog.04960

Subject Editor: David Nogués-Bravo

Editor-in-Chief: Miguel Araújo

Accepted 30 April 2020

Conceptualisation

1

2

Data preparation

3

Model fitting

4

**Assessment of
model**

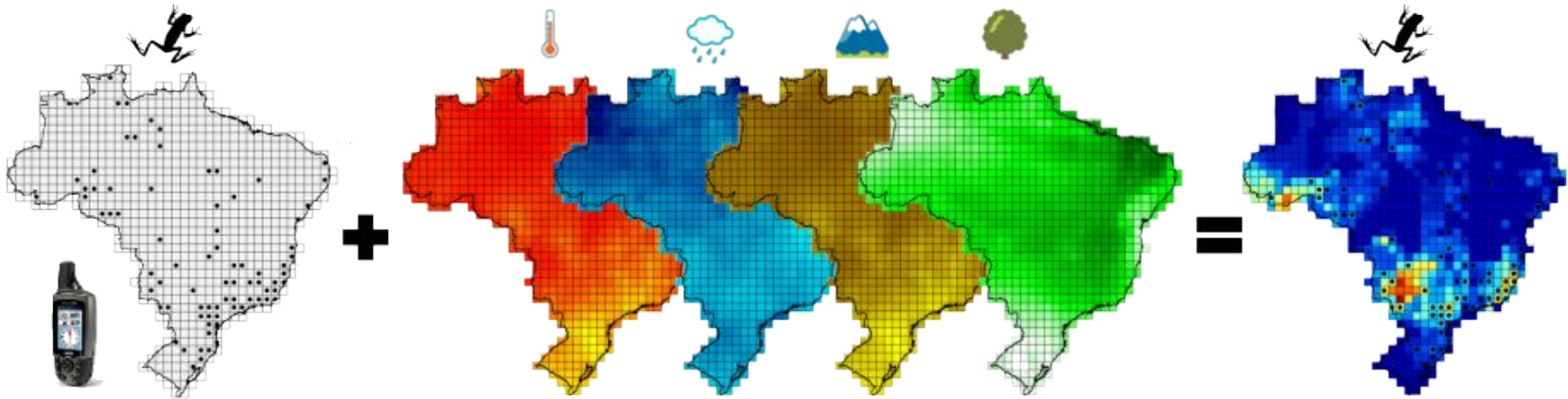
5

**Predictions in
space and time**

<https://doi.org/10.1111/ecog.04960>

Modelos de Nicho Ecológico (ENMs)

Preparação dos dados



"Ocorrências"

Variáveis ambientais

Adequabilidade

species	lon	lat
sp1	-40.2	-23.4
sp1	-38.8	-20.3
sp1	-43.3	-19.9

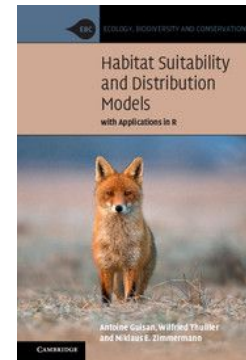
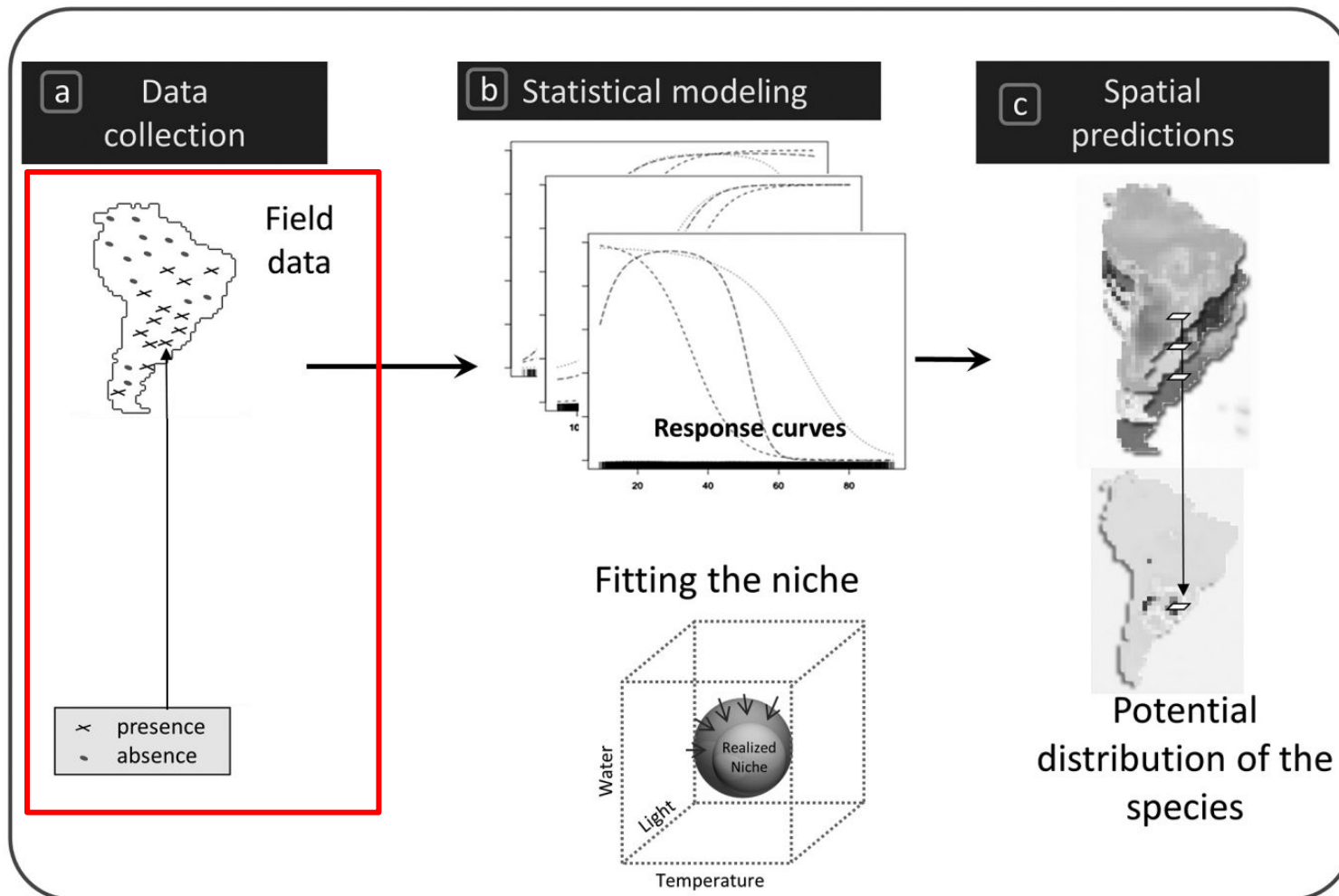
variaveis
temperatura
precipitação
relevo

valores
0
até
1

5. Dados de entrada: ocorrências e variáveis

Ocorrências

Visão geral



Guisan et al. (2017)

Ocorrências

Fontes

1. Coletas em campo



Ocorrências

Fontes

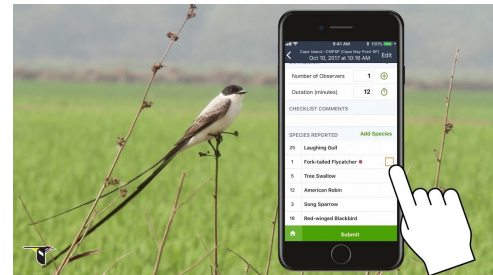
1. Coletas em campo
2. Literatura (artigos, data papers, ...)



Ocorrências

Fontes

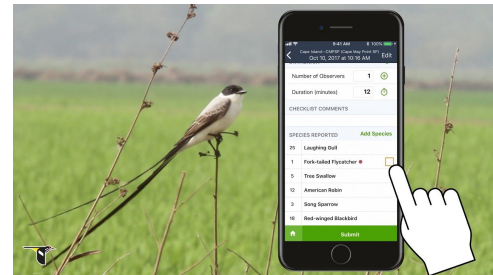
1. Coletas em campo
2. Literatura (artigos, data papers, ...)
3. Naturalistas e ciência cidadã (e-Bird, iNaturalist, ...)



Ocorrências

Fontes

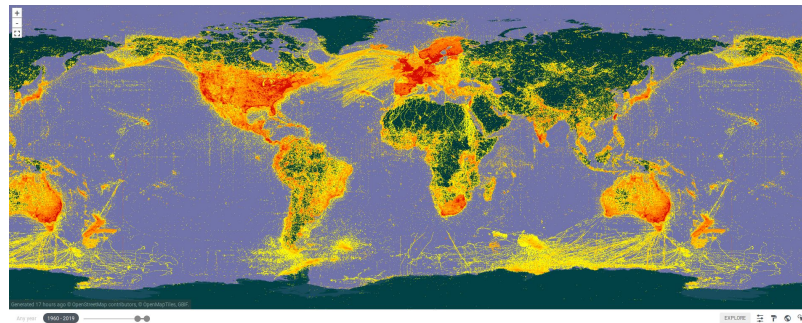
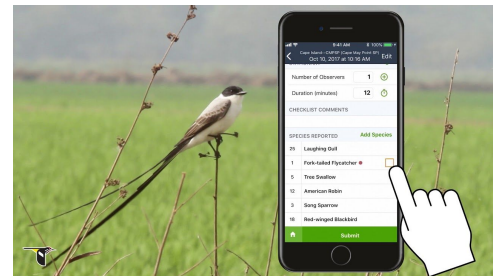
1. Coletas em campo
2. Literatura (artigos, data papers, ...)
3. Naturalistas e ciência cidadã (e-Bird, iNaturalist, ...)
4. Coleções científicas e museus (Museu Nacional, MZUSP, CFHB, ...)



Ocorrências

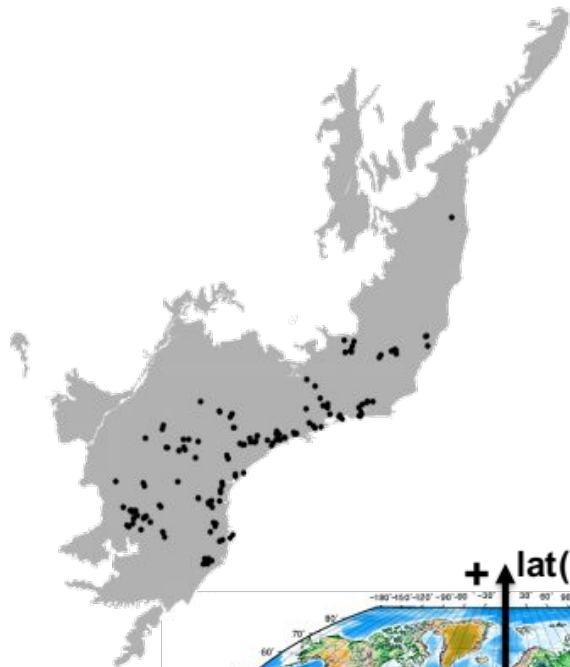
Fontes

1. Coletas em campo
2. Literatura (artigos, data papers, ...)
3. Naturalistas e ciência cidadã (e-Bird, iNaturalist, ...)
4. Coleções científicas e museus (Museu Nacional, MZUSP, CFHB, ...)
5. Banco de dados (GBIF, SpeciesLink, ...)

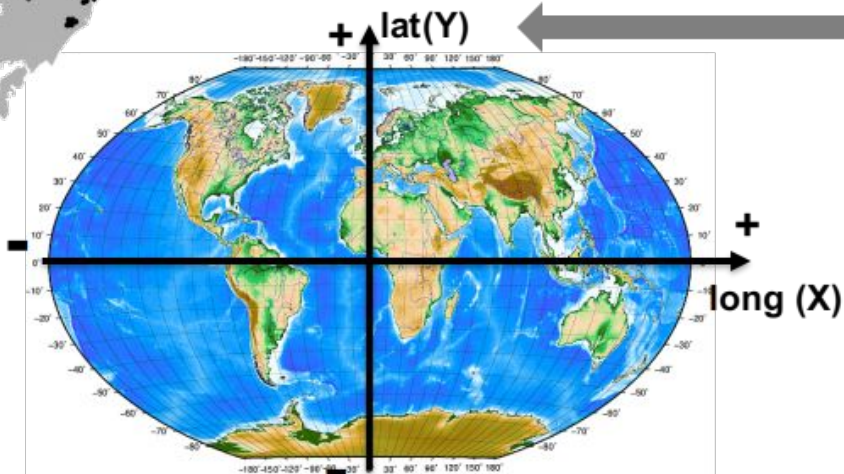


Ocorrências

Formato



sp	long	lat
vitreorana_uranoscopa	-52.8300	-26.4400
vitreorana_uranoscopa	-52.6836	-27.1253
vitreorana_uranoscopa	-52.5569	-26.5642
vitreorana_uranoscopa	-52.4500	-26.5667
vitreorana_uranoscopa	-52.4489	-27.0689
vitreorana_uranoscopa	-52.4147	-26.8667

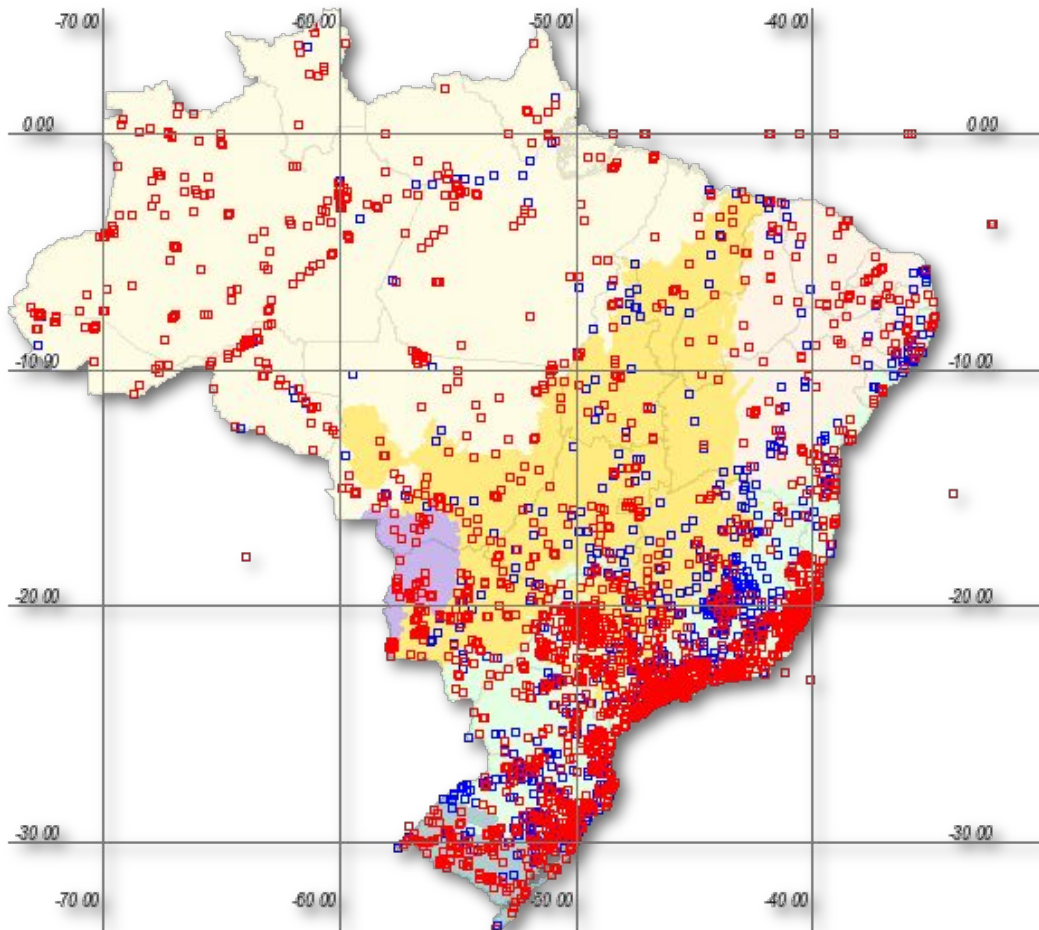


Desafios:

Viés de amostragem

Ocorrências

Viés de amostragem

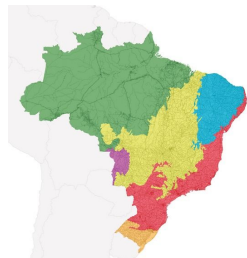


Boana faber

*species*link

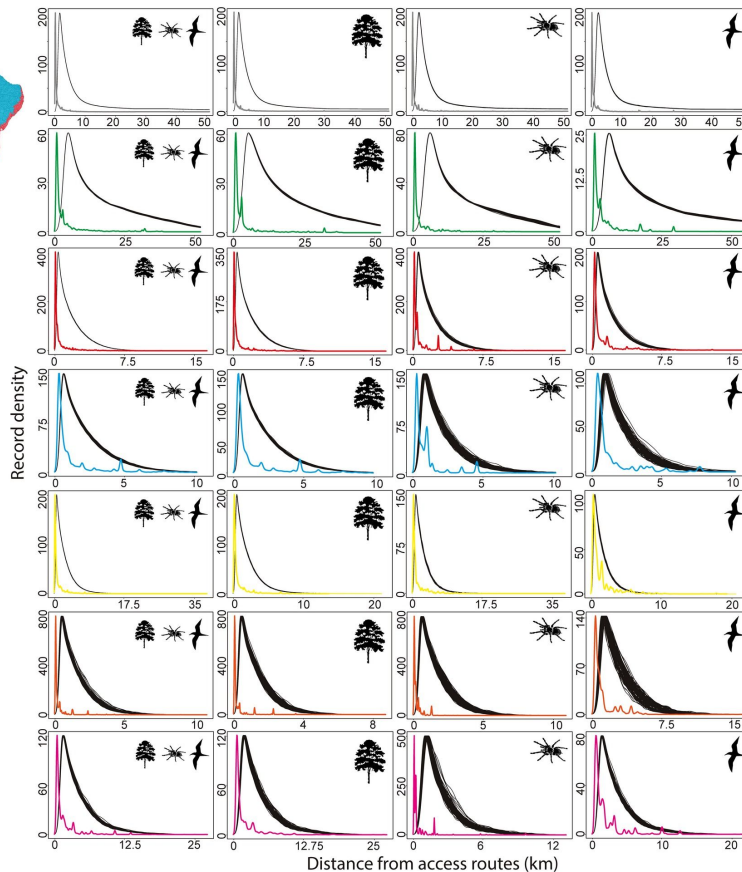
Ocorrências

Viés de amostragem



■ Amazonia
■ Atlantic Rainforest
■ Caatinga
■ Cerrado
■ Pampa
■ Pantanal
— Access routes

● ● ● ● ● ● ●
 All groups
 Angiosperms
 Arthropods
 Vertebrates

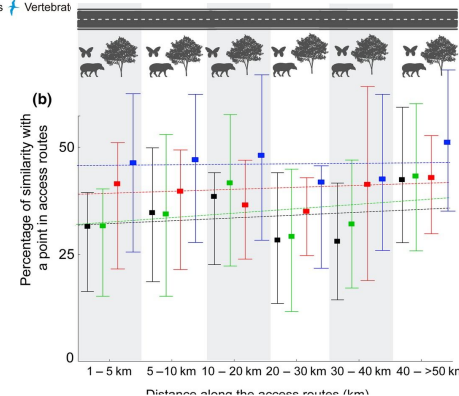
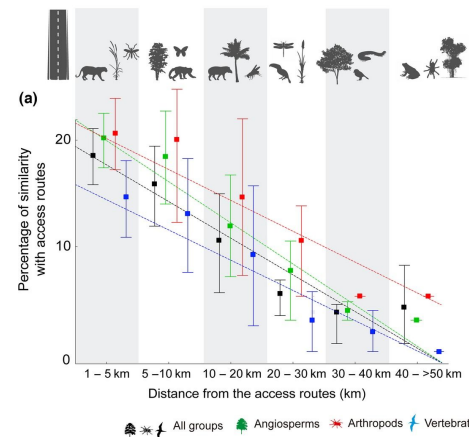


Diversity and Distributions, (Diversity Distrib.) (2016) 22, 1232–1244



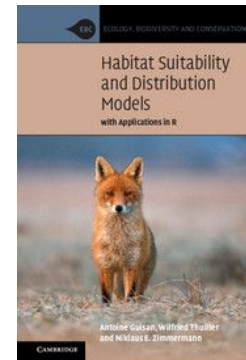
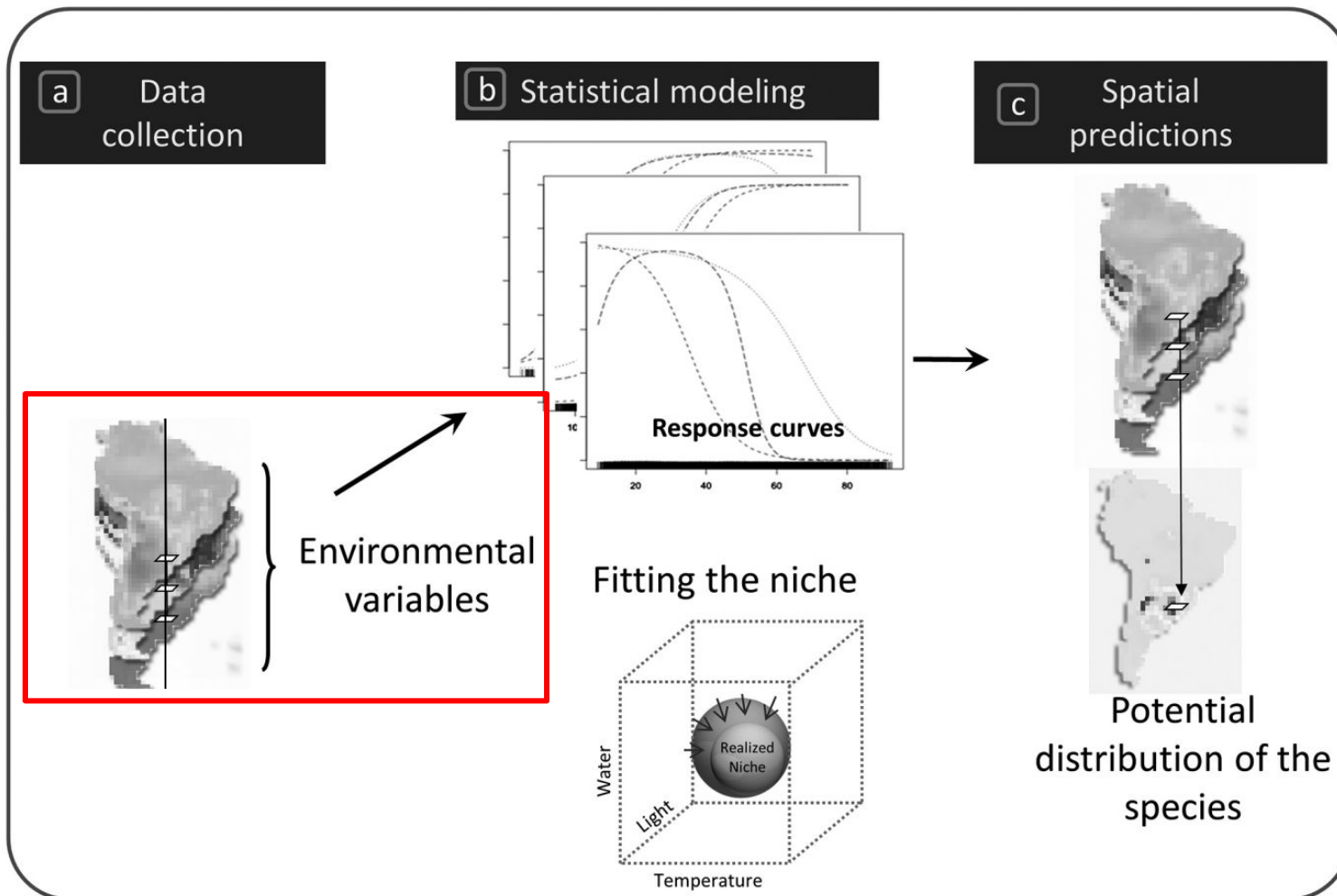
The strong influence of collection bias on biodiversity knowledge shortfalls of Brazilian terrestrial biodiversity

Ubirajara Oliveira^{1,2*}, Adriano Pereira Paglia³, Antonio D. Brescovit⁴,
 Claudio J. B. de Carvalho⁵, Daniel Paiva Silva⁶, Daniella T. Rezende⁷,
 Felipe Sá Fortes Leite⁸, João Aguiar Nogueira Batista⁹,
 João Paulo Peixoto Pena Barbosa⁴, João Renato Stehmann⁹,
 John S. Ascher¹⁰, Marcelo Ferreira de Vasconcelos^{11,12},
 Paulo De Marco Jr¹³, Peter Löwenberg-Neto¹⁴, Priscila Guimarães Dias¹⁵,
 Viviane Gianluppi Ferro¹³ and Adalberto J. Santos²



Variáveis ambientais

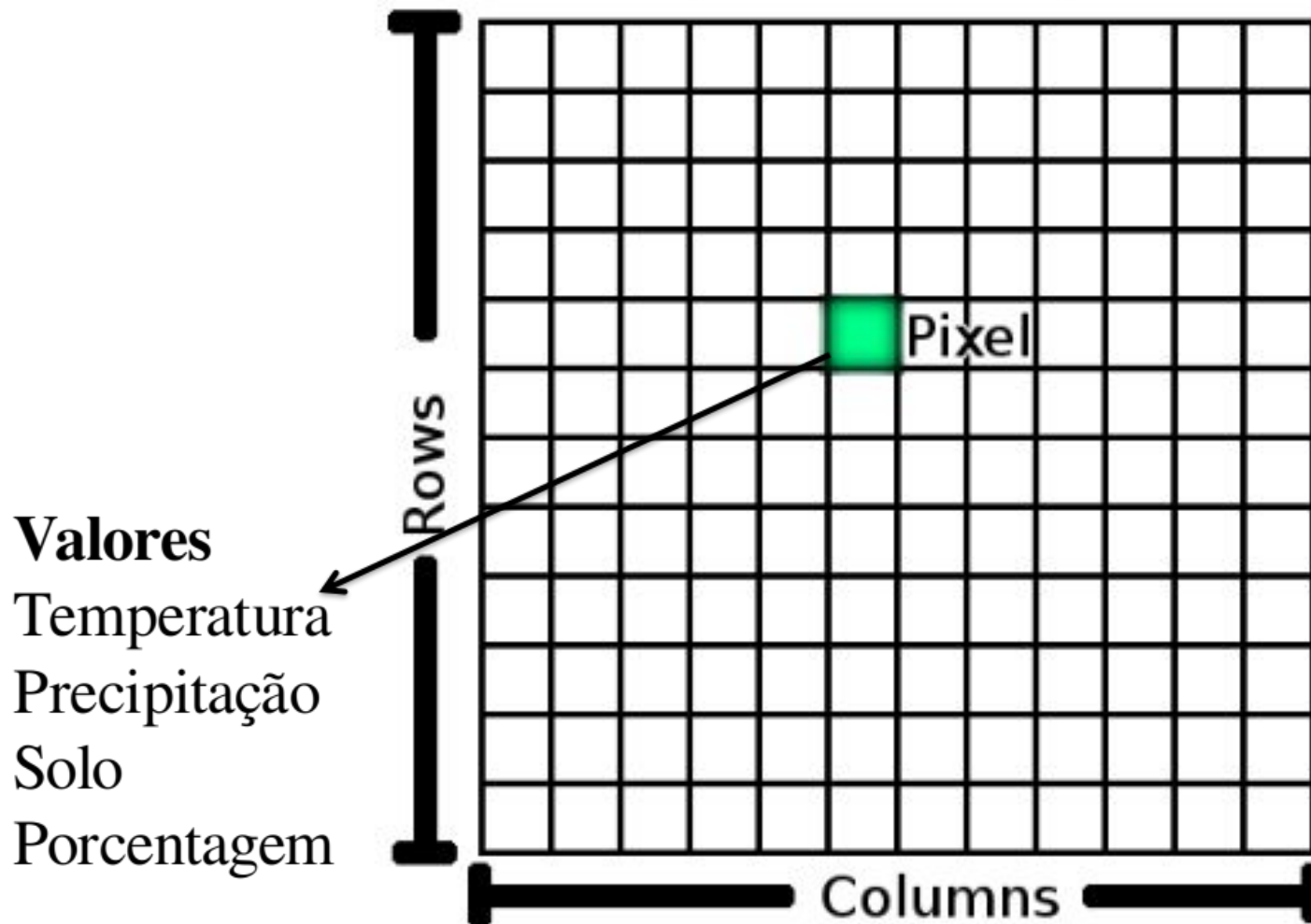
Visão geral



Guisan et al. (2017)

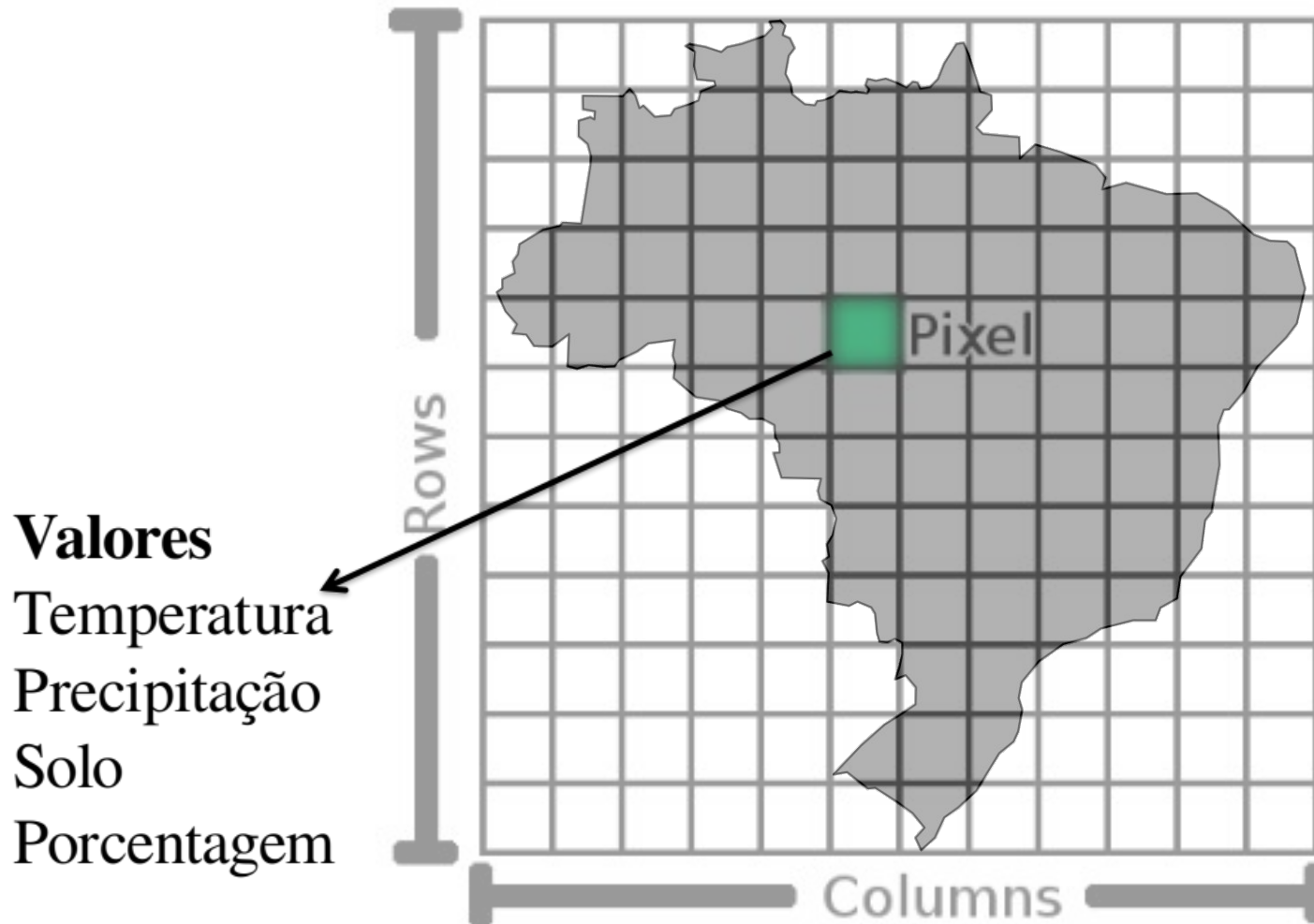
Variáveis ambientais

Raster - Extensão e resolução



Variáveis ambientais

Raster - Extensão e resolução



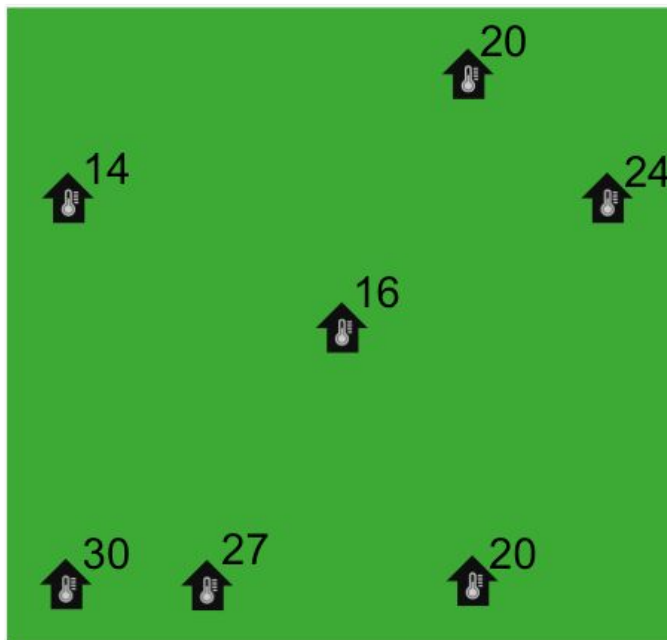
Variáveis ambientais

Raster - Interpolação

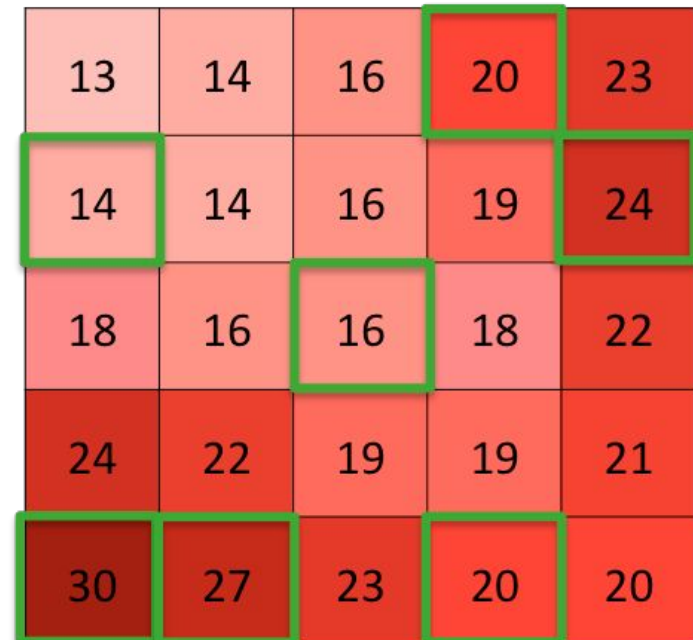


<https://support.bccvl.org.au/support/home>

Temperature (°C) at stations



Temperature (°C) interpolated



Adapted from http://planet.botany.uwc.ac.za/nisl/GIS/spatial/chap_1_11.h

Variáveis ambientais

WorldClim - Bioclimáticas

WorldClim - Global Climate Data

Free climate data for ecological modeling and GIS

Contact

Home

Bioclimatic variables

Bioclimatic variables are derived from the monthly temperature and rainfall values in order to generate more biologically meaningful variables. These are often used in [species distribution modeling](#) and related ecological modeling techniques. The bioclimatic variables represent annual trends (e.g., mean annual temperature, annual precipitation) seasonality (e.g., annual range in temperature and precipitation) and extreme or limiting environmental factors (e.g., temperature of the coldest and warmest month, and precipitation of the wet and dry quarters). A quarter is a period of three months (1/4 of the year).

They are coded as follows:

BIO1 = Annual Mean Temperature
BIO2 = Mean Diurnal Range (Mean of monthly (max temp - min temp))
BIO3 = Isothermality ($BIO2/BIO7$) (* 100)
BIO4 = Temperature Seasonality (standard deviation * 100)
BIO5 = Max Temperature of Warmest Month
BIO6 = Min Temperature of Coldest Month
BIO7 = Temperature Annual Range (BIO5-BIO6)
BIO8 = Mean Temperature of Wettest Quarter
BIO9 = Mean Temperature of Driest Quarter
BIO10 = Mean Temperature of Warmest Quarter
BIO11 = Mean Temperature of Coldest Quarter
BIO12 = Annual Precipitation
BIO13 = Precipitation of Wettest Month
BIO14 = Precipitation of Driest Month
BIO15 = Precipitation Seasonality (Coefficient of Variation)
BIO16 = Precipitation of Wettest Quarter
BIO17 = Precipitation of Driest Quarter
BIO18 = Precipitation of Warmest Quarter
BIO19 = Precipitation of Coldest Quarter

BIO01 = Temperatura média anual
BIO02 = Variação Diurna Média de Temperatura (Média mensal (Tmax-Tmin))
BIO03 = Isotermalidade ((BIO2/BIO7) (* 100))
BIO04 = Sazonalidade da Temperatura (desvio padrão * 100)
BIO05 = Temperatura máxima do mês mais quente
BIO06 = Temperatura mínima do mês mais frio
BIO07 = Amplitude térmica anual (BIO5-BIO6)
BIO08 = Temperatura média do trimestre mais úmido
BIO09 = Temperatura média do trimestre mais seco
BIO10 = Temperatura média do trimestre mais quente
BIO11 = Temperatura média do trimestre mais frio

Temperatura

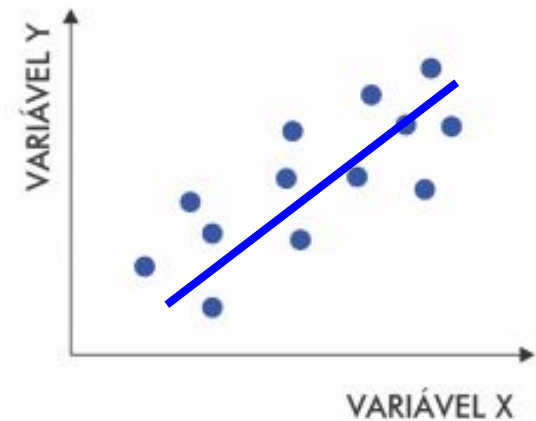
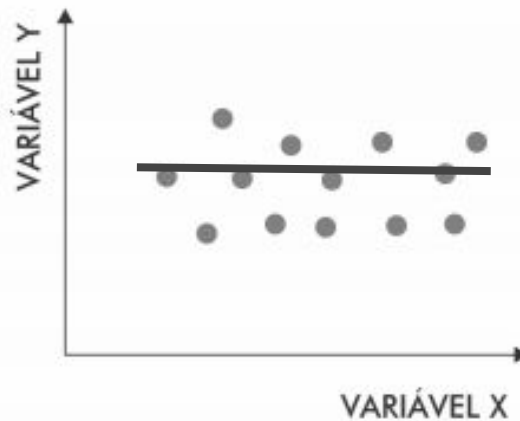
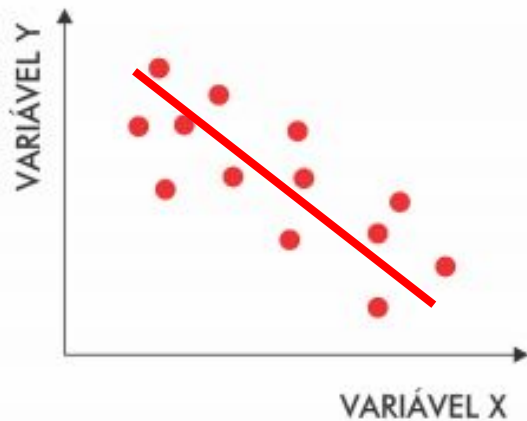
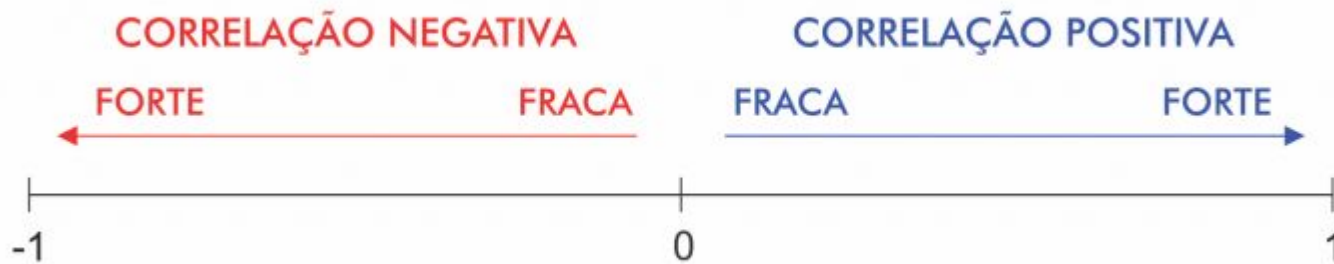
BIO12 = Precipitação Anual
BIO13 = Precipitação do mês mais chuvoso
BIO14 = Precipitação do mês mais seco
BIO15 = Sazonalidade da Precipitação (coeficiente de variação)
BIO16 = Precipitação do trimestre mais chuvoso
BIO17 = Precipitação do trimestre mais seco
BIO18 = Precipitação do trimestre mais quente
BIO19 = Precipitação do trimestre mais frio

Precipitação

Desafios: Colinearidade

Variáveis ambientais

Colinearidade - Correlação



SDM passo a passo

Estrutura dos ENMs

ECOGRAPHY

Review and synthesis

A standard protocol for reporting species distribution models

Damaris Zurell, Janet Franklin, Christian König, Phil J. Bouchet, Carsten F. Dormann, Jane Elith, Guillermo Fandos, Xiao Feng, Gurrutza Guillera-Aroita, Antoine Guisan, José J. Lahoz-Monfort, Pedro J. Leitão, Daniel S. Park, A. Townsend Peterson, Giovanni Rapacciuolo, Dirk R. Schmatz, Boris Schröder, Josep M. Serra-Díaz, Wilfried Thuiller, Katherine L. Yates, Niklaus E. Zimmermann and Cory Merow

Ecography

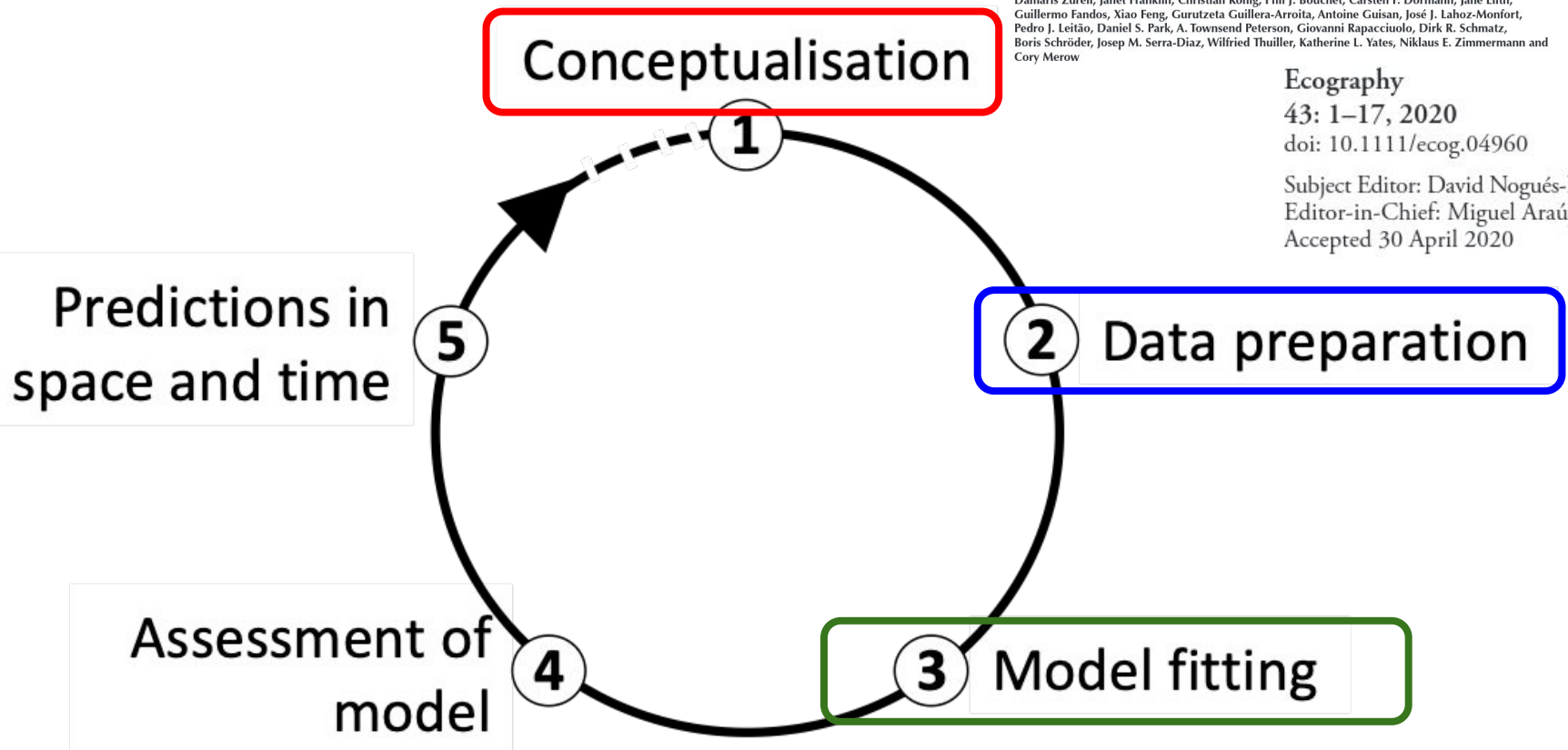
43: 1–17, 2020

doi: 10.1111/ecog.04960

Subject Editor: David Nogués-Bravo

Editor-in-Chief: Miguel Araújo

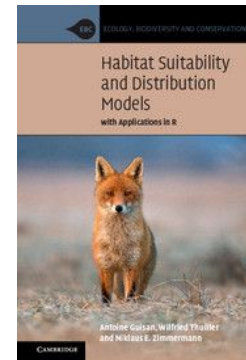
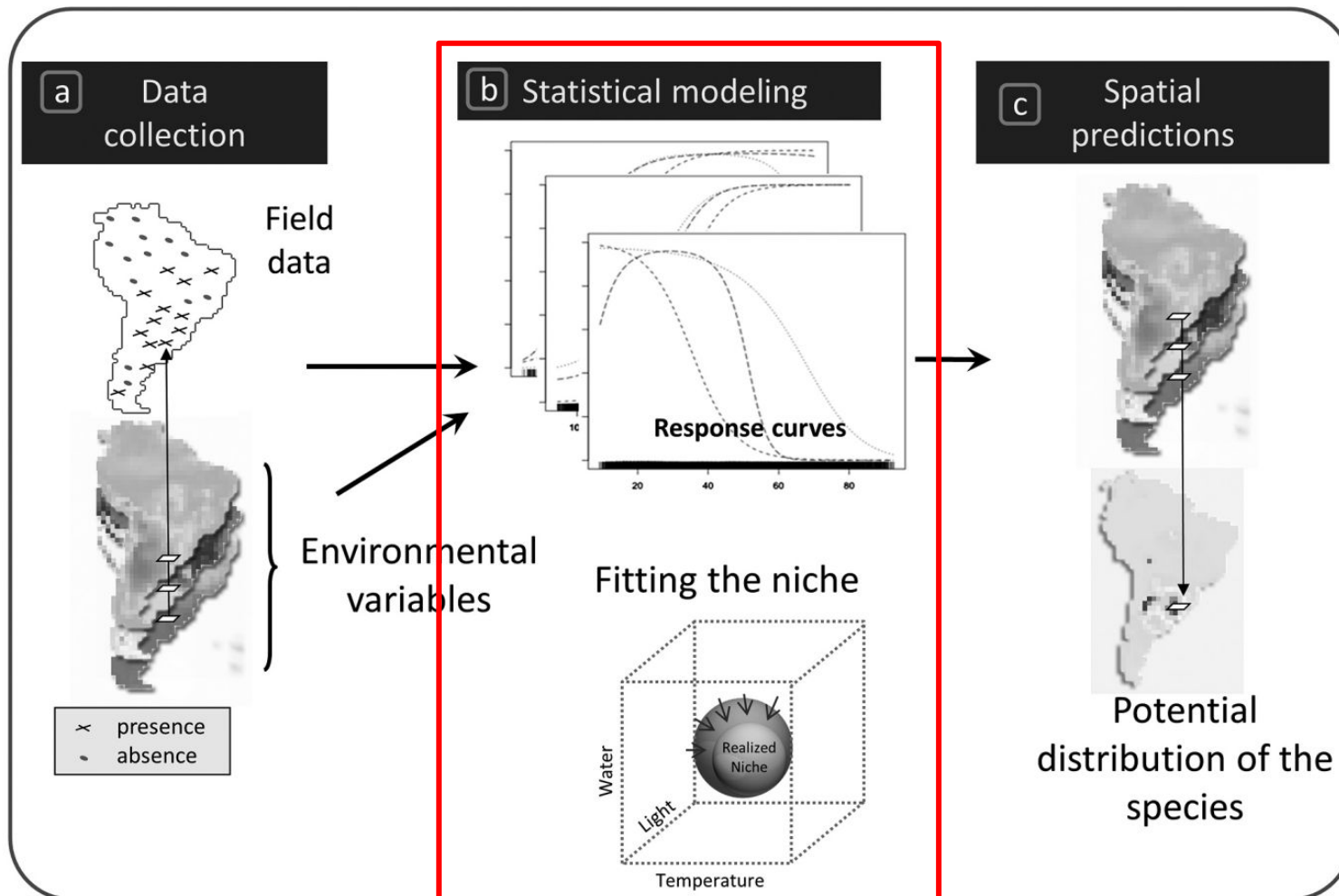
Accepted 30 April 2020



6. Ajuste dos modelos

Ajuste dos ENMs

Algoritmos estimam o nicho realizado



Guisan et al. (2017)

Ajuste dos ENMs

Muitos tipos de algoritmos

Lima-Ribeiro &
Diniz-Filho (2013)



Apenas presença

Bioclim

Aquário

Dist. Euclidiana

Dist. Mahalanobis

Domain (dist. Gower)

ENFA (ecological niche factor analysis)

Presença/Background

GARP (genetic algorithm for rule-set production)

Maxent (maximum entropy)

SVM (support vector machine)

Aprendizado de Máquina
(machine learning)
"cofre"

Presença/Ausência

Estatístico ("turbina")

GLMZ (generalized linear model)

GAM (generalized additive model)

FDA (flexible discriminant analysis)

MARS (multivariate adaptive reg. splines)

BRT (boosted regression trees)

→ **GBM** (gradient boosting machine)

CART (classification and regression trees)

RDNFOR (random forest)

NNET (neural networks)

→ **ANN** (artificial neural networks)

Ajuste dos ENMs

Mais utilizado - MaxEnt



Lima-Ribeiro &
Diniz-Filho (2013)

Apenas presença

Aquário

Bioclim
Dist. Euclidiana
Dist. Mahalanobis
Domain (dist. Gower)
ENFA (ecological niche factor analysis)

Presença/Background

GARP (genetic algorithm for rule-set production)

Maxent (maximum entropy)

SVM (support vector machine)

Aprendizado de Máquina
(machine learning)
"cofre"

Presença/Ausência

Estatístico ("turbina")

GLMZ (generalized linear model)

GAM (generalized additive model)

FDA (flexible discriminant analysis)

MARS (multivariate adaptive reg. splines)

BRT (boosted regression trees)

→ GBM (gradient boosting machine)

CART (classification and regression trees)

RDNFOR (random forest)

NNET (neural networks)

→ ANN (artificial neural networks)

Ajuste dos ENMs

Apenas Presença

Lima-Ribeiro &
Diniz-Filho (2013)



Apenas presença

Bioclim

Aquário

Dist. Euclidiana

Dist. Mahalanobis

Domain (dist. Gower)

ENFA (ecological niche factor analysis)

Presença/Background

GARP (genetic algorithm for rule-set production)

Maxent (maximum entropy)

SVM (support vector machine)

Aprendizado de Máquina
(machine learning)
"cofre"

Presença/Ausência

Estatístico ("turbina")

GLMZ (generalized linear model)

GAM (generalized additive model)

FDA (flexible discriminant analysis)

MARS (multivariate adaptive reg. splines)

BRT (boosted regression trees)

→ GBM (gradient boosting machine)

CART (classification and regression trees)

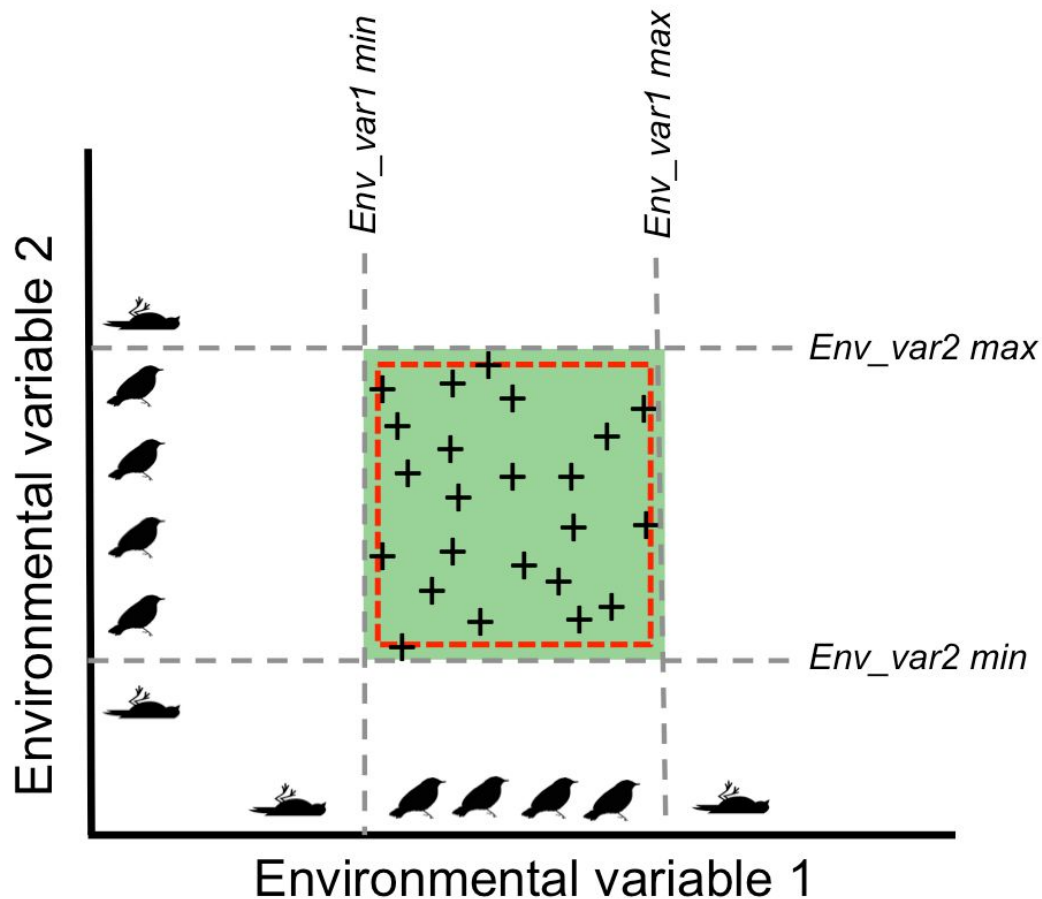
RDNFOR (random forest)

NNET (neural networks)

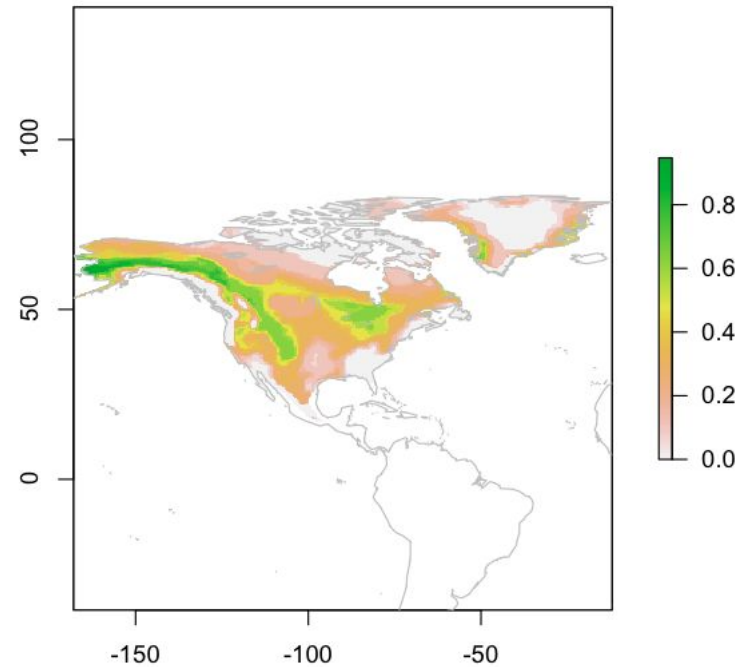
→ ANN (artificial neural networks)

Ajuste dos ENMs

BIOCLIM - Envelope Climático



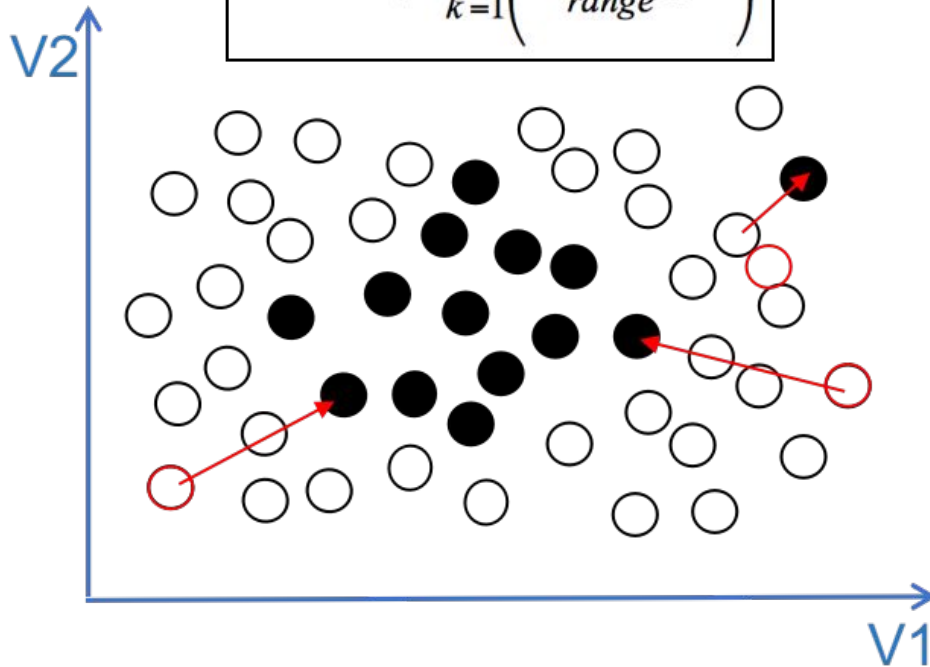
Lima-Ribeiro &
Diniz-Filho (2013)



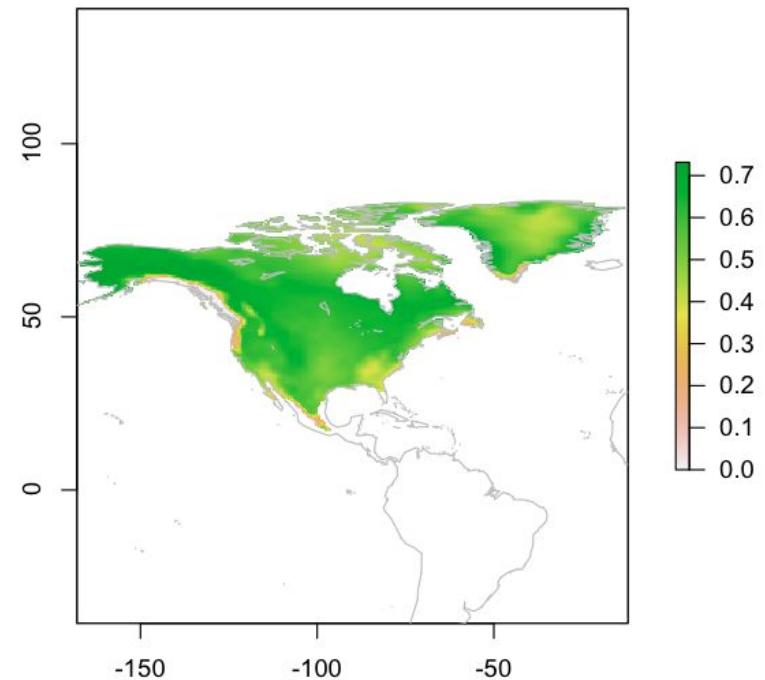
Ajuste dos ENMs

DOMAIN - Distância de Gower

$$d_{AB} = \frac{1}{V} \sum_{k=1}^V \left(\frac{|A_K - B_K|}{\text{range}^K} \right)$$



Lima-Ribeiro &
Diniz-Filho (2013)



Ajuste dos ENMs

Presença/Background (plano de fundo)

Lima-Ribeiro &
Diniz-Filho (2013)



Apenas presença

Bioclim

Aquário

Dist. Euclidiana

Dist. Mahalanobis

Domain (dist. Gower)

ENFA (ecological niche factor analysis)

Presença/Background

GARP (genetic algorithm for rule-set production)

Maxent (maximum entropy)

SVM (support vector machine)

Aprendizado de Máquina
(machine learning)
"cofre"

Presença/Ausência

Estatístico ("turbina")

GLMZ (generalized linear model)

GAM (generalized additive model)

FDA (flexible discriminant analysis)

MARS (multivariate adaptive reg. splines)

BRT (boosted regression trees)

→ **GBM** (gradient boosting machine)

CART (classification and regression trees)

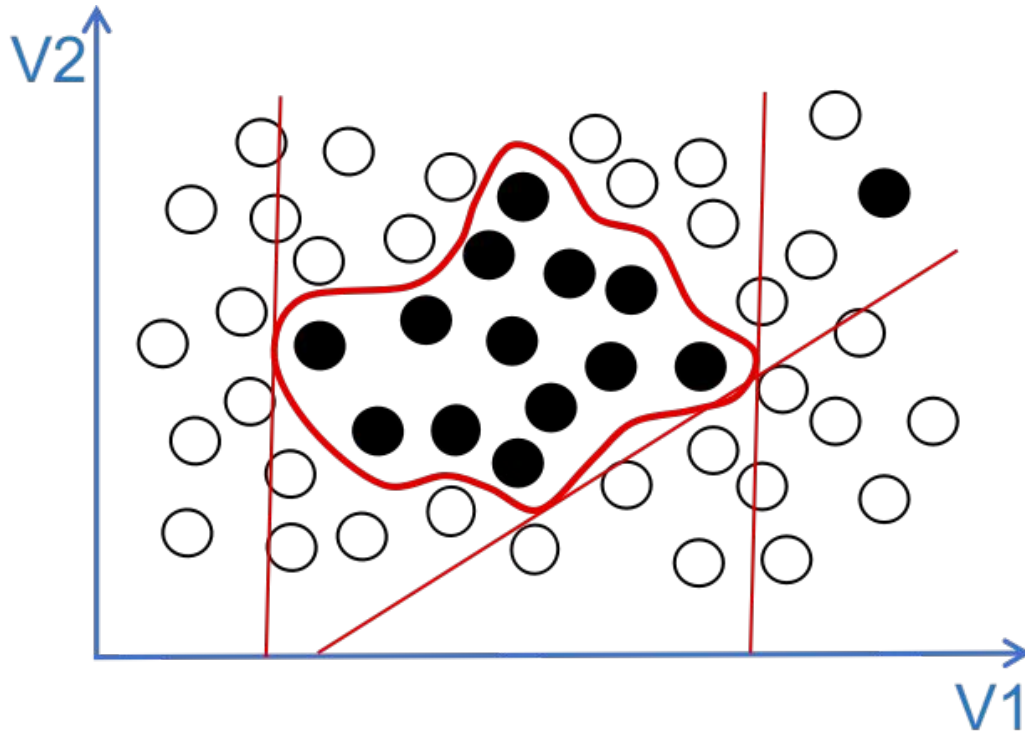
RDNFOR (random forest)

NNET (neural networks)

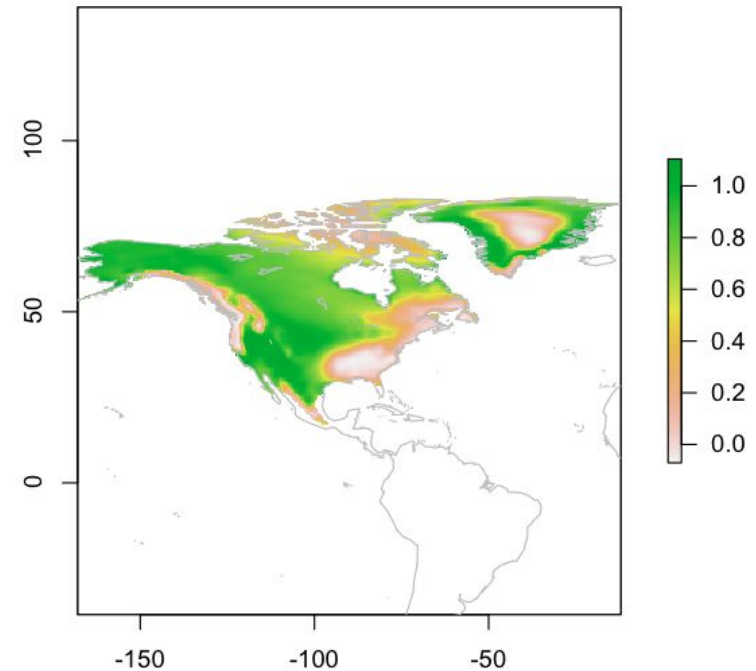
→ **ANN** (artificial neural networks)

Ajuste dos ENMs

Support Vector Machine (SVM)

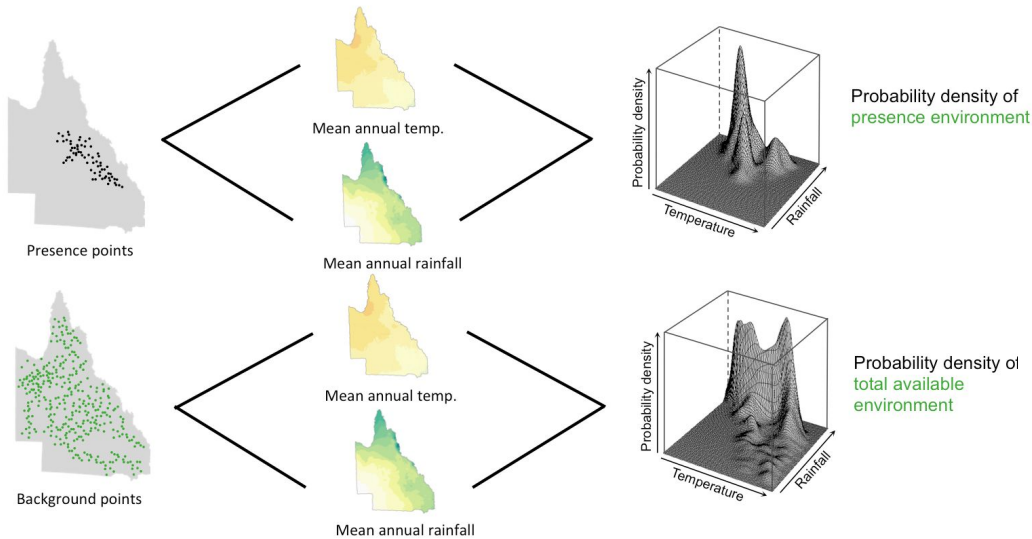


Lima-Ribeiro &
Diniz-Filho (2013)



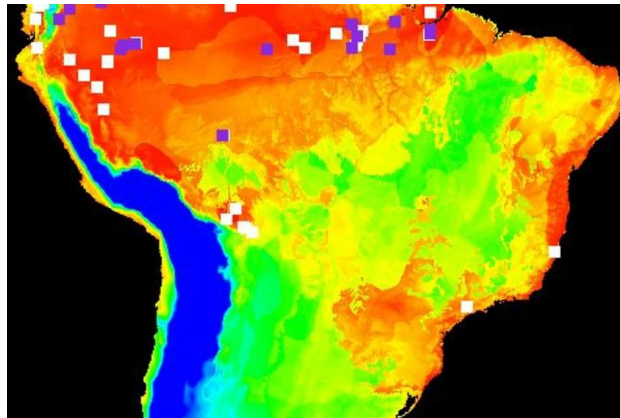
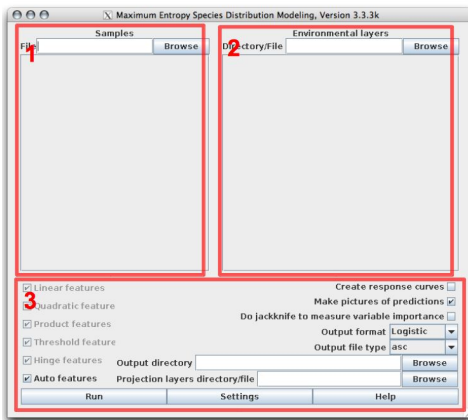
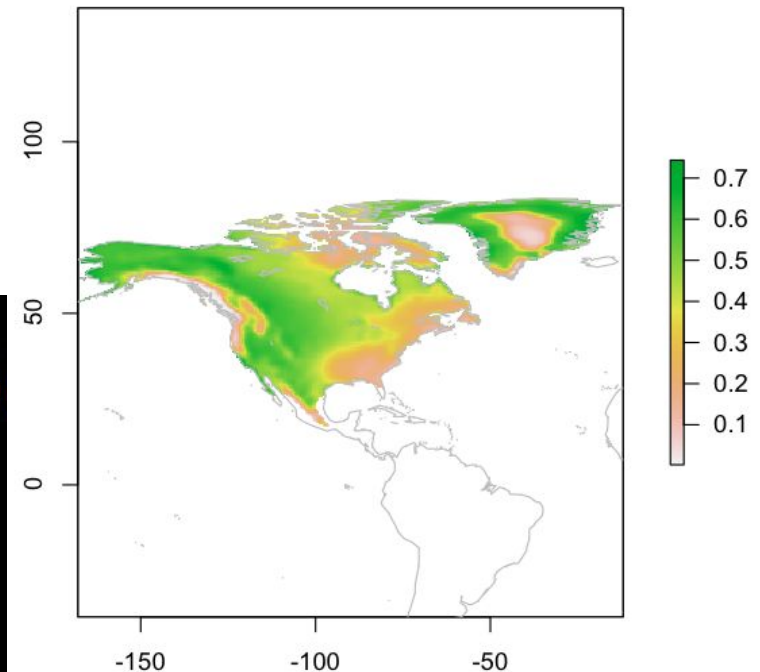
Ajuste dos ENMs

Maximum Entropy (MaxEnt)



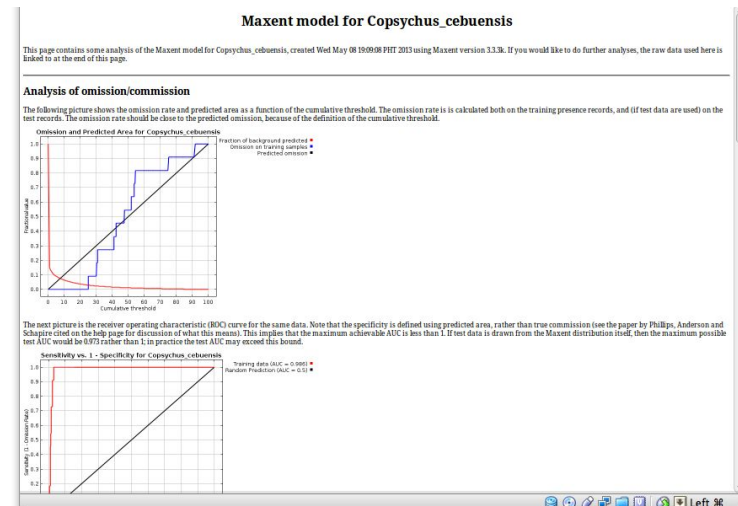
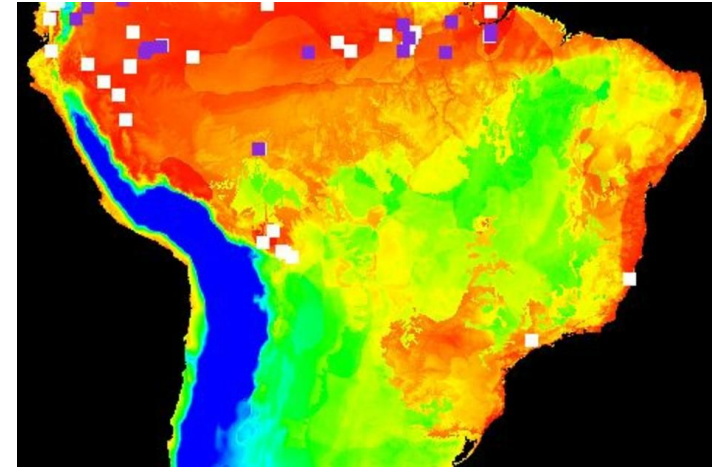
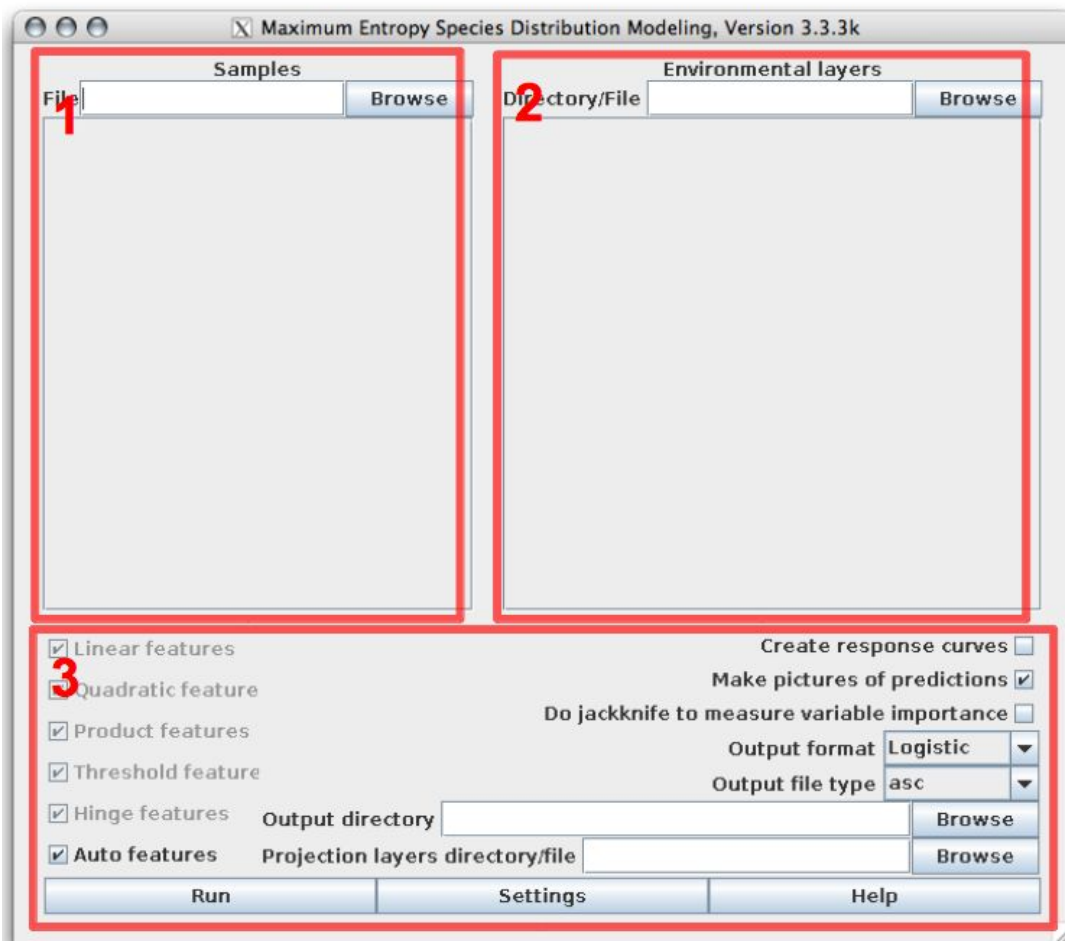
Adapted from Elith et al. (2011) *A statistical explanation of MaxEnt for ecologists*. *Diversity and Distributions*, 17, 43-57.

Lima-Ribeiro & Diniz-Filho (2013)



Ajuste dos ENMs

Maximum Entropy (MaxEnt)



Ajuste dos ENMs

Presença e ausência

Lima-Ribeiro &
Diniz-Filho (2013)



Apenas presença

Aquário

Bioclim
Dist. Euclidiana
Dist. Mahalanobis
Domain (dist. Gower)
ENFA (ecological niche factor analysis)

Presença/Background

GARP (genetic algorithm for rule-set production)
Maxent (maximum entropy)
SVM (support vector machine)

Aprendizado de Máquina
(machine learning)
"cofre"

Presença/Ausência

Estatístico ("turbina")

GLMZ (generalized linear model)
GAM (generalized additive model)
FDA (flexible discriminant analysis)
MARS (multivariate adaptive reg. splines)
BRT (boosted regression trees)
→ GBM (gradient boosting machine)
CART (classification and regression trees)
RDNFOR (random forest)
NNET (neural networks)
→ ANN (artificial neural networks)

Onde encontrar dados de
ausência?

Ajuste dos ENMs

Ausência “real” (modelos de ocupação)

Modelling of species distributions, range dynamics and communities under imperfect detection: advances, challenges and opportunities

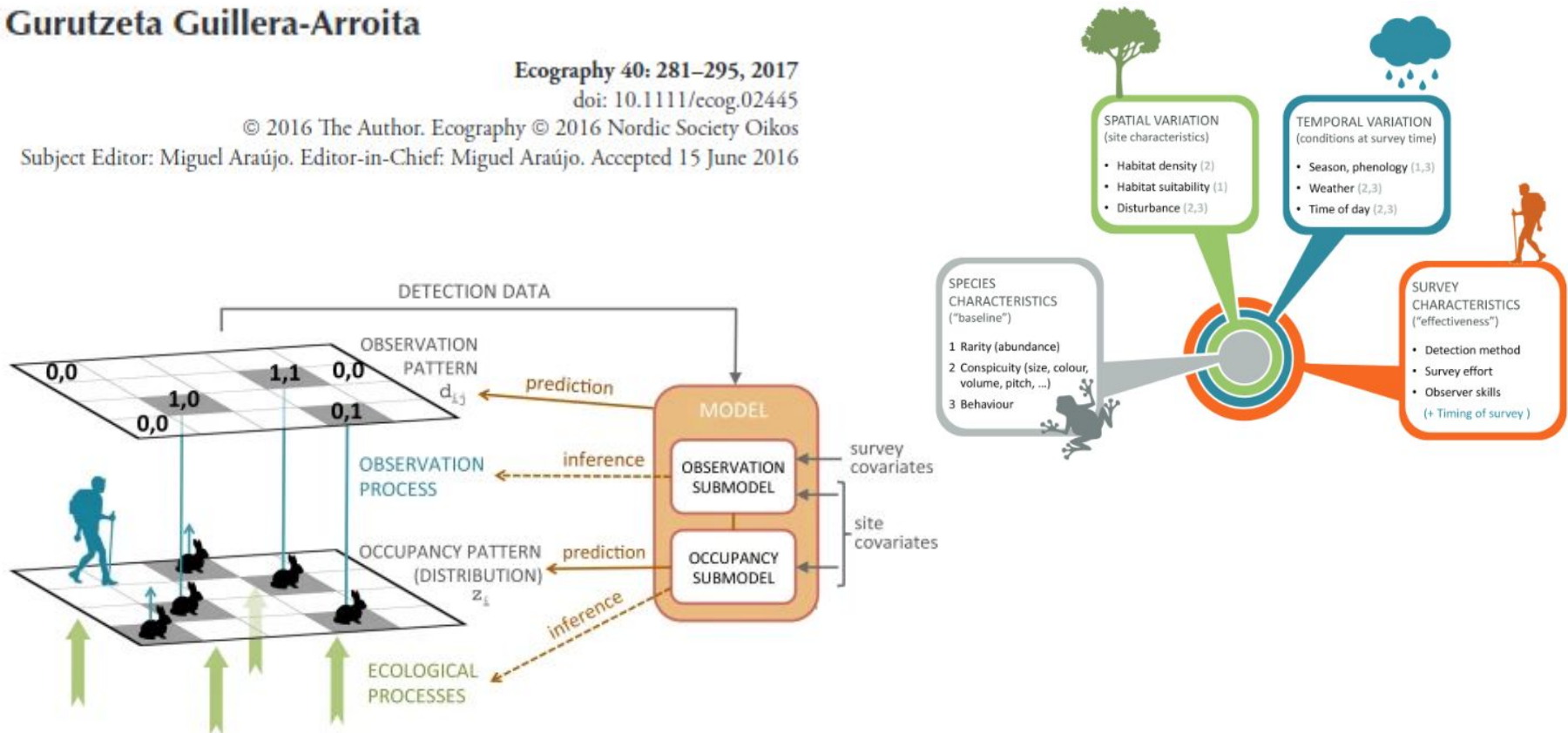
Gurutzeta Guillera-Arroita

Ecography 40: 281–295, 2017

doi: 10.1111/ecog.02445

© 2016 The Author. Ecography © 2016 Nordic Society Oikos

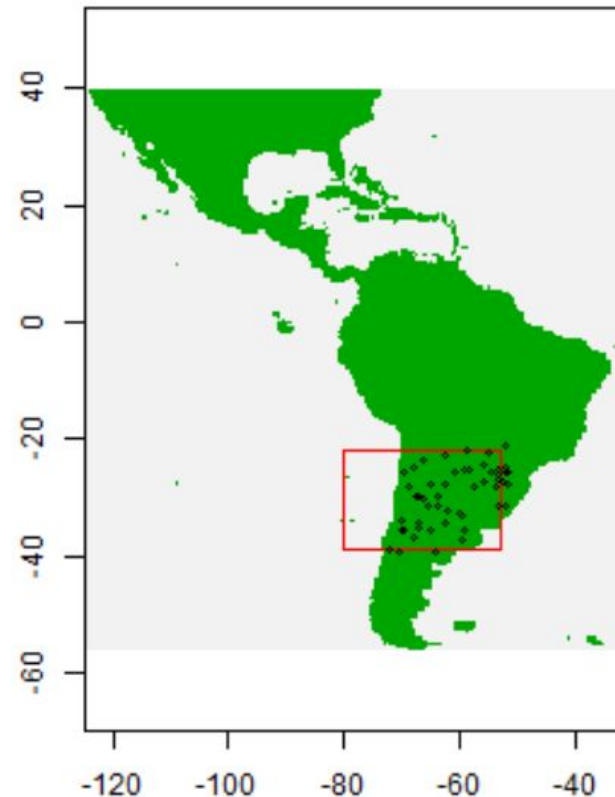
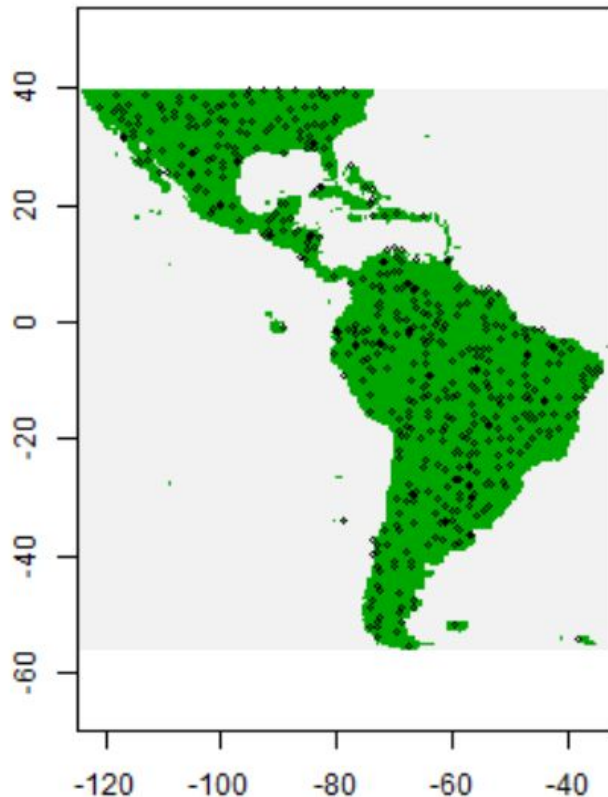
Subject Editor: Miguel Araújo. Editor-in-Chief: Miguel Araújo. Accepted 15 June 2016



Ajuste dos ENMs

Pseudo-ausência

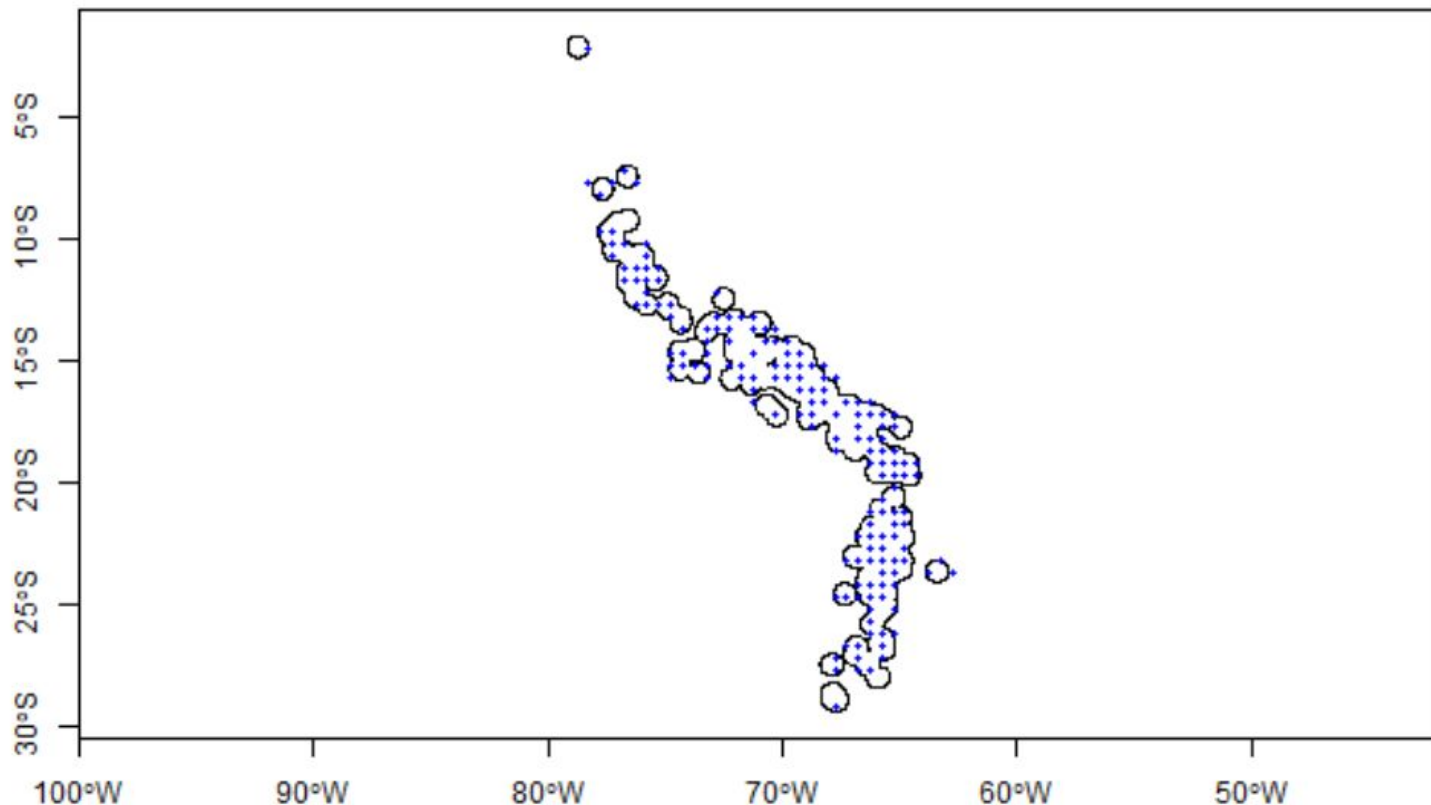
Sorteio de **pontos aleatórios** (sem **padrão espacial**) para serem considerados como **ausência verdadeira**



Ajuste dos ENMs

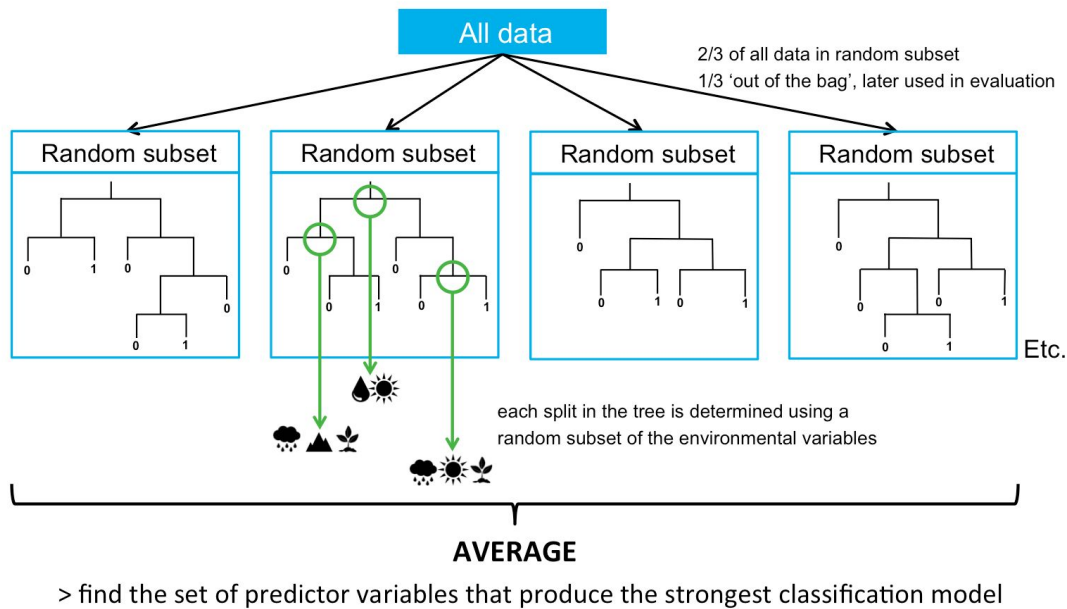
Pseudo-ausência

Sorteio de **pontos aleatórios** (com **padrão espacial**) para serem considerados como **ausência verdadeira**

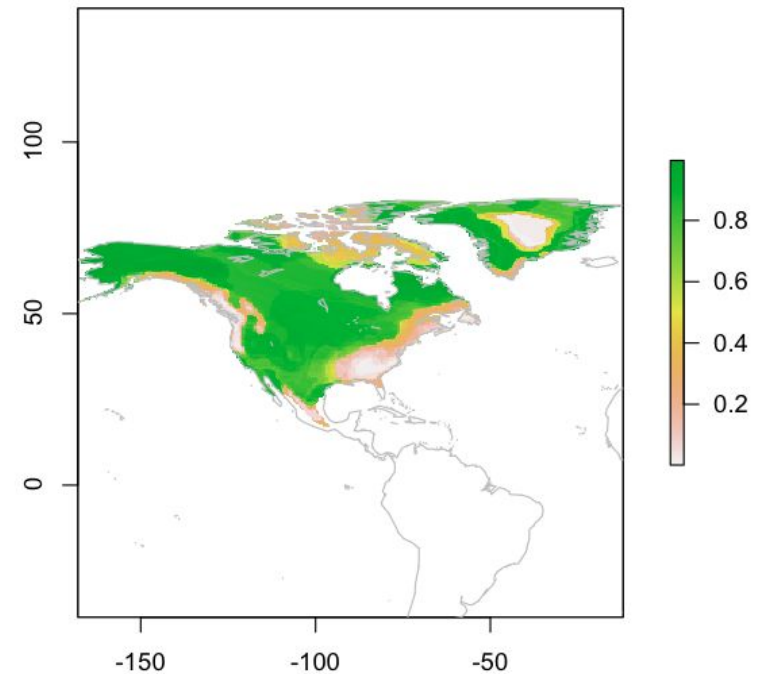


Ajuste dos ENMs

Random Forest

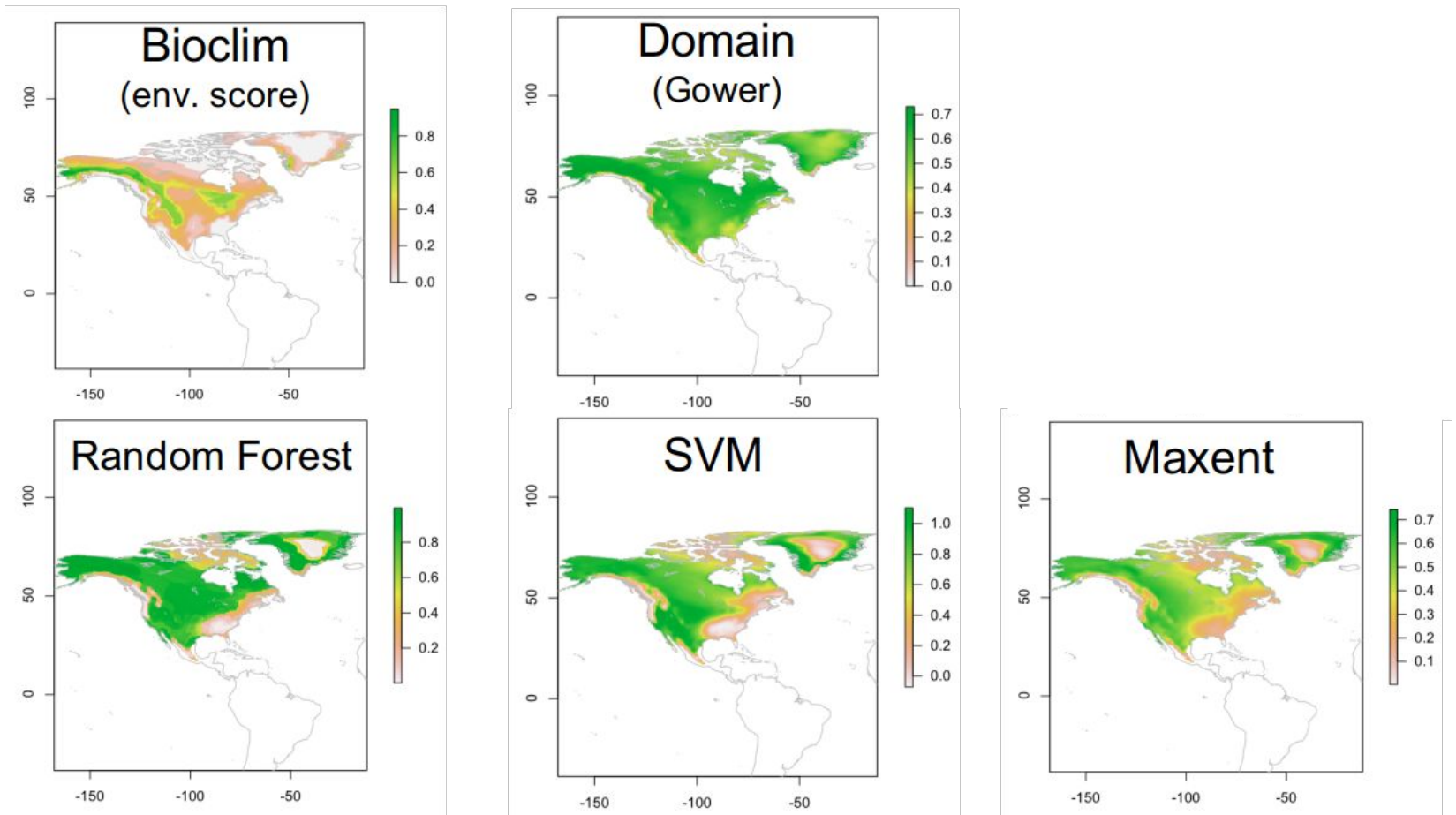


Lima-Ribeiro &
Diniz-Filho (2013)



Ajuste dos ENMs

Qual algoritmo usar?



Ajuste dos ENMs

Consenso (*Ensemble*)



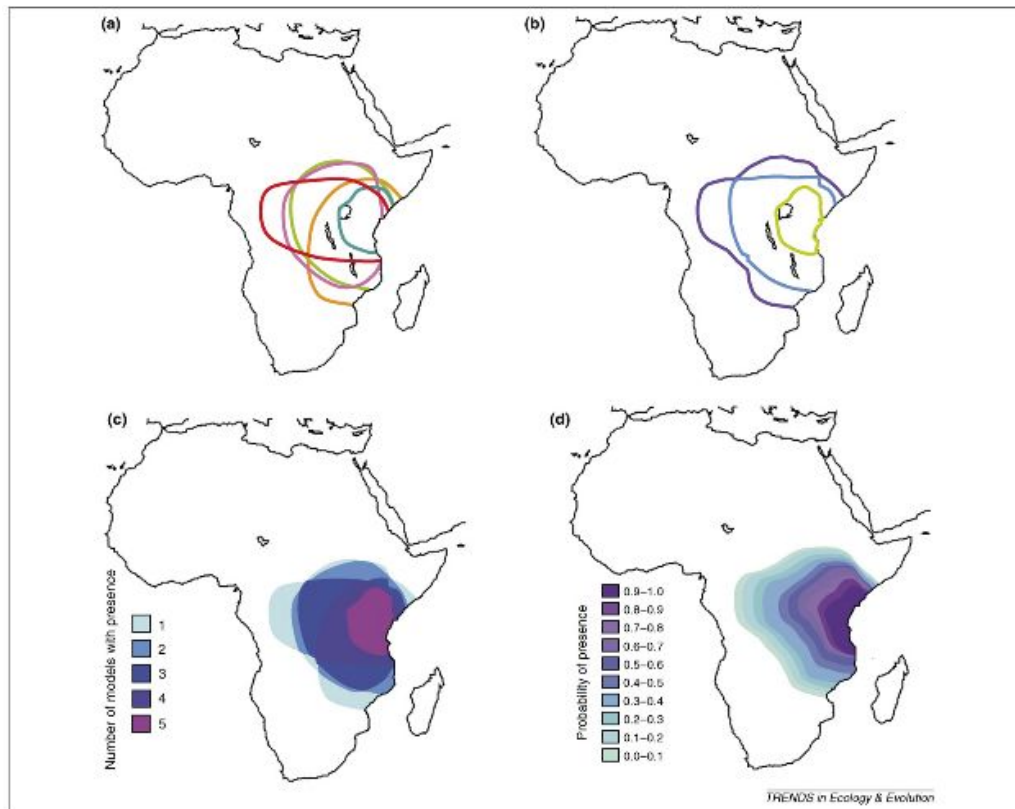
Review

TRENDS in Ecology and Evolution Vol.22 No.1

Full text provided by www.sciencedirect.com
ScienceDirect

Ensemble forecasting of species distributions

Miguel B. Araújo¹ and Mark New²



SDM passo a passo

Estrutura dos ENMs

ECOGRAPHY

Review and synthesis

A standard protocol for reporting species distribution models

Damaris Zurell, Janet Franklin, Christian König, Phil J. Bouchet, Carsten F. Dormann, Jane Elith, Guillermo Fandos, Xiao Feng, Gurrutza Guillera-Aroita, Antoine Guisan, José J. Lahoz-Monfort, Pedro J. Leitão, Daniel S. Park, A. Townsend Peterson, Giovanni Rapacciuolo, Dirk R. Schmatz, Boris Schröder, Josep M. Serra-Díaz, Wilfried Thuiller, Katherine L. Yates, Niklaus E. Zimmermann and Cory Merow

Ecography

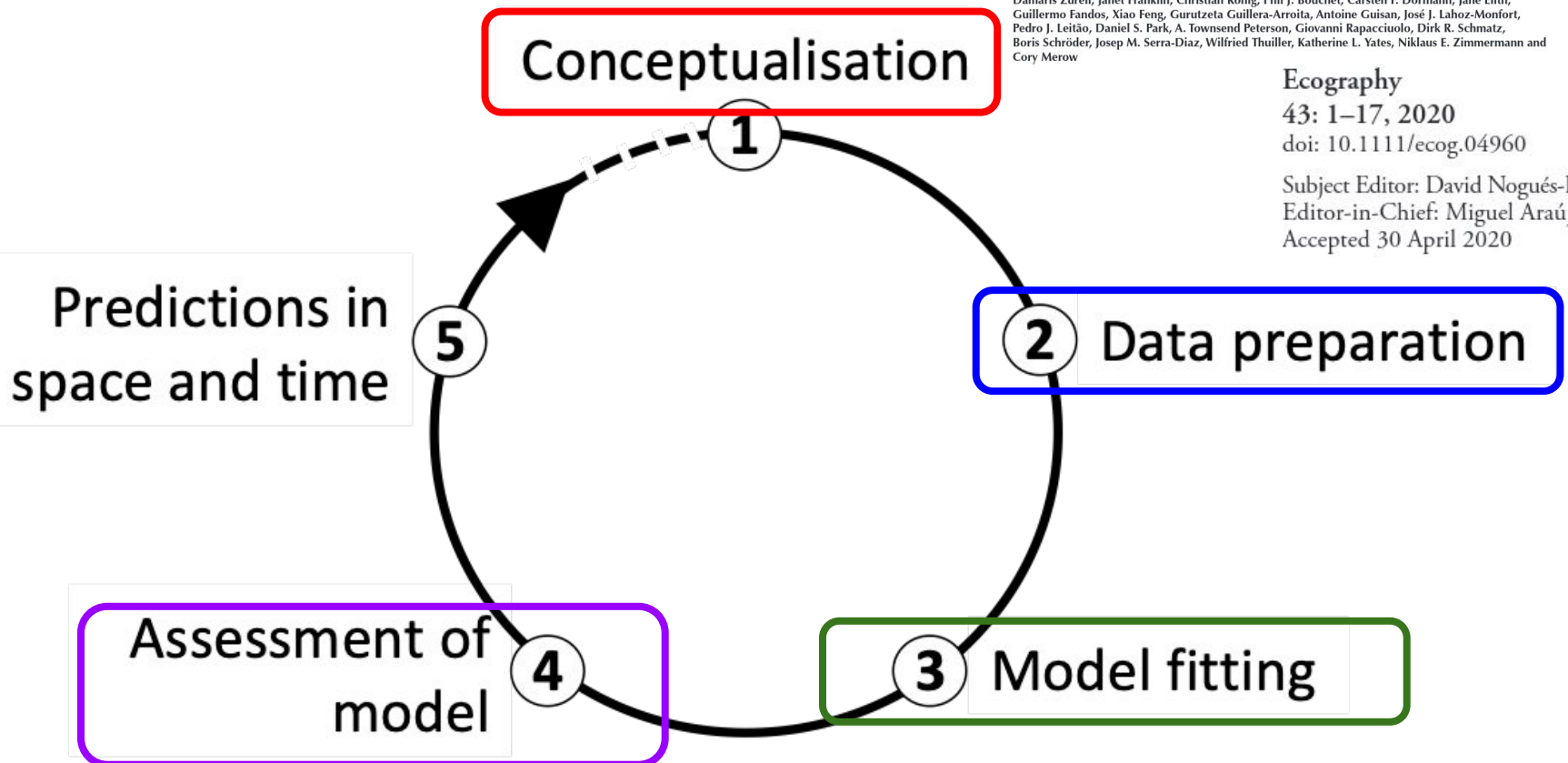
43: 1–17, 2020

doi: 10.1111/ecog.04960

Subject Editor: David Nogués-Bravo

Editor-in-Chief: Miguel Araújo

Accepted 30 April 2020



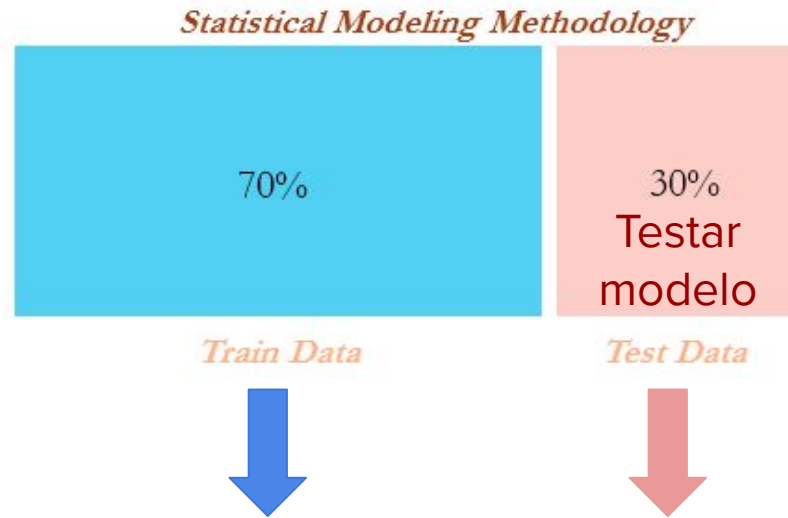
<https://doi.org/10.1111/ecog.04960>

7. Avaliação dos modelos

Como saber se meu modelo se
aproxima da realidade?

Avaliação dos ENMs

Partição dos dados em **treino** e **teste**



"Ocorrências"

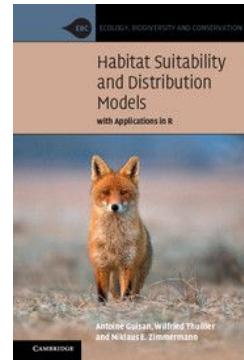
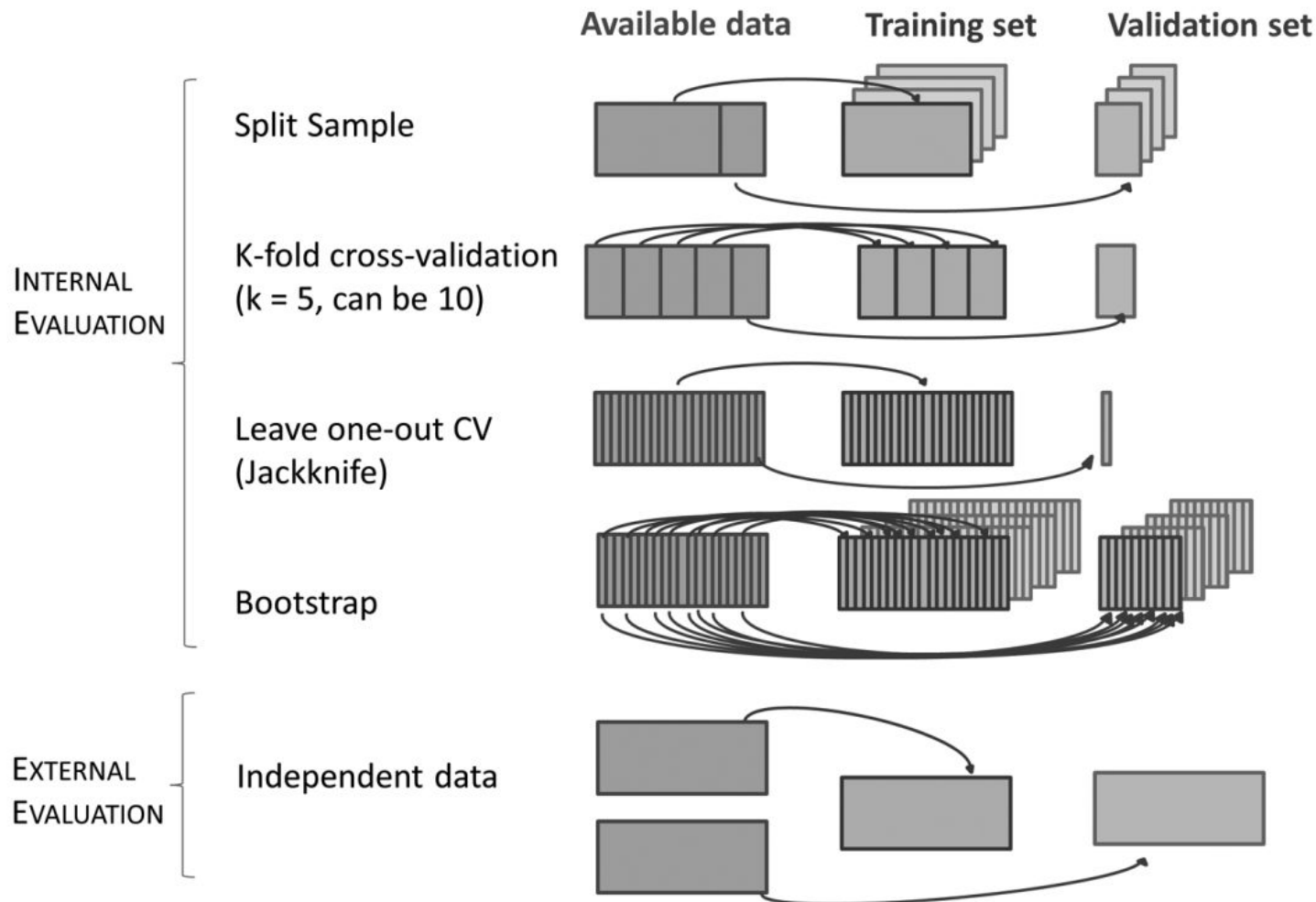
ATENÇÃO!!!

Presenças
e
Pseudo-ausências

species	lon	lat	
sp1	-40.2	-23.4	Treino
sp1	-38.8	-20.3	Teste
sp1	-43.3	-19.9	Treino

Avaliação dos ENMs

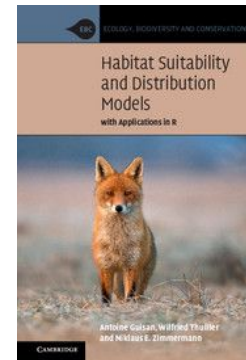
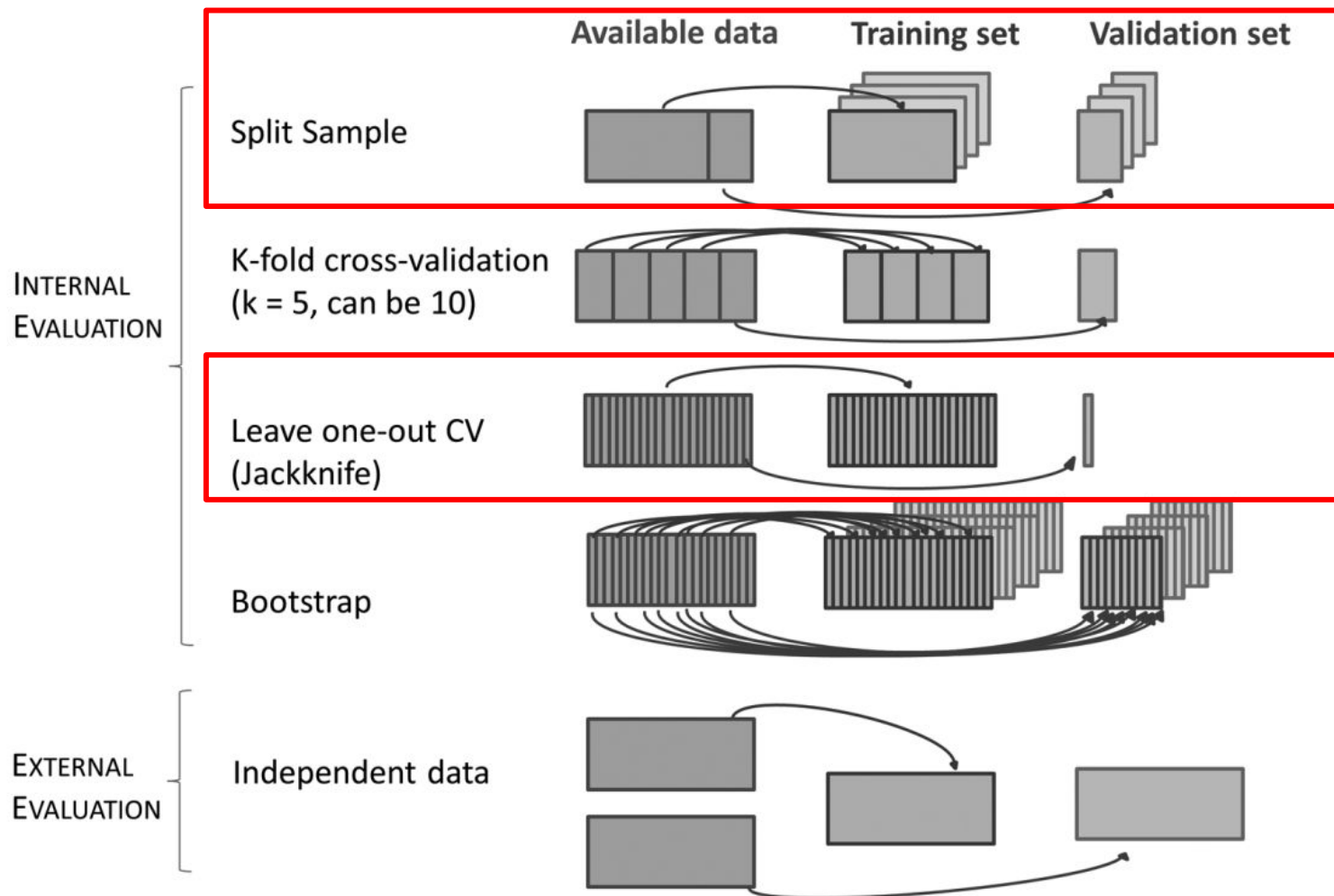
Tipos de avaliação



Guisan et al. (2017)

Avaliação dos ENMs

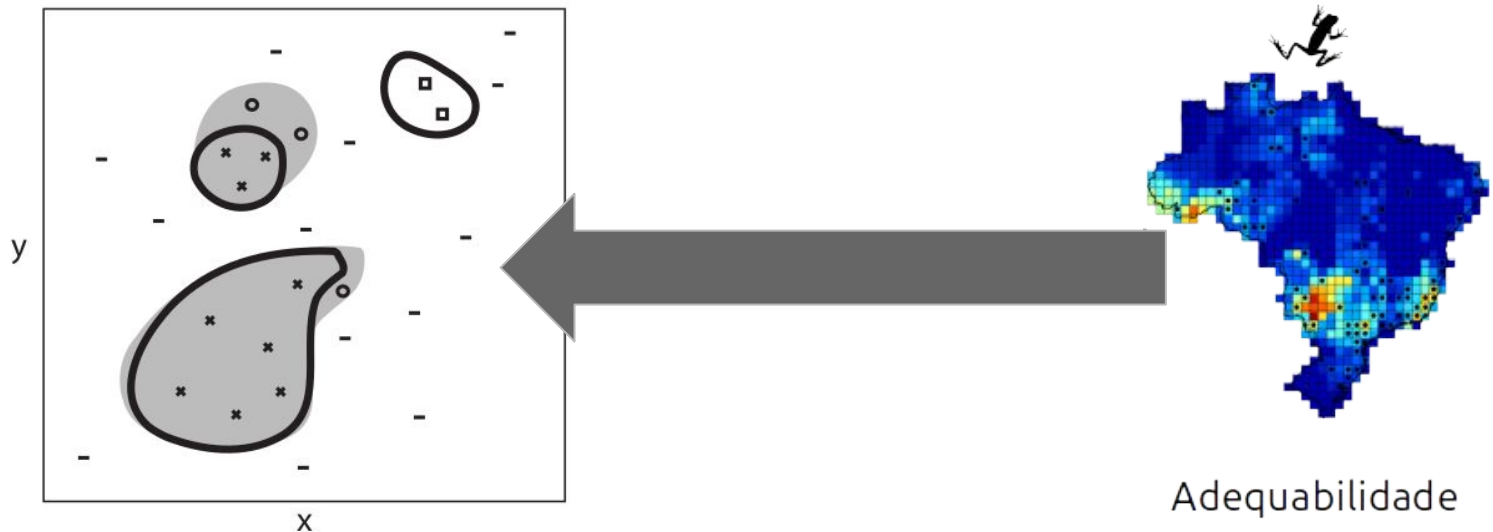
Tipos de avaliação



Guisan et al. (2017)

Avaliação dos ENMs

Como saber se o modelo acerta a realidade?



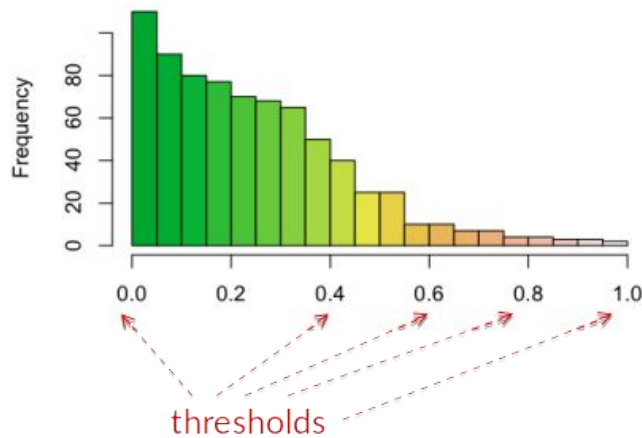
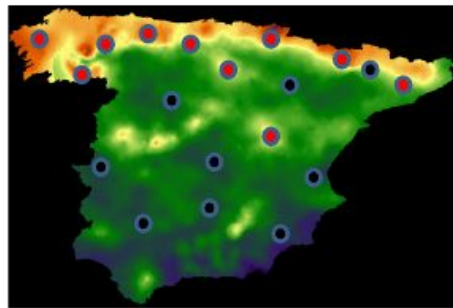
Adequabilidade

valores
0
até
1

- Occupied distributional area, G_O
- Areas predicted by an ecological niche model
- × True positive
- True negative
- False negative
- False positive

Avaliação dos ENMs

Limiares (*Thresholds*) - transformar em 1 e 0



Threshold
0



0.4



0.6



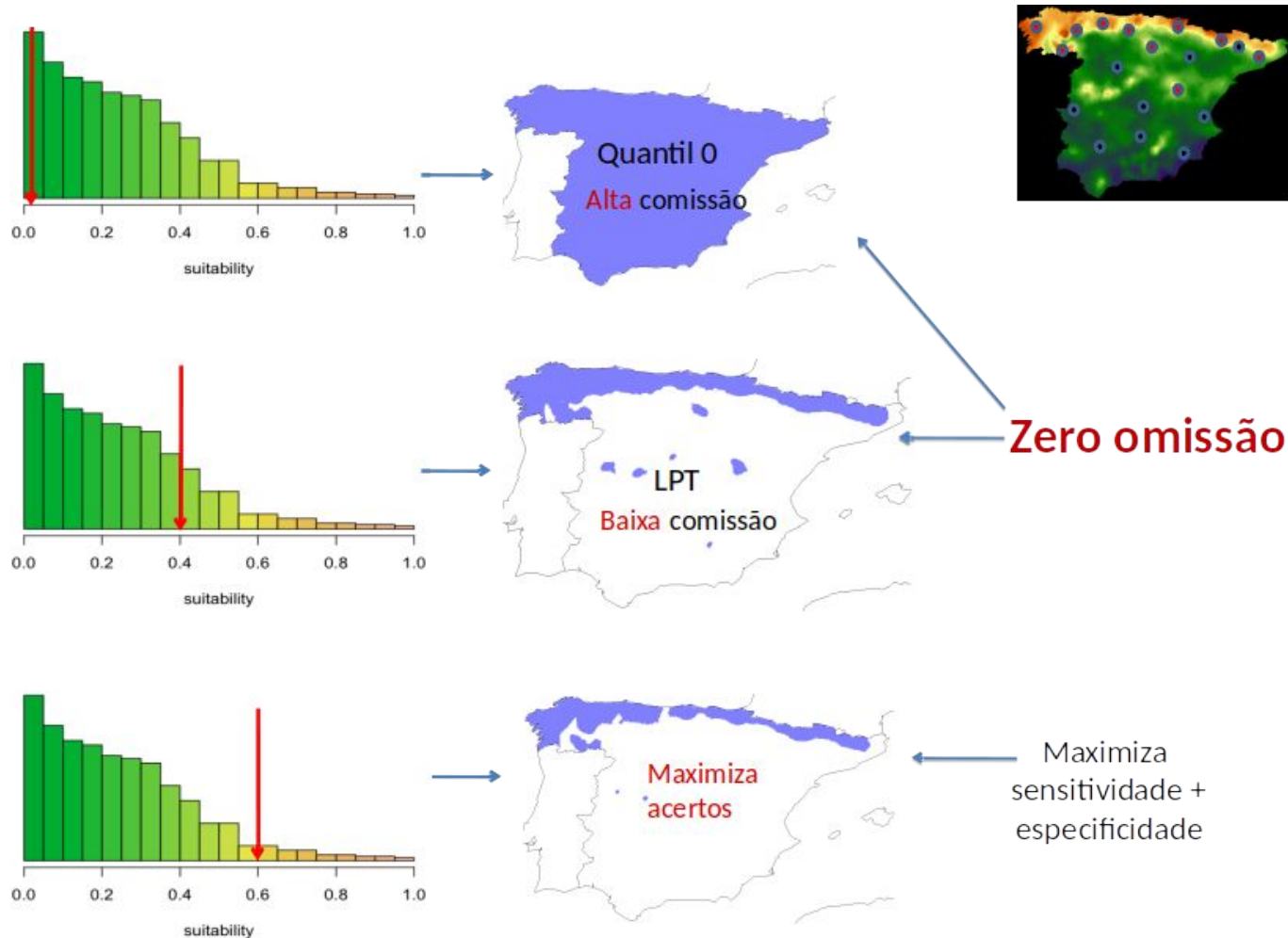
0.8



1

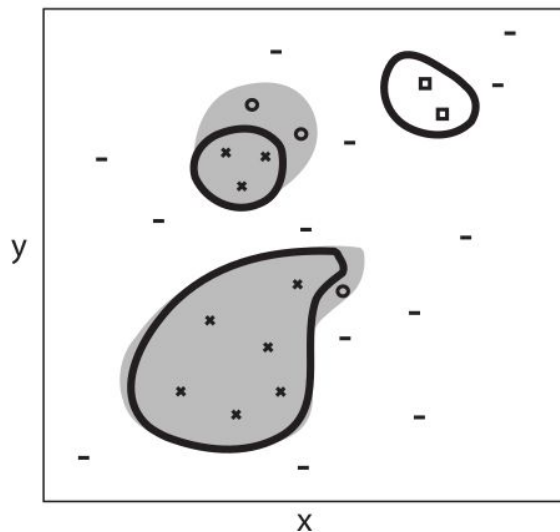
Avaliação dos ENMs

Limiares (*Thresholds*)



Avaliação dos ENMs

Matriz de confusão - para os **dados de teste**

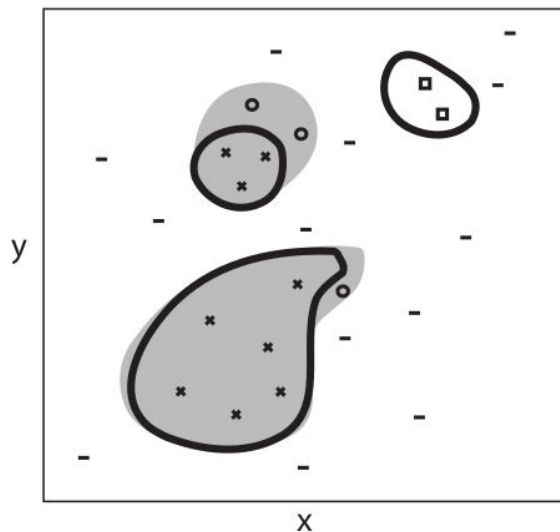


- Occupied distributional area, G_O
- Areas predicted by an ecological niche model
- × True positive
- True negative
- False negative
- False positive

		Observation	
		Present	Absent
Prediction	Present	True positive	False positive
	Absent	False negative	True negative

Avaliação dos ENMs

Matriz de confusão - para os **dados de teste**



		Observation	
		Present	Absent
Prediction	Present	X True positive	False positive
	Absent	False negative	True negative

● Occupied distributional area, G_O

○ Areas predicted by an ecological niche model

x True positive

- True negative

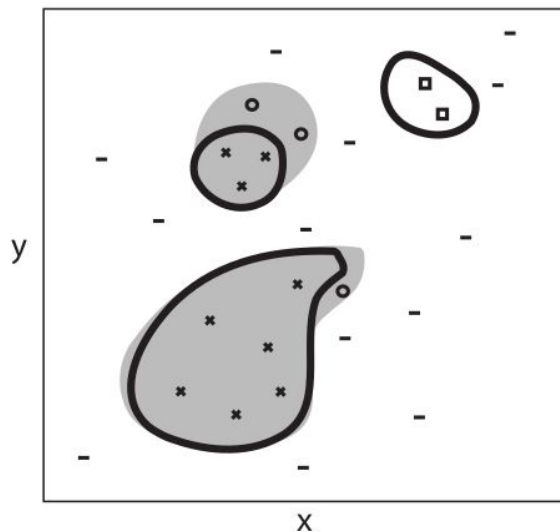
o False negative

□ False positive

**→ Ocorrência que o modelo previu
como presença (acerto)**

Avaliação dos ENMs

Matriz de confusão - para os **dados de teste**



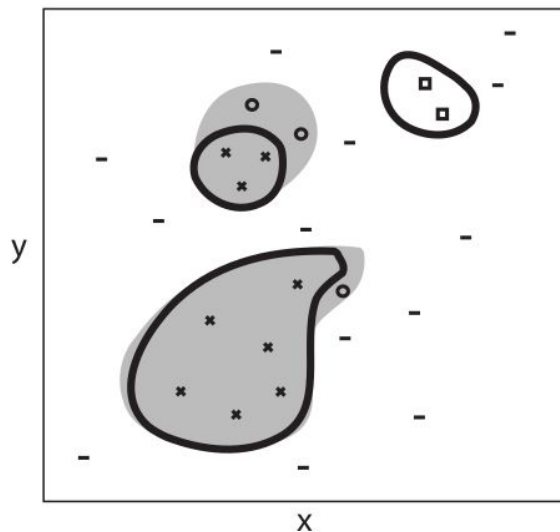
		Observation	
		Present	Absent
Prediction	Present	True positive	False positive
	Absent	False negative	True negative

- Occupied distributional area, G_O
- Areas predicted by an ecological niche model
- × True positive
- True negative
- False negative
- False positive

Pseudo-ausência que o modelo previu como **ausência (acerto)**

Avaliação dos ENMs

Matriz de confusão - para os **dados de teste**



		Observation	
		Present	Absent
Prediction	Present	True positive	False positive
	Absent	0 False negative	True negative

● Occupied distributional area, G_O

○ Areas predicted by an ecological niche model

× True positive

- True negative

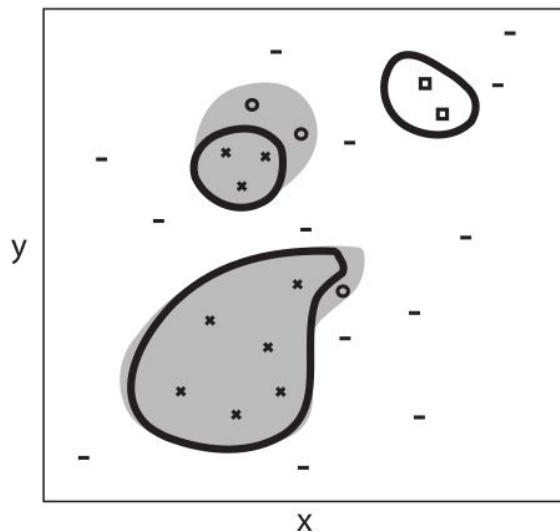
○ False negative

□ False positive

Ocorrência que o modelo previu
como **ausência (erro de omissão)**

Avaliação dos ENMs

Matriz de confusão - para os **dados de teste**



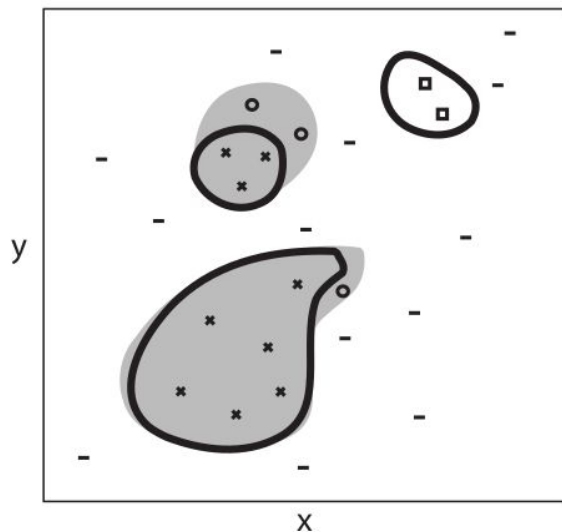
		Observation	
		Present	Absent
Prediction	Present	True positive	False positive 
	Absent	False negative	True negative

-  Occupied distributional area, G_O
-  Areas predicted by an ecological niche model
- $\times$ True positive
- $-$ True negative
- o False negative
- $\square$ False positive

Pseudo-ausência que o modelo previu
como **presença (erro de comissão)**

Avaliação dos ENMs

Matriz de confusão - para os **dados de teste**



- Occupied distributional area, G_O
- Areas predicted by an ecological niche model
- × True positive
- True negative
- False negative
- False positive

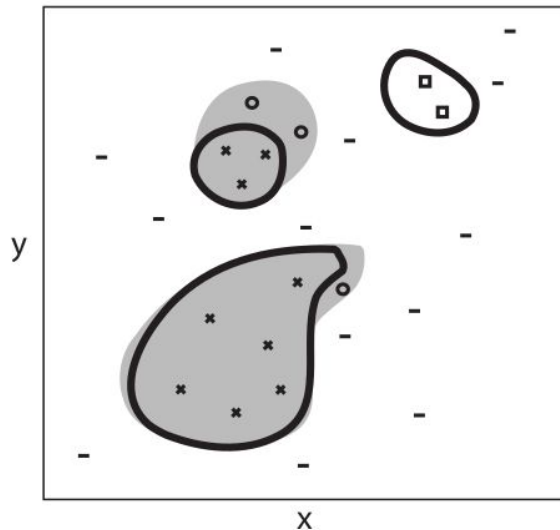
		Observation	
		Present	Absent
Prediction	Present	True positive	False positive
	Absent	False negative	True negative



**Sensitividade: presenças corretas
total de presenças**

Avaliação dos ENMs

Matriz de confusão - para os **dados de teste**



- Occupied distributional area, G_O
- Areas predicted by an ecological niche model
- × True positive
- True negative
- False negative
- False positive

		Observation	
		Present	Absent
Prediction	Present	True positive	False positive
	Absent	False negative	True negative

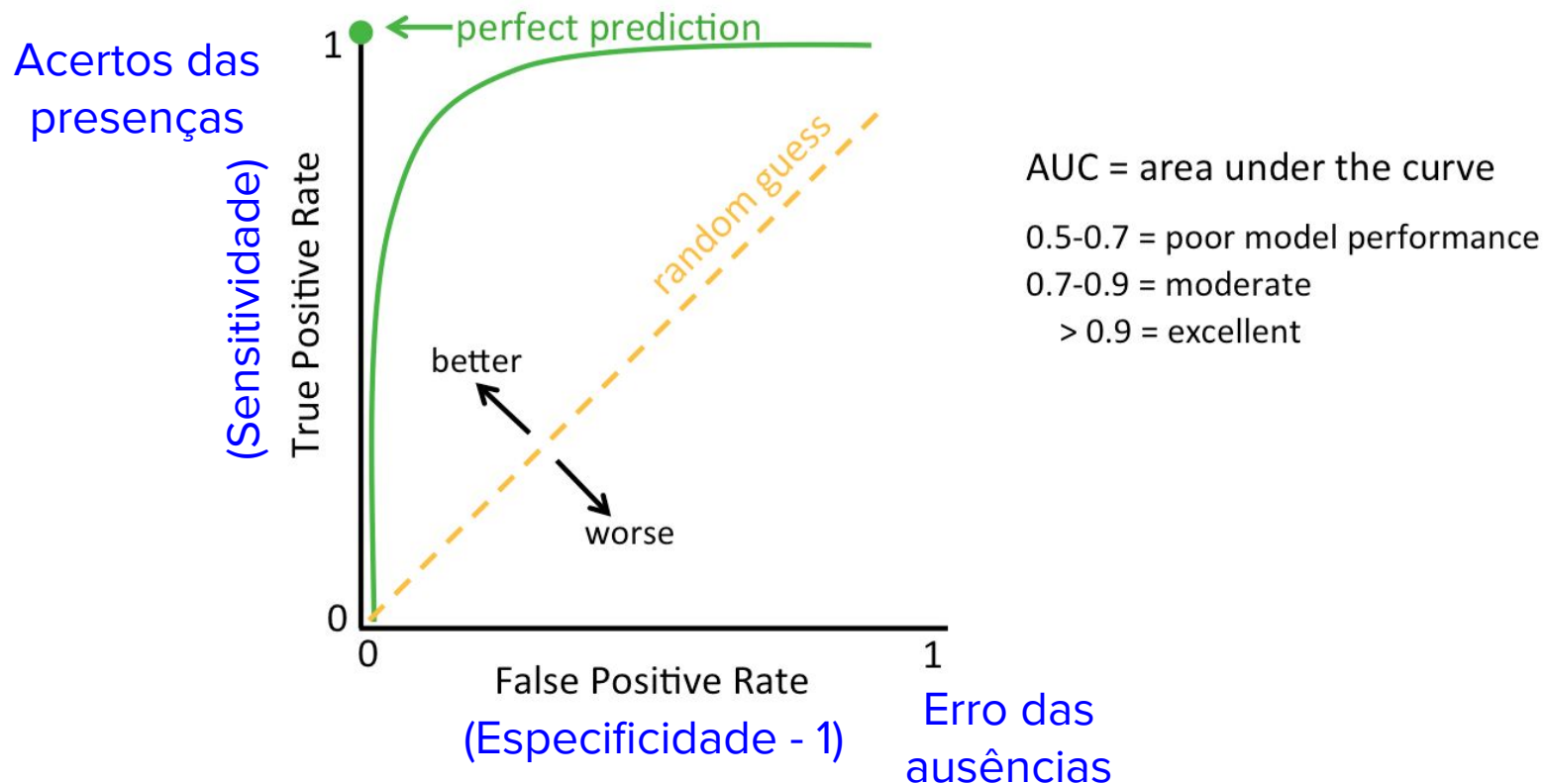


**Especificidade: pseudo-ausências corretas
total de pseudo-ausências**

Avaliação dos ENMs

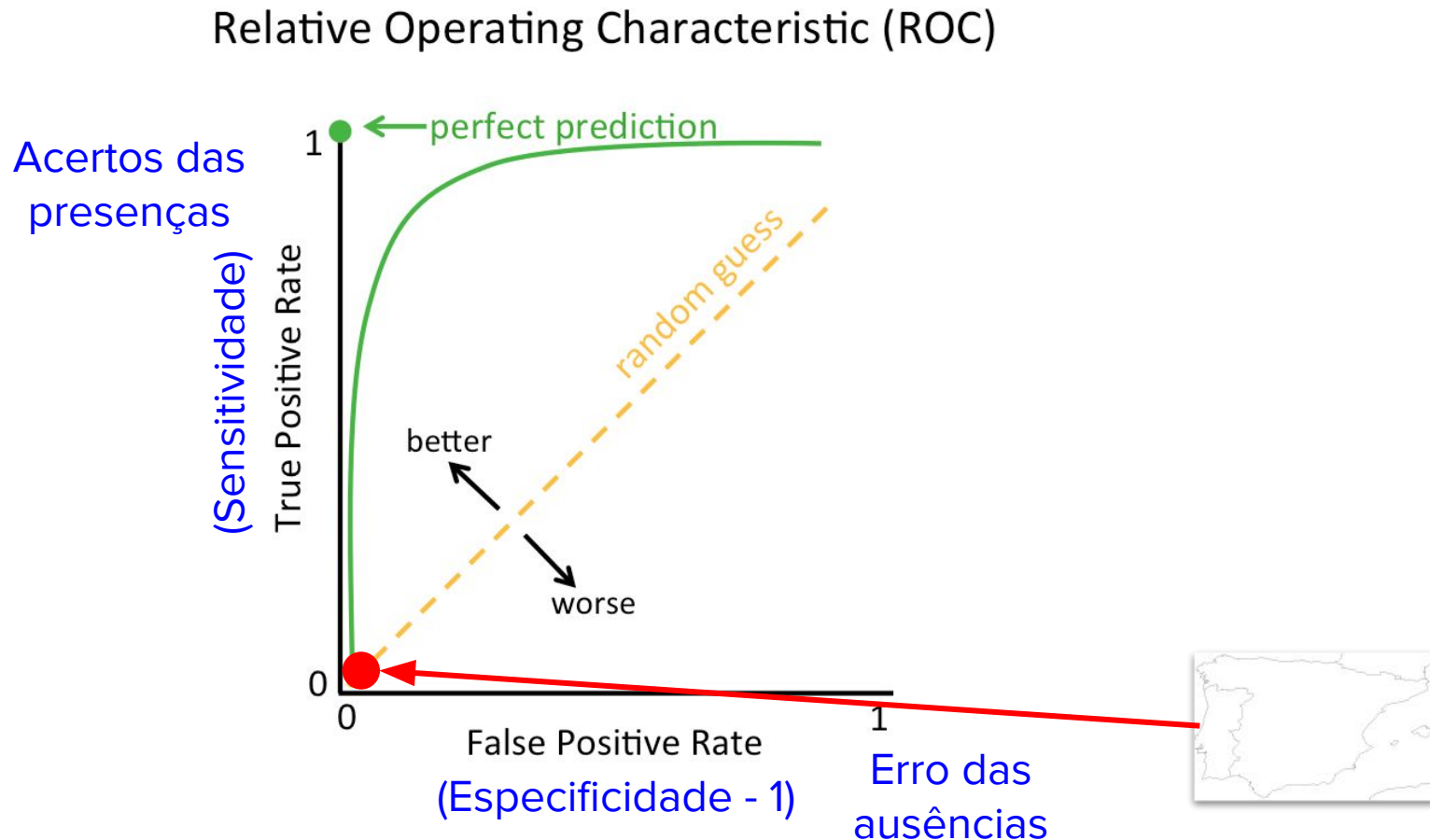
Curva ROC

Relative Operating Characteristic (ROC)



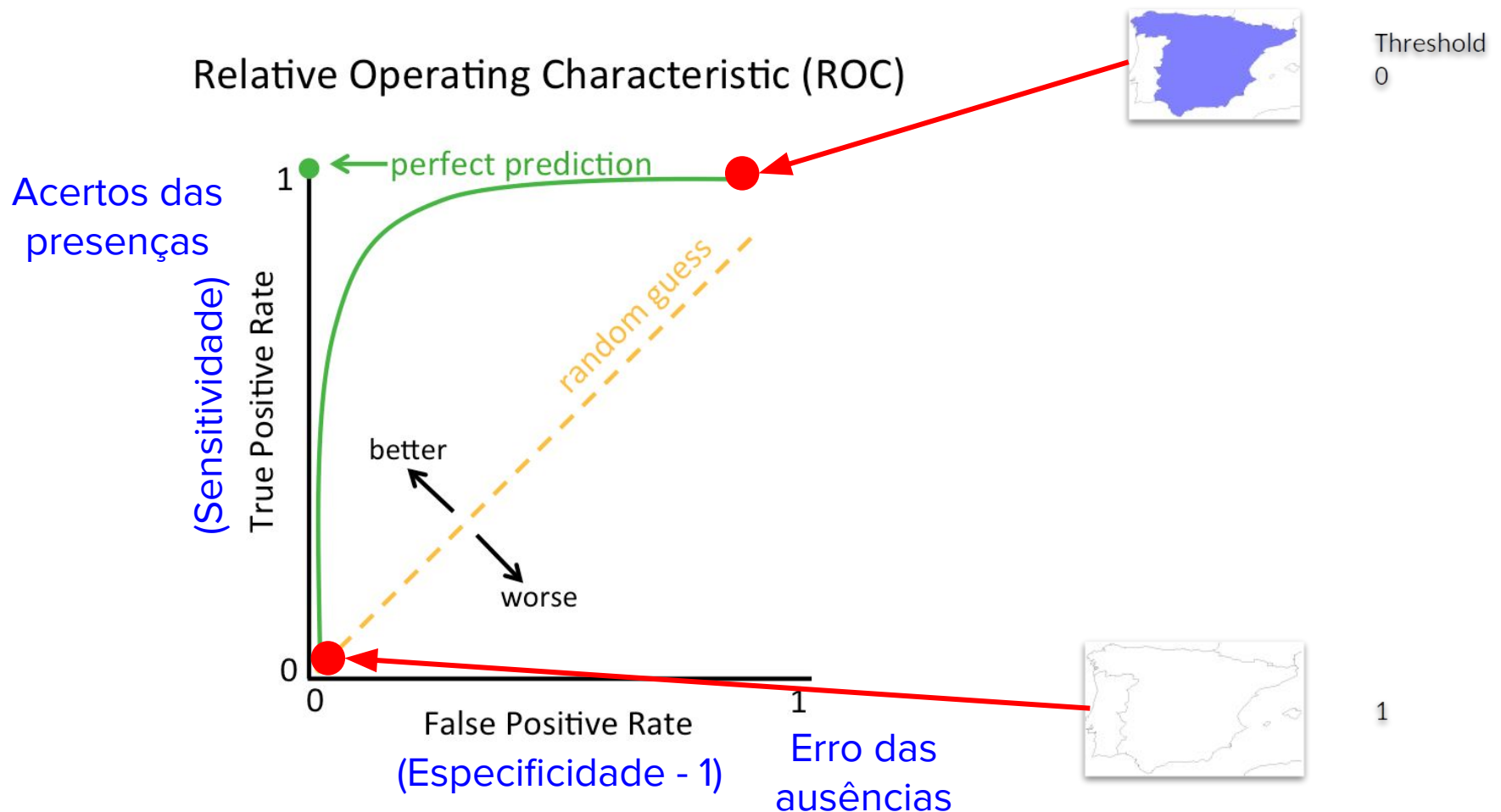
Avaliação dos ENMs

Curva ROC



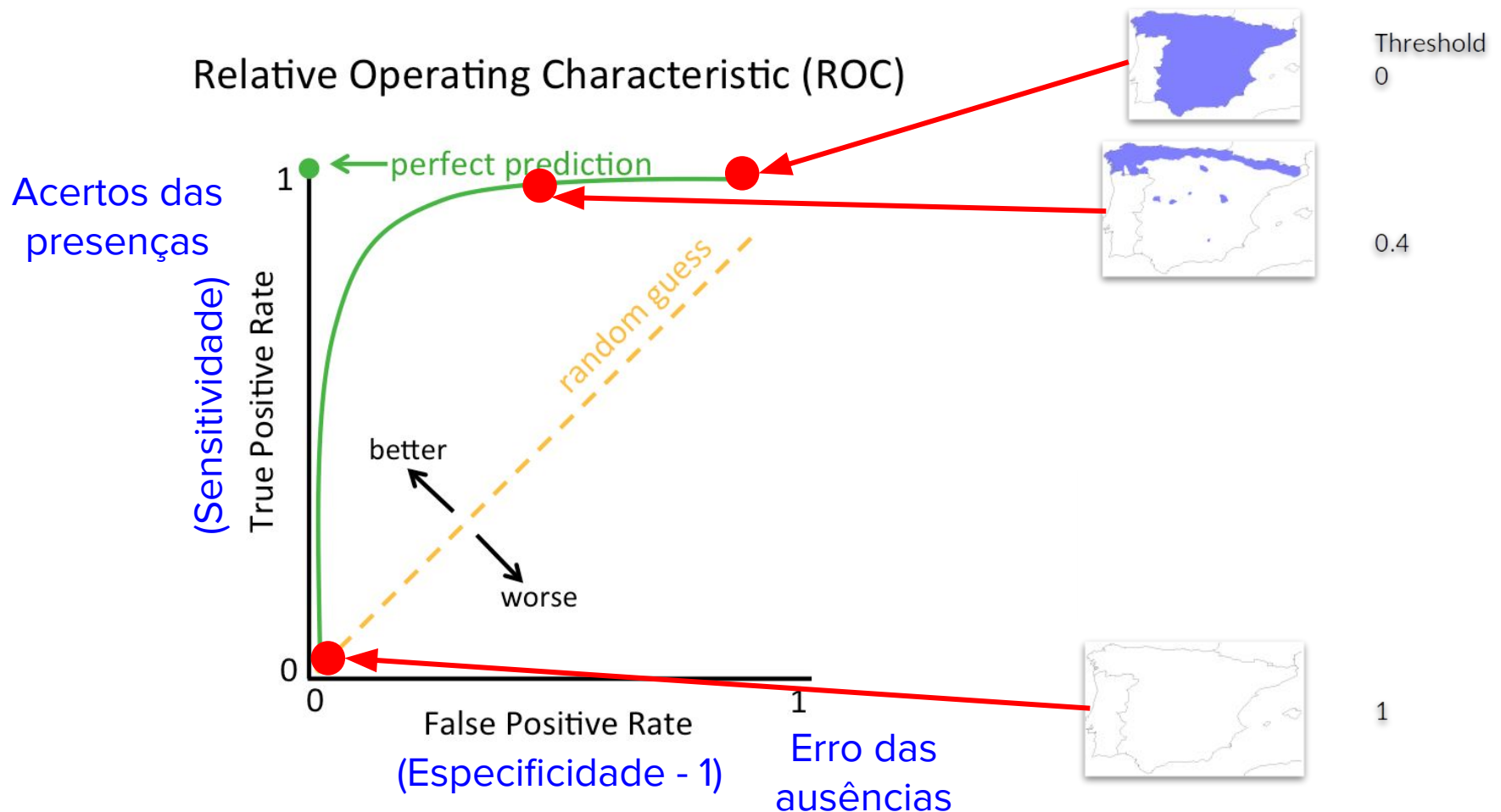
Avaliação dos ENMs

Curva ROC



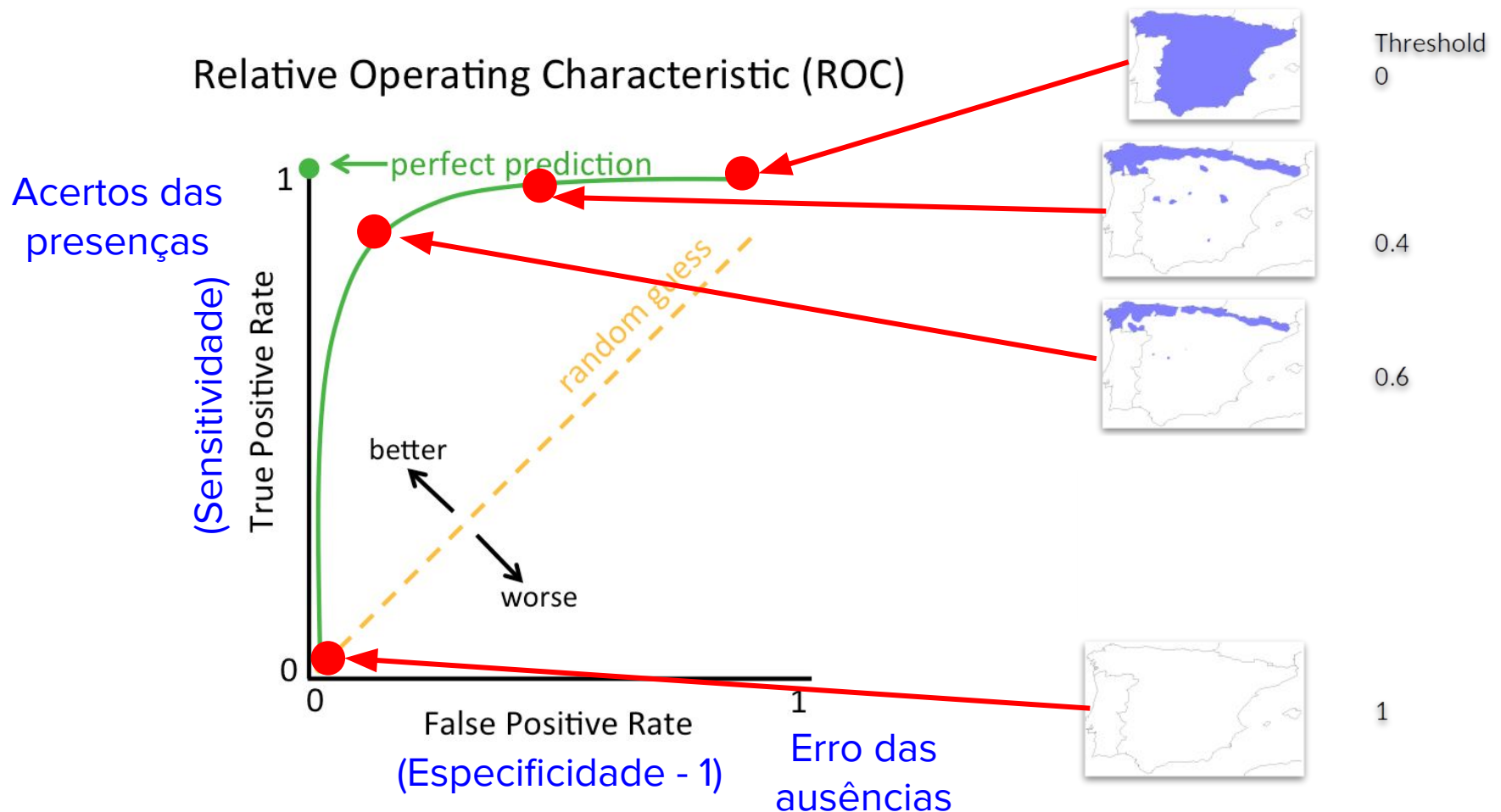
Avaliação dos ENMs

Curva ROC



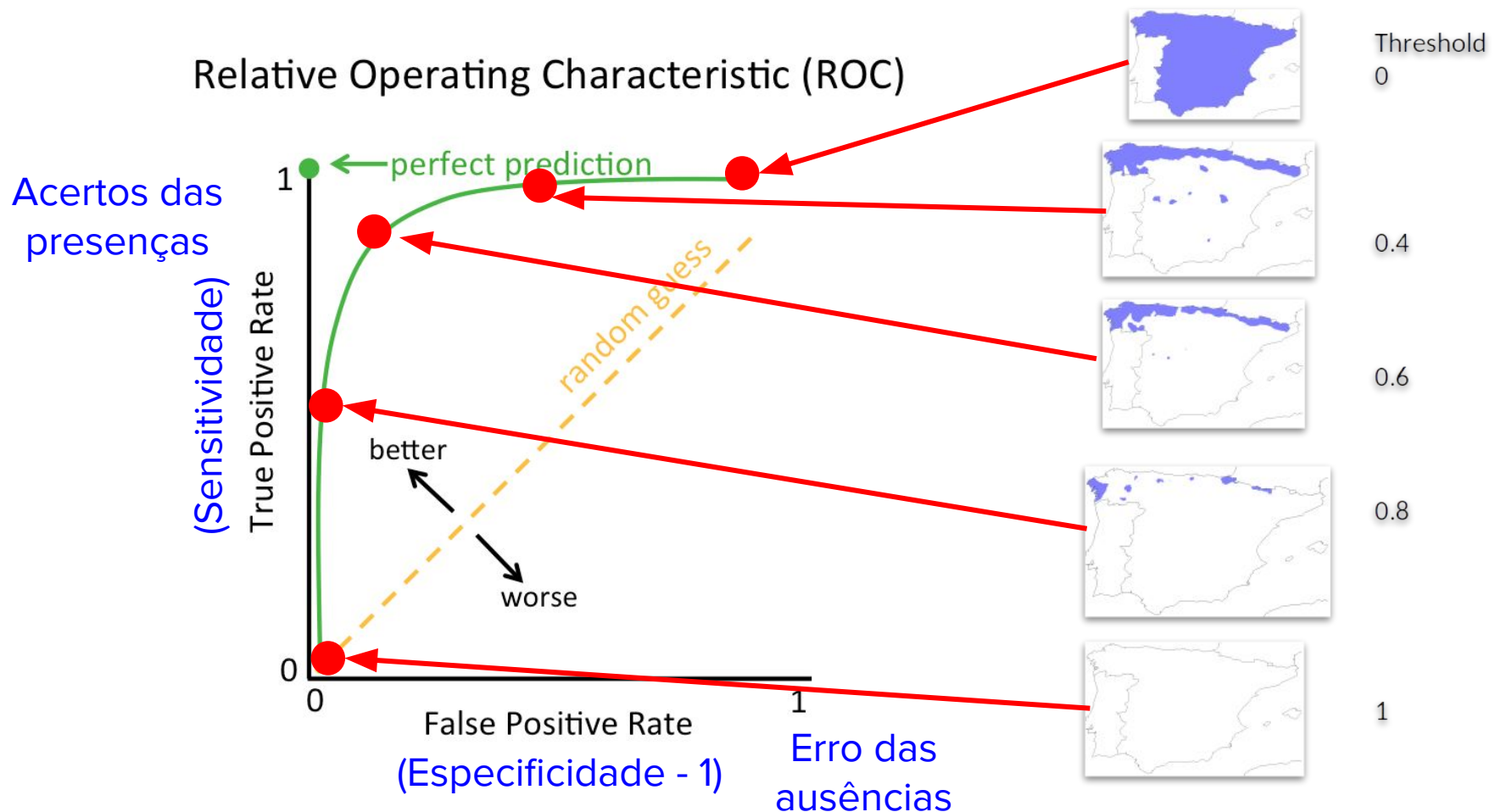
Avaliação dos ENMs

Curva ROC



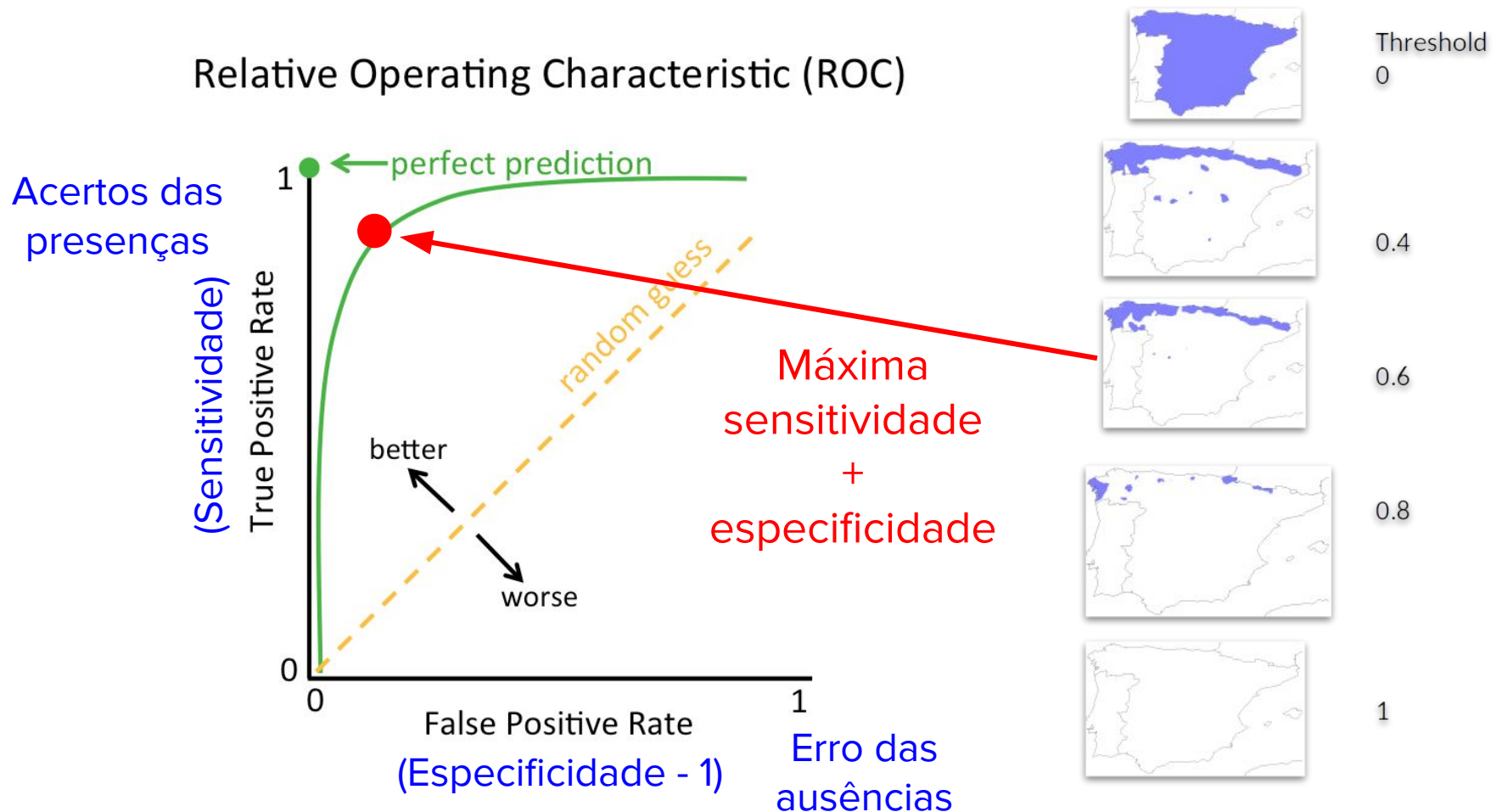
Avaliação dos ENMs

Curva ROC



Avaliação dos ENMs

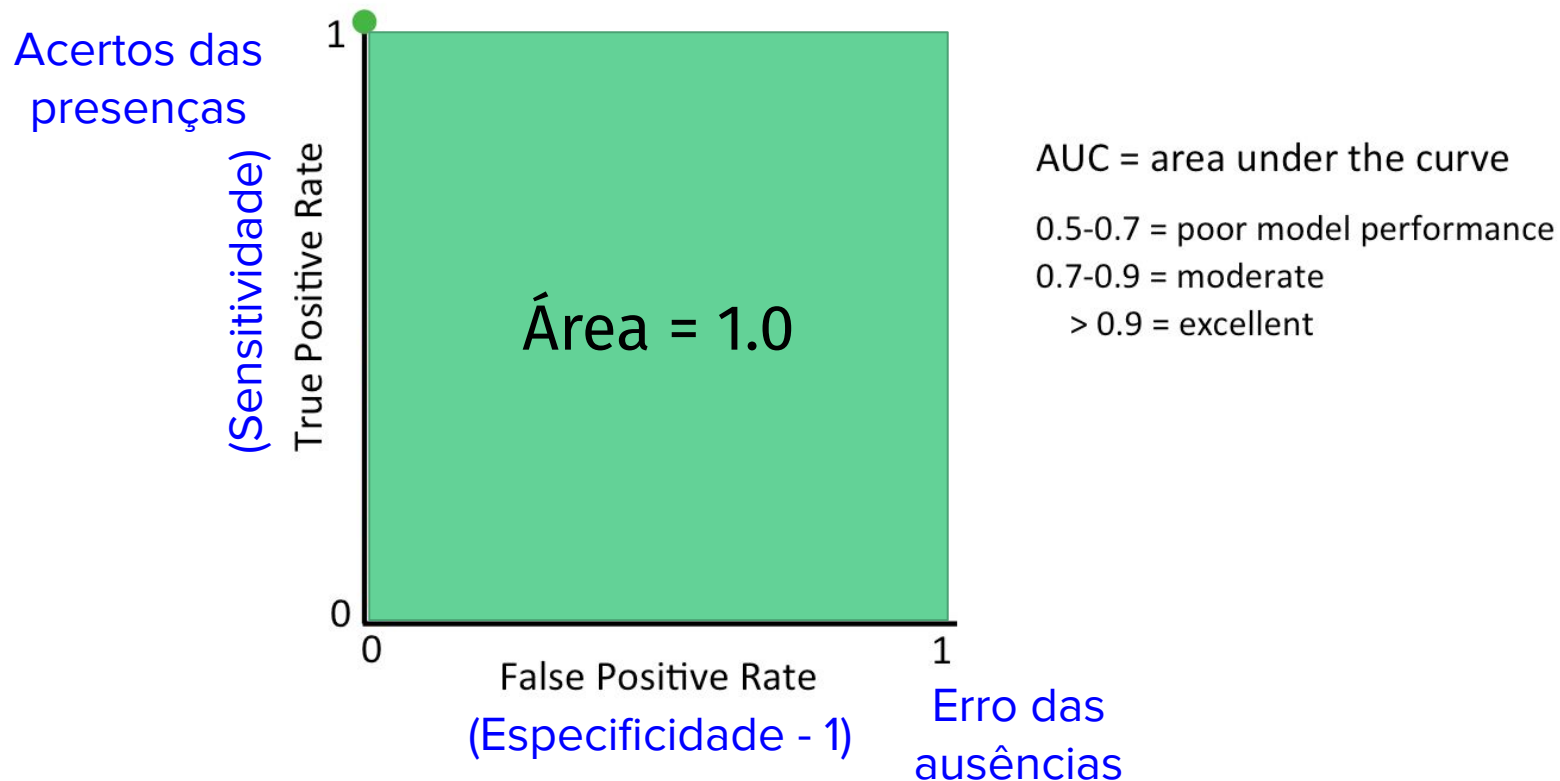
Curva ROC



Avaliação dos ENMs

AUC

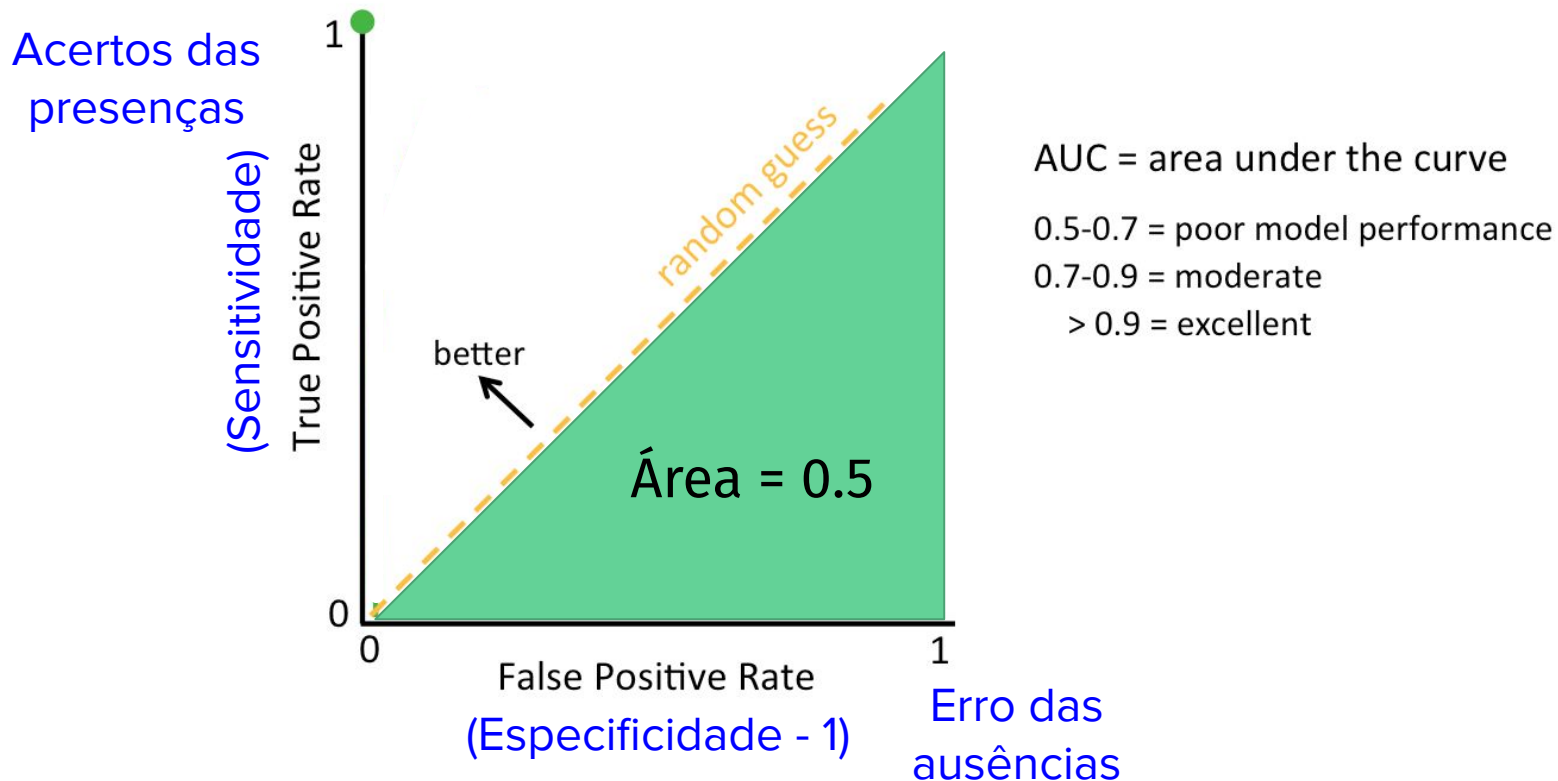
Relative Operating Characteristic (ROC)



Avaliação dos ENMs

AUC

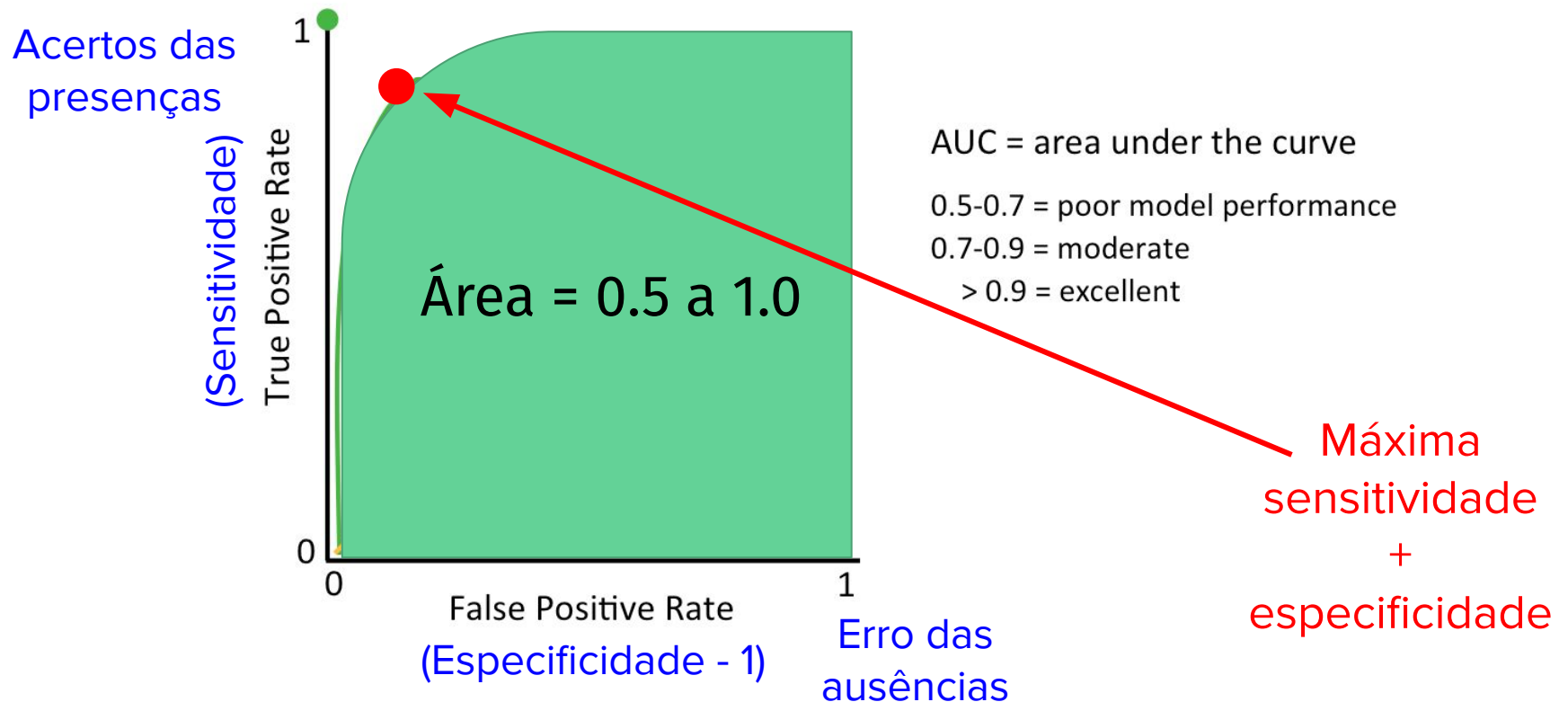
Relative Operating Characteristic (ROC)



Avaliação dos ENMs

AUC

Relative Operating Characteristic (ROC)



Avaliação dos ENMs

TSS (*True skill statistic*)

Número de sucessos menos o número de sucessos aleatórios

Varia de -1 to 1. Valores próximos a 0 modelos não diferentes do aleatórios

Depende de um valor de corte (*threshold*)

$$\textbf{TSS} = \textbf{sensitividade} + \textbf{especificidade} - 1$$

SDM passo a passo

Estrutura dos ENMs

ECOGRAPHY

Review and synthesis

A standard protocol for reporting species distribution models

Damaris Zurell, Janet Franklin, Christian König, Phil J. Bouchet, Carsten F. Dormann, Jane Elith, Guillermo Fandos, Xiao Feng, Gurrutza Guillera-Aroita, Antoine Guisan, José J. Lahoz-Monfort, Pedro J. Leitão, Daniel S. Park, A. Townsend Peterson, Giovanni Rapacciuolo, Dirk R. Schmatz, Boris Schröder, Josep M. Serra-Díaz, Wilfried Thuiller, Katherine L. Yates, Niklaus E. Zimmermann and Cory Merow

Ecography

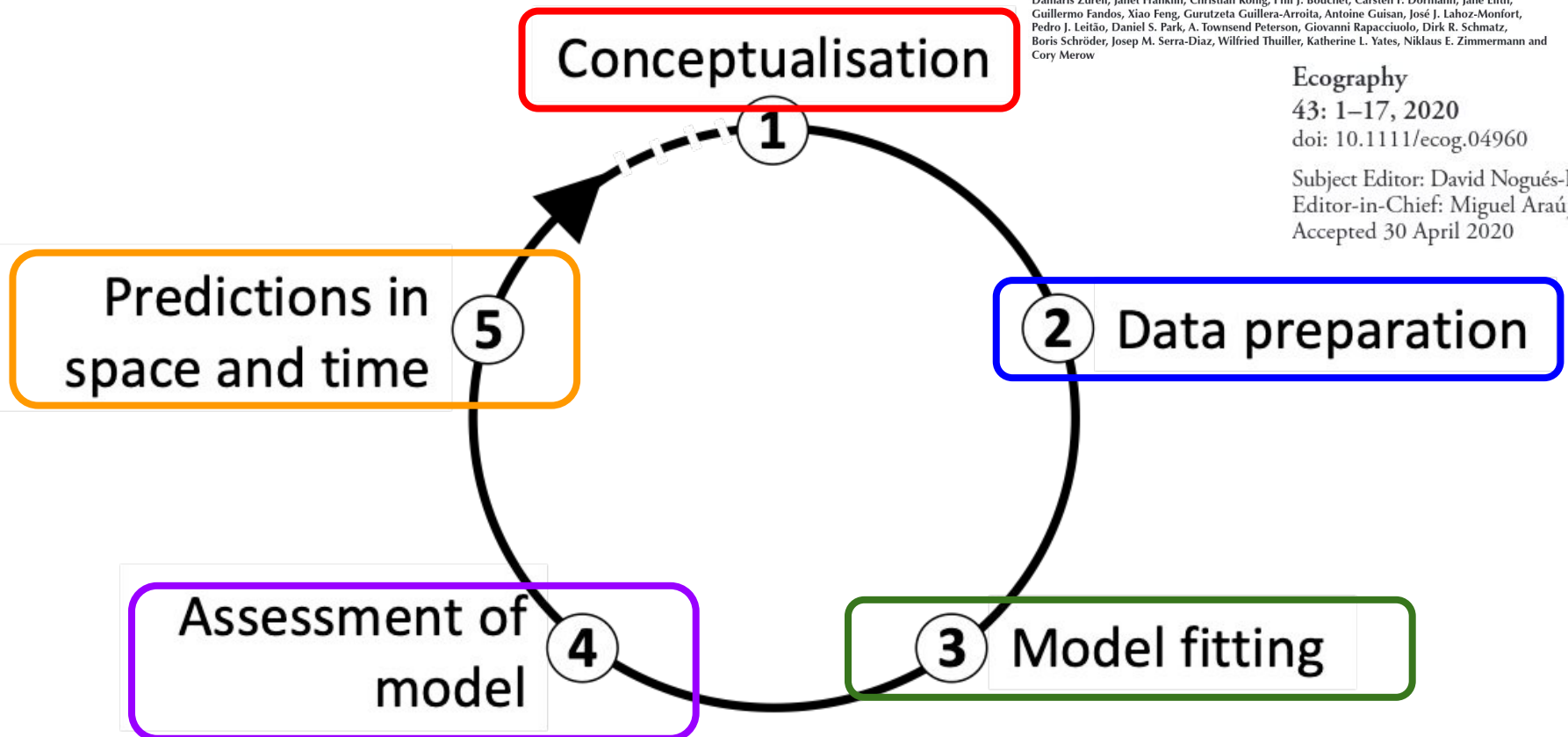
43: 1–17, 2020

doi: 10.1111/ecog.04960

Subject Editor: David Nogués-Bravo

Editor-in-Chief: Miguel Araújo

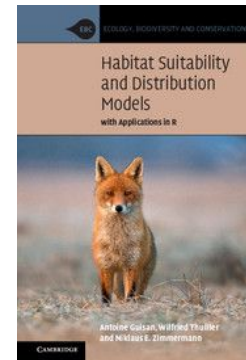
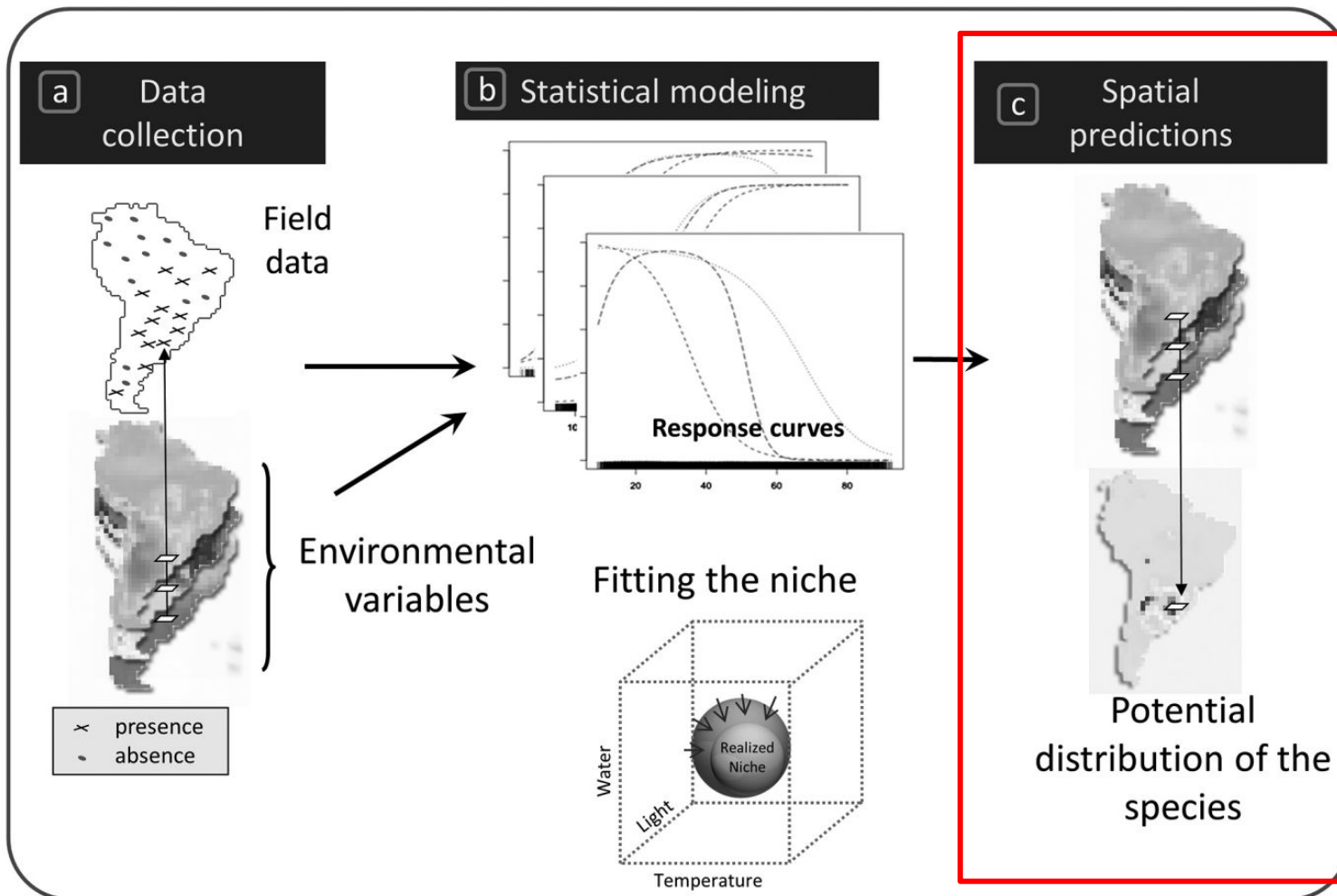
Accepted 30 April 2020



8. Predições no espaço e no tempo

Predições no espaço e no tempo

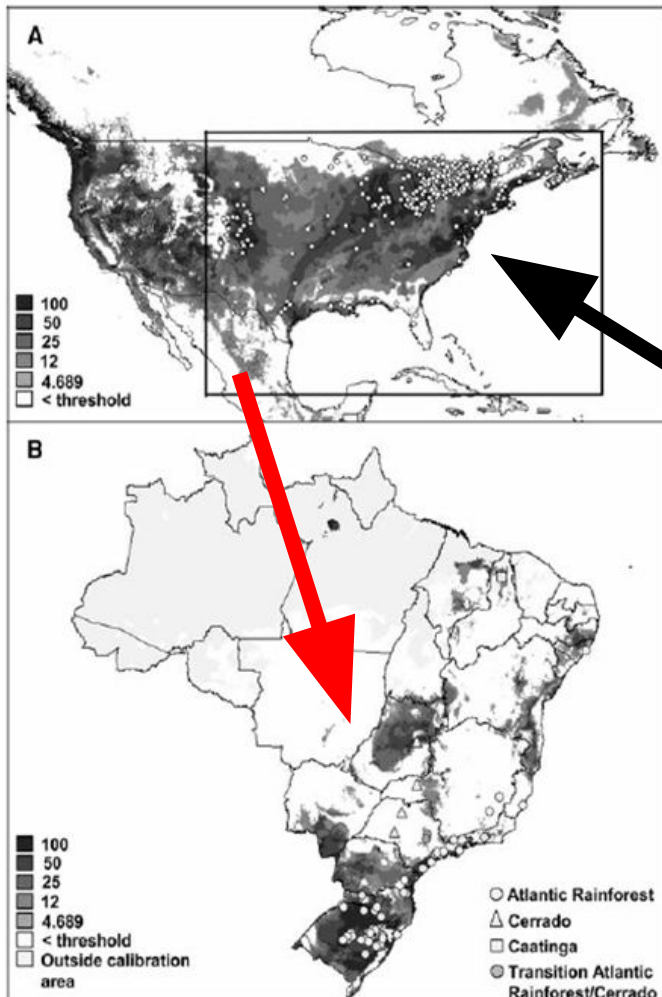
Predições (espaço e no tempo)



Guisan et al. (2017)

Predições no espaço e no tempo

Espaço - Espécies invasoras



Biol Invasions
DOI 10.1007/s10530-007-9154-5

ORIGINAL PAPER

Predicting the potential distribution of the alien invasive American bullfrog (*Lithobates catesbeianus*) in Brazil

João G. R. Giovanelli · Célio F. B. Haddad ·
João Alexandrino

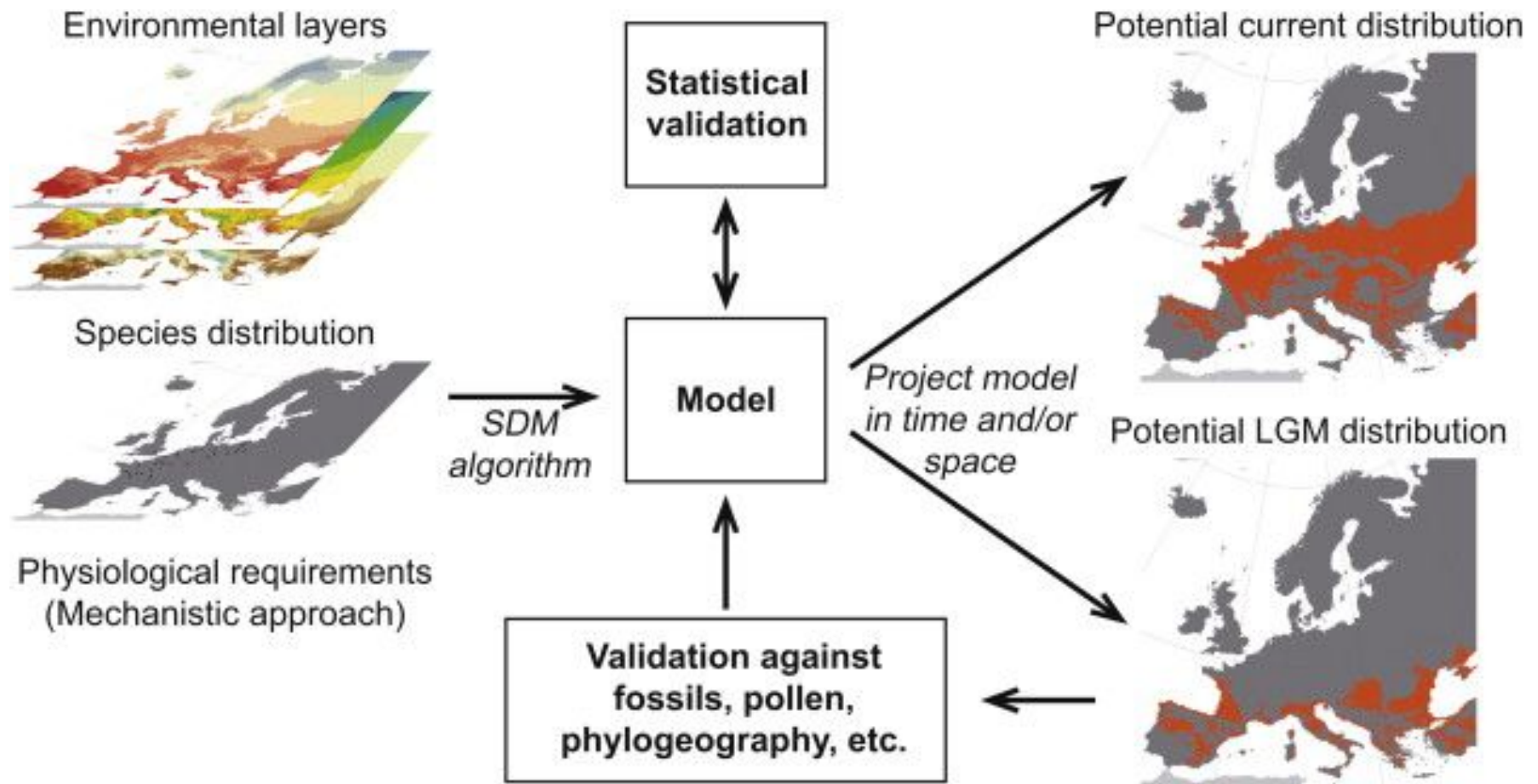


Foto: Carl D. Howe

Giovanelli et al., 2008. Biological Invasions

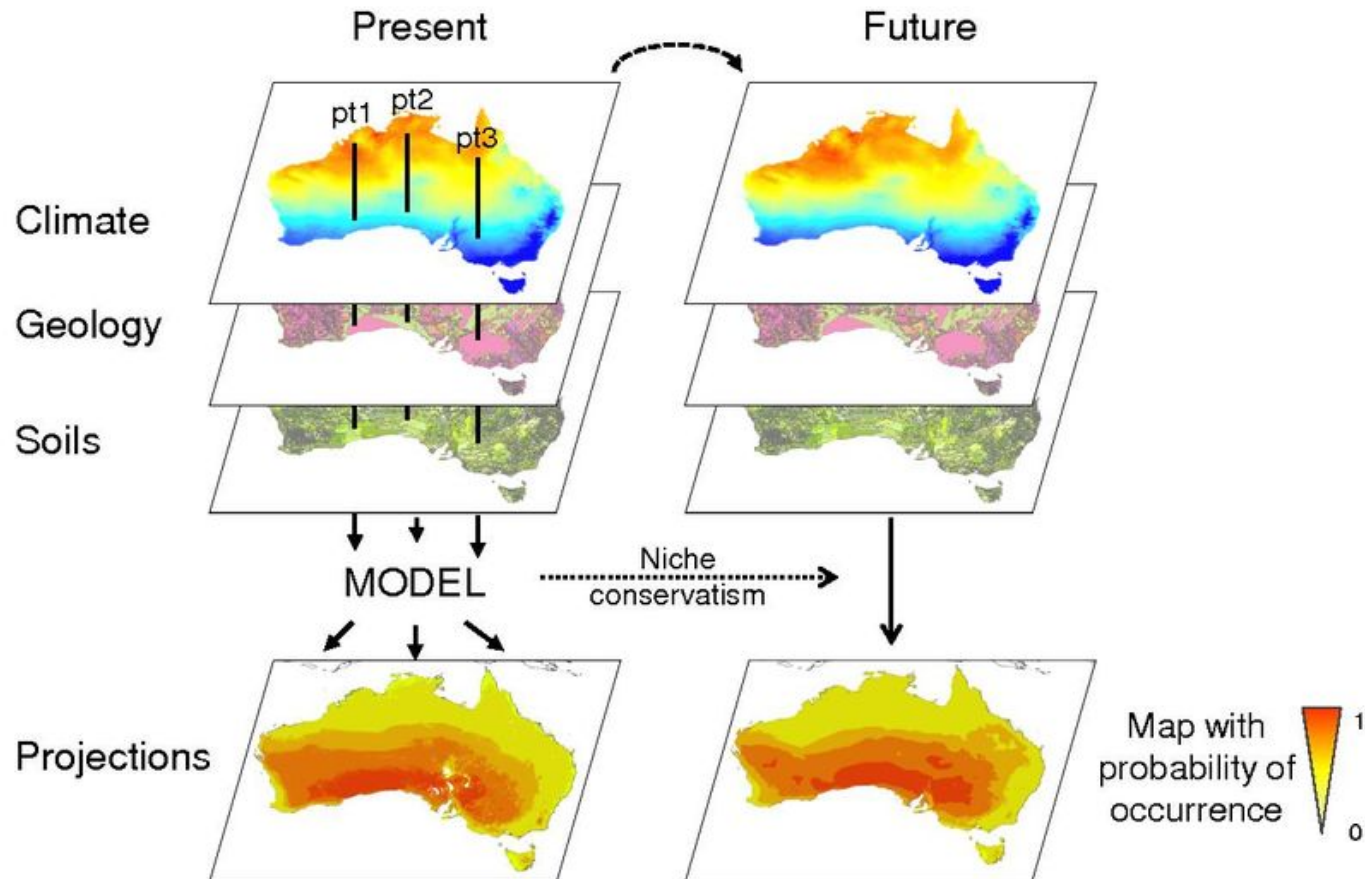
Predições no espaço e no tempo

Tempo - passado



Predições no espaço e no tempo

Tempo - futuro

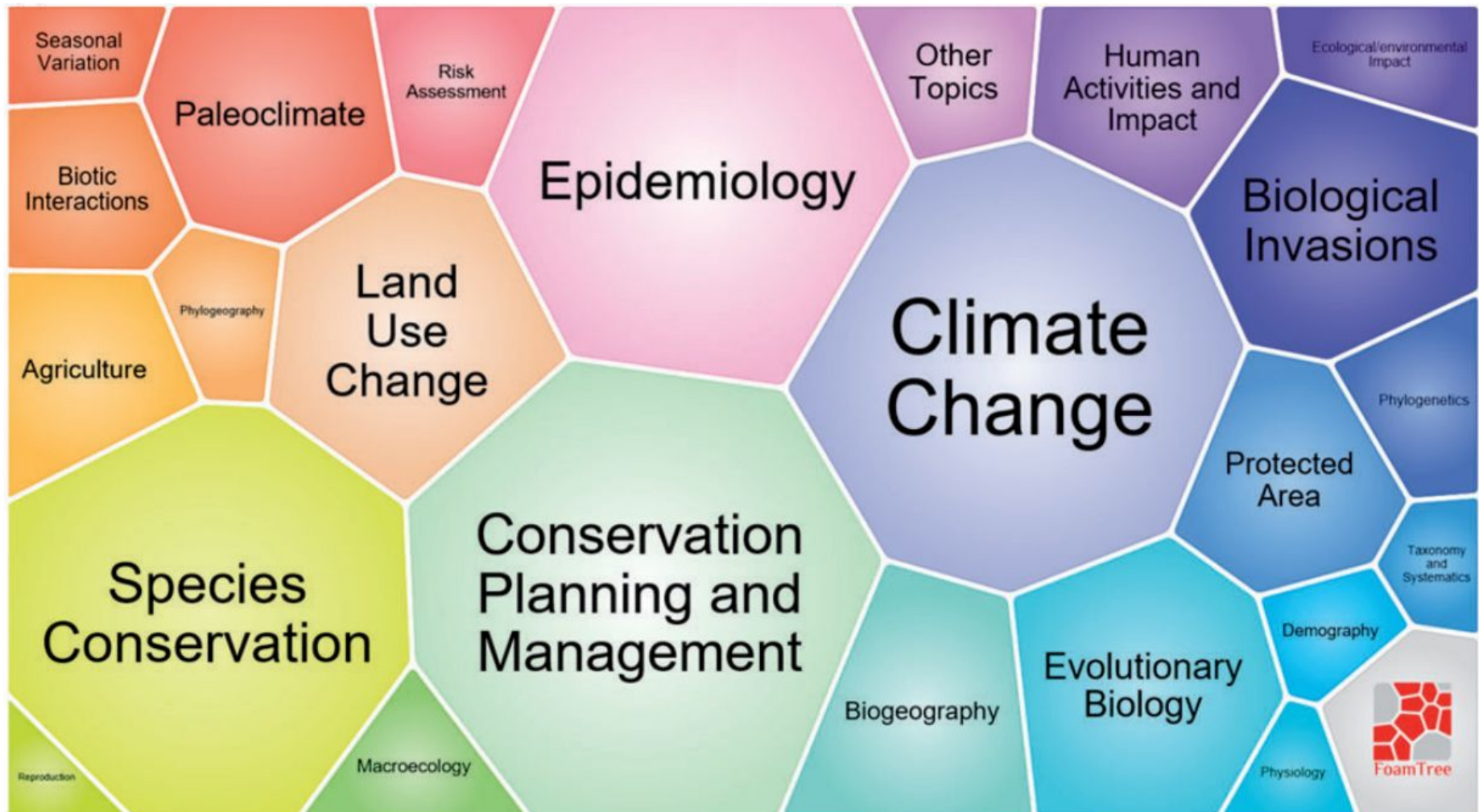


9. Aplicações e mais informações

Aplicações

Áreas de aplicação

Urbina-Cardona, N. et al. "Species Distribution Modeling in Latin America: A 25-Year Retrospective Review." *Tropical Conservation Science* 12 (2019).



Diferentes respostas às mudanças climáticas de duas palmeiras de buritis na América do Sul

Perspectives in Ecology and Conservation

Marcones Ferreira Costa

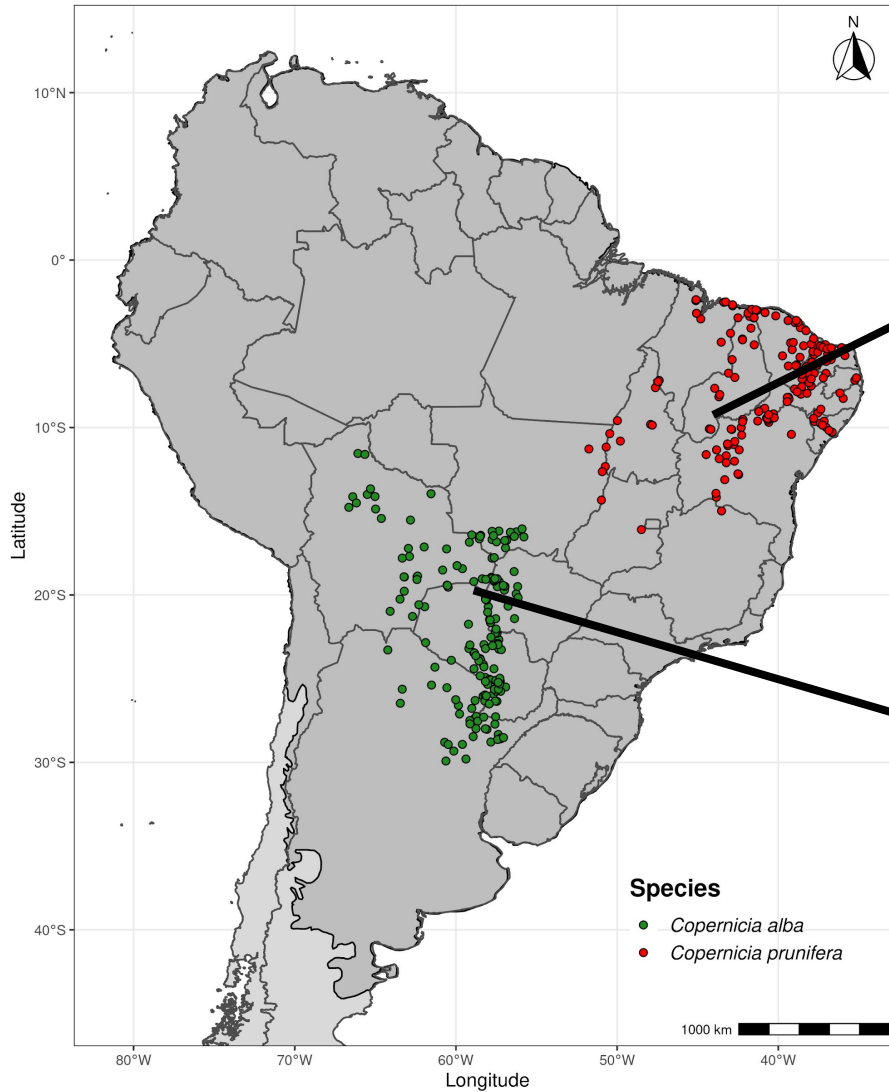
Maurício H. Vancine

Maria Imaculada Zucchi

07/07/2019



Ocorrências e limite



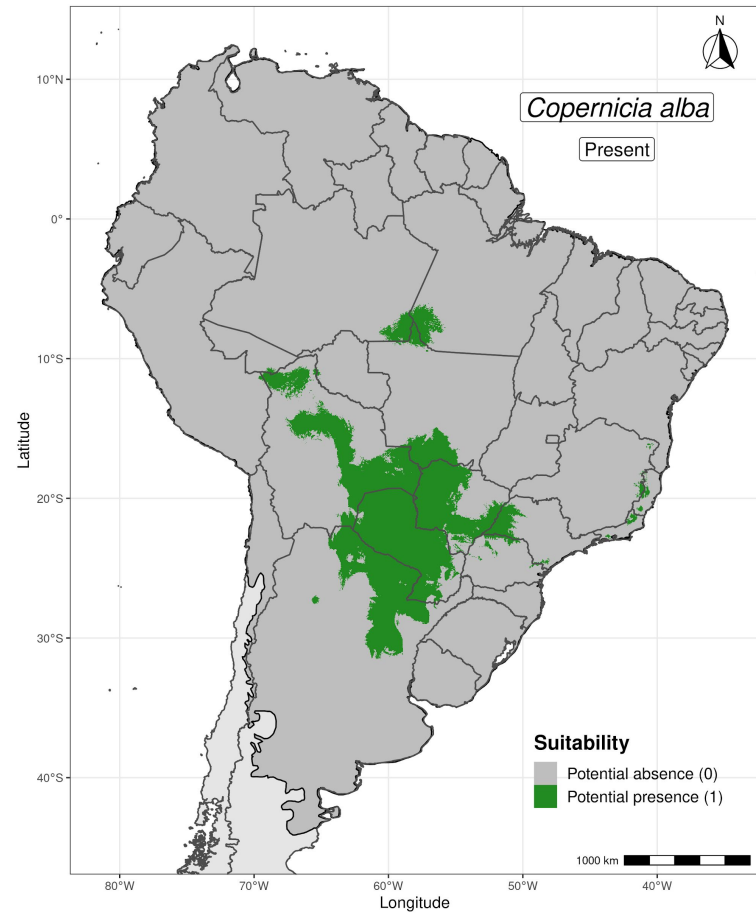
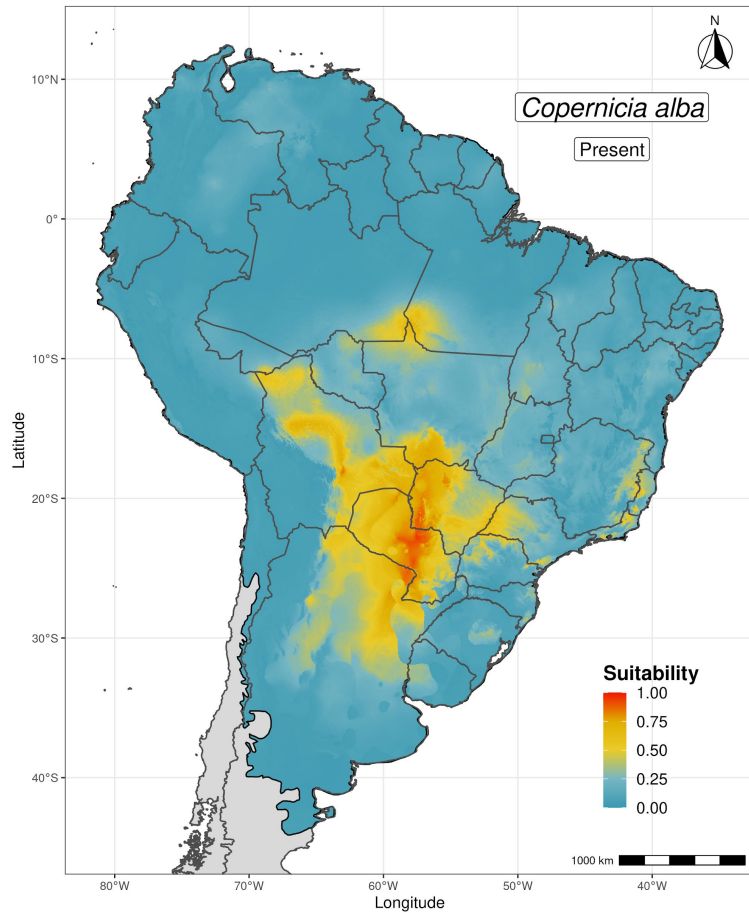
Copernicia prunifera



Copernicia alba

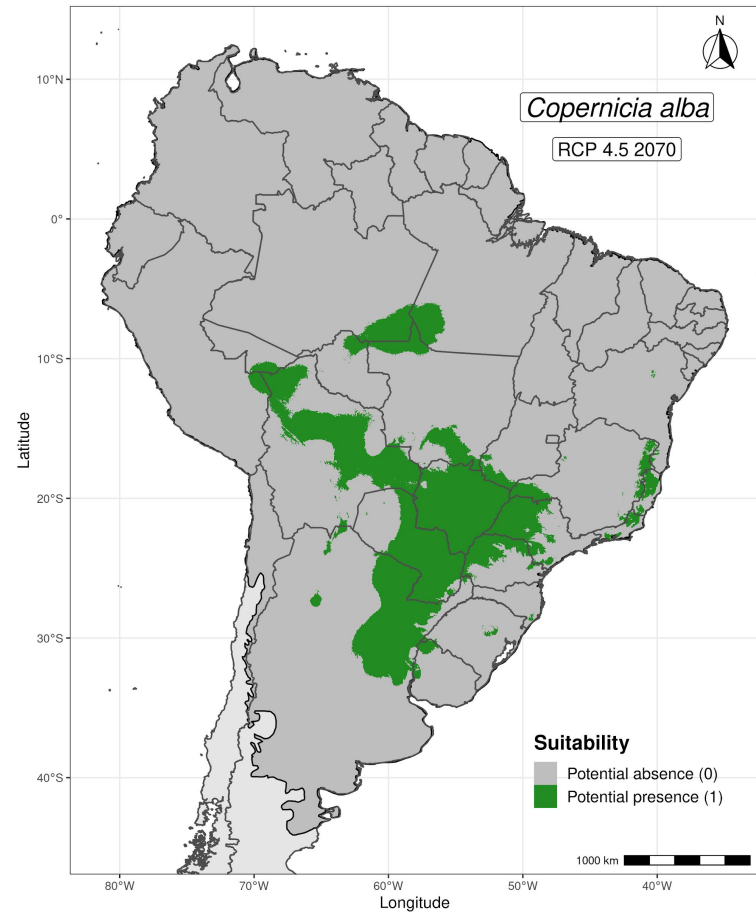
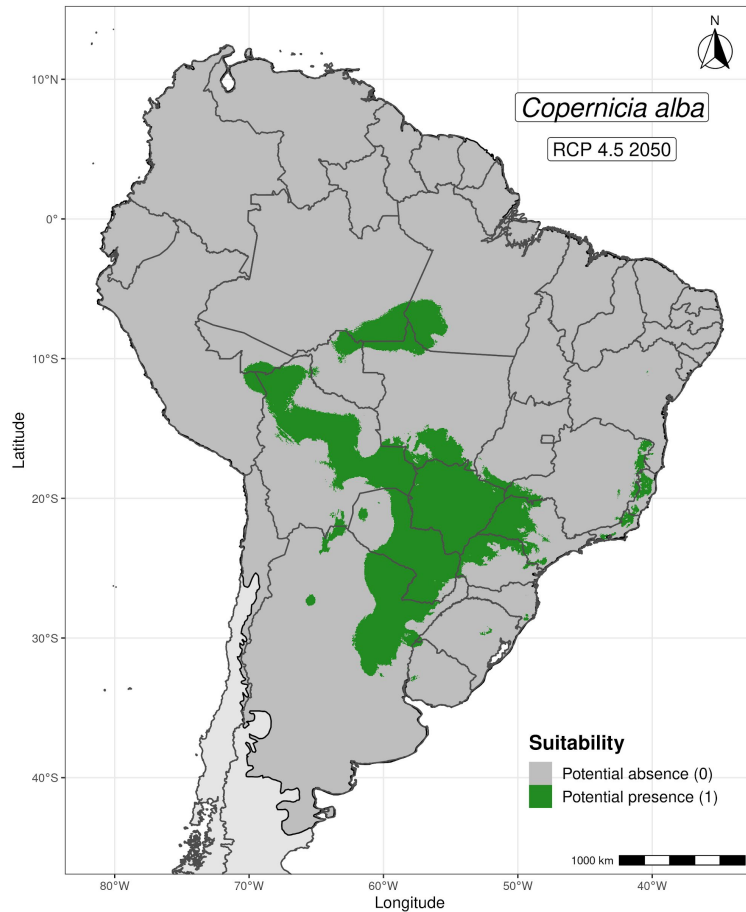
ENMs - Presente

Copernicia alba



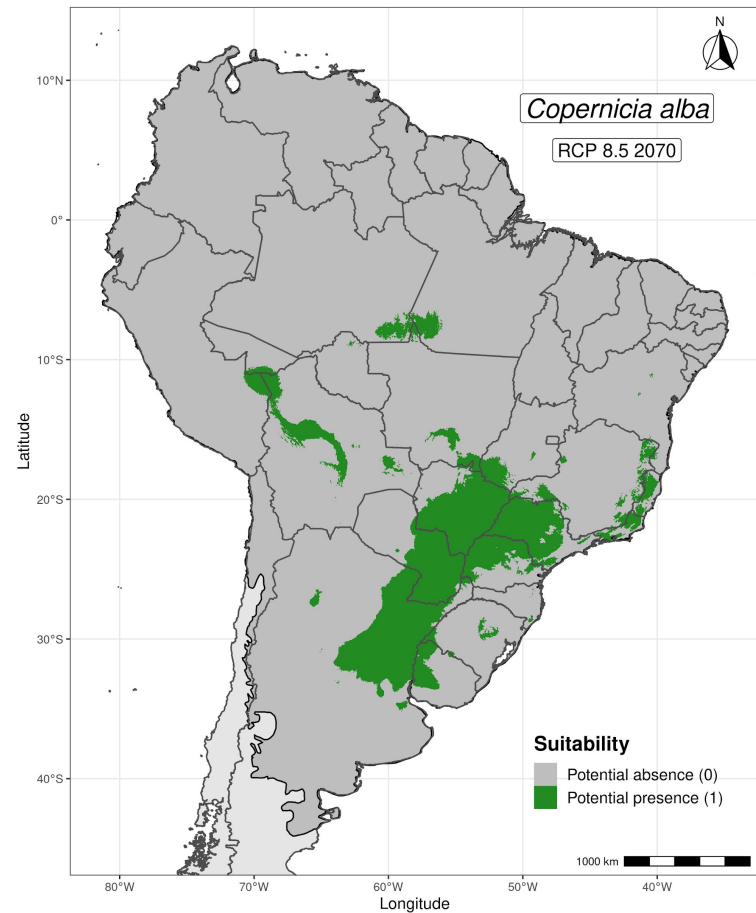
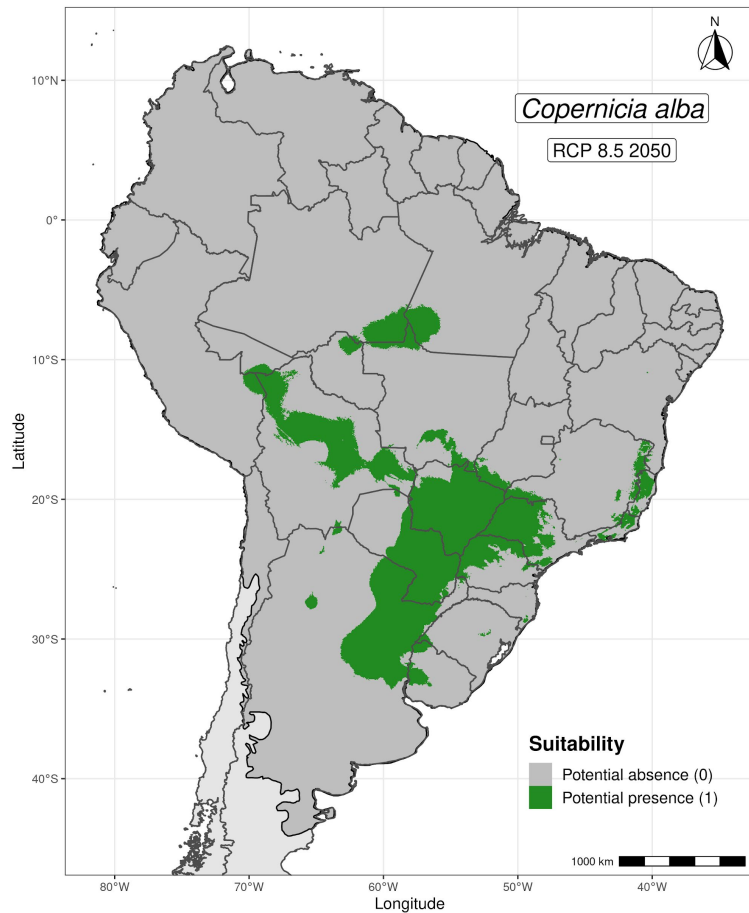
ENMs - Cenário otimista (RCP 4.5)

Copernicia alba



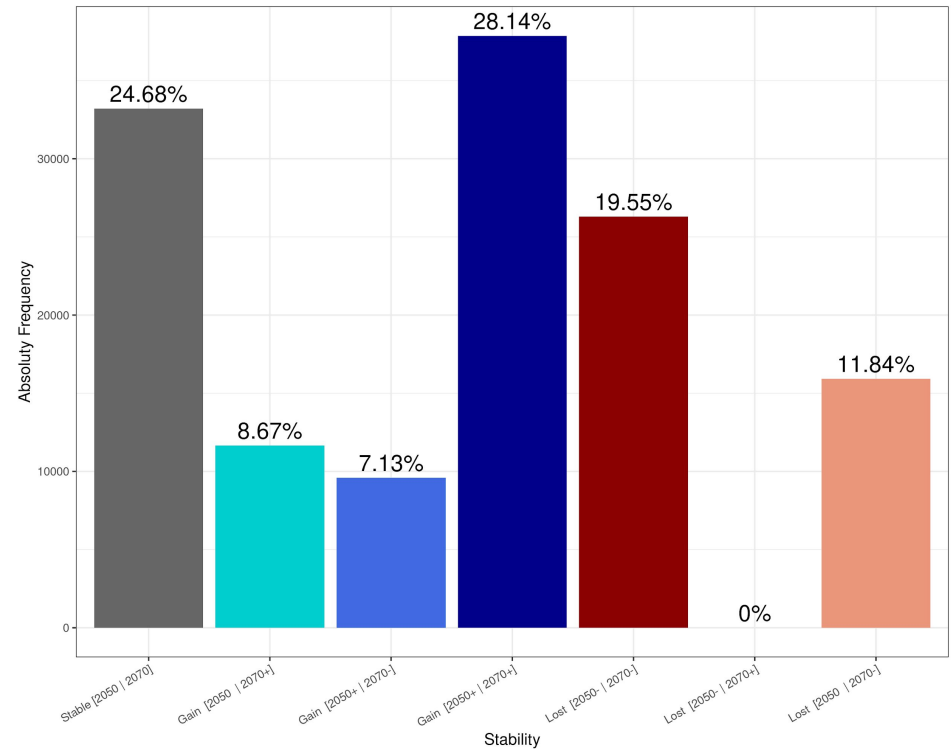
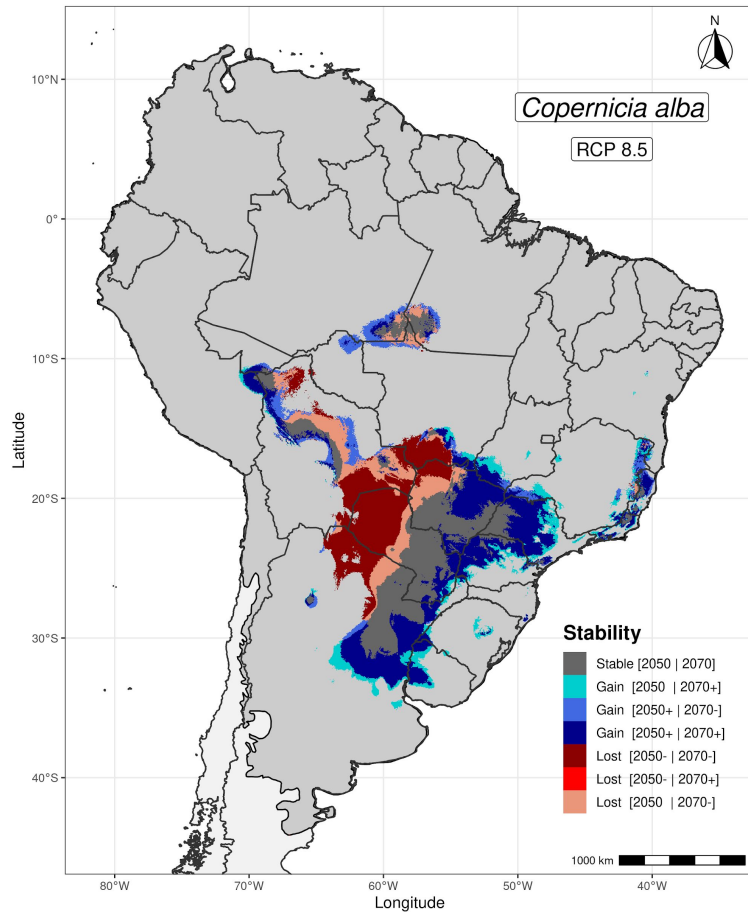
ENMs - Cenário pessimista (RCP 8.5)

Copernicia alba



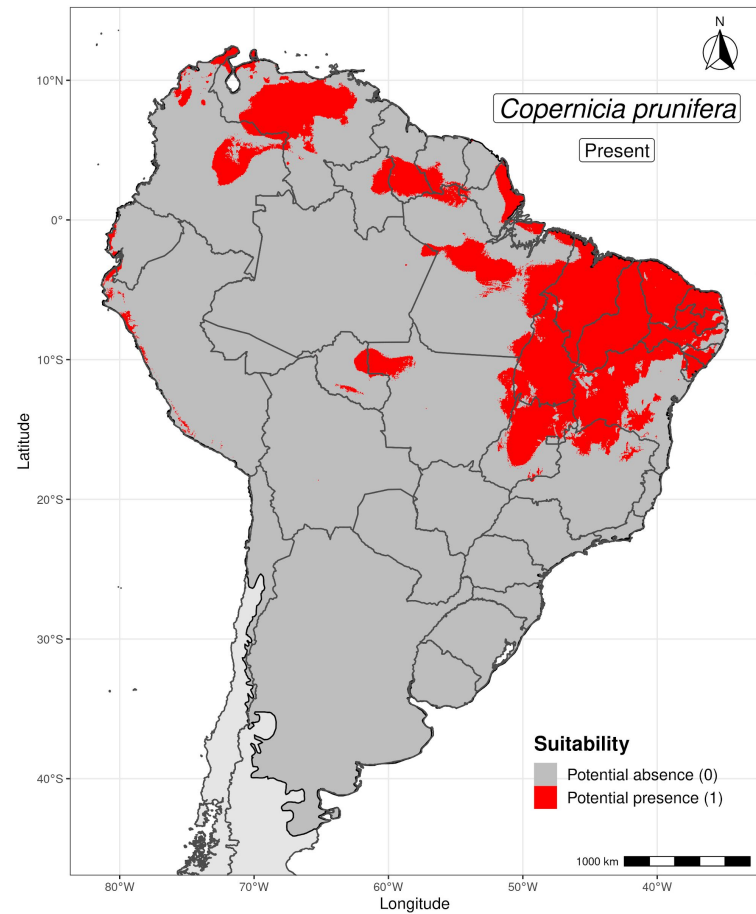
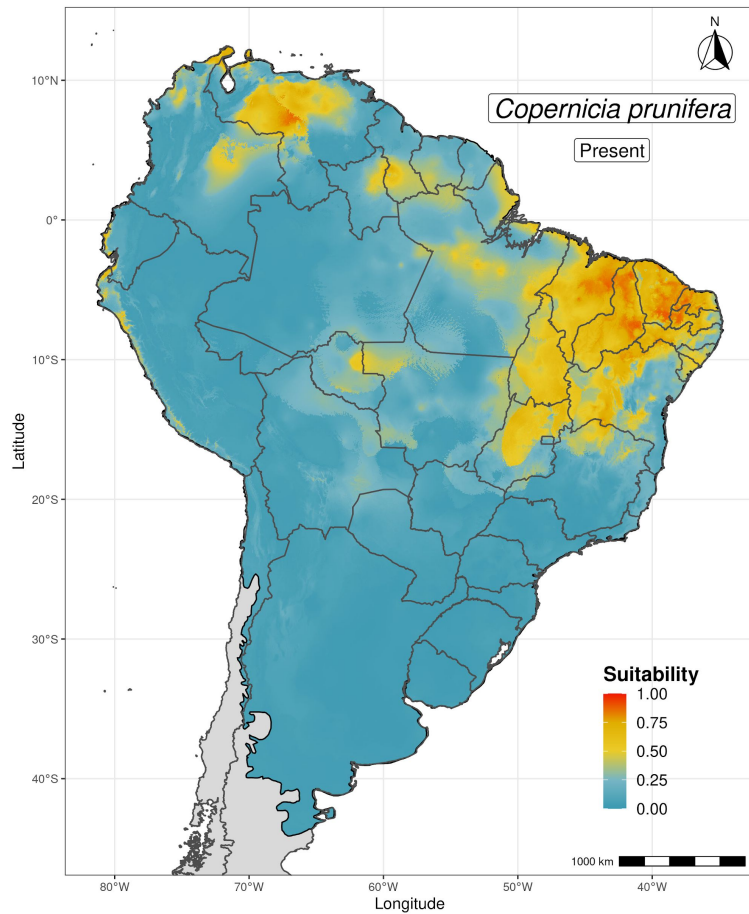
ENMs - Estabilidade (RCP 8.5)

Copernicia alba



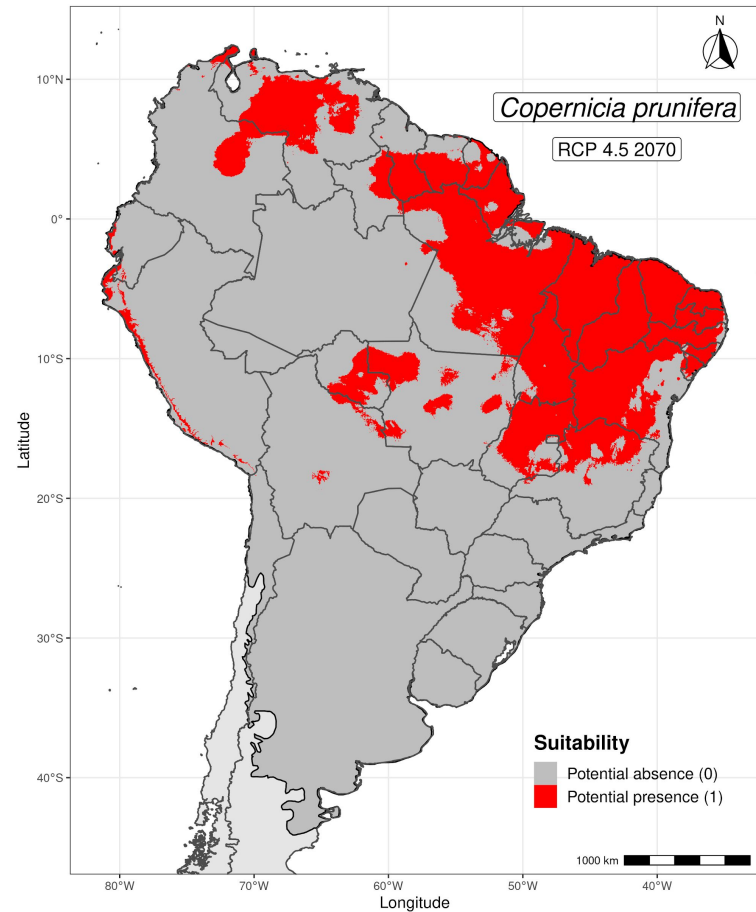
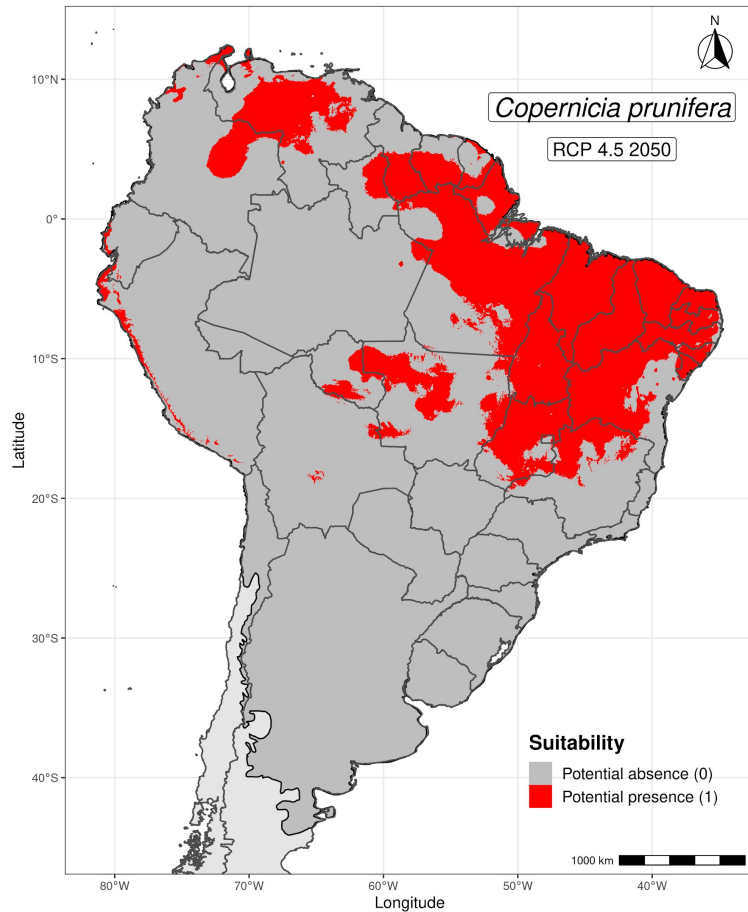
ENMs - Presente

Copernicia prunifera



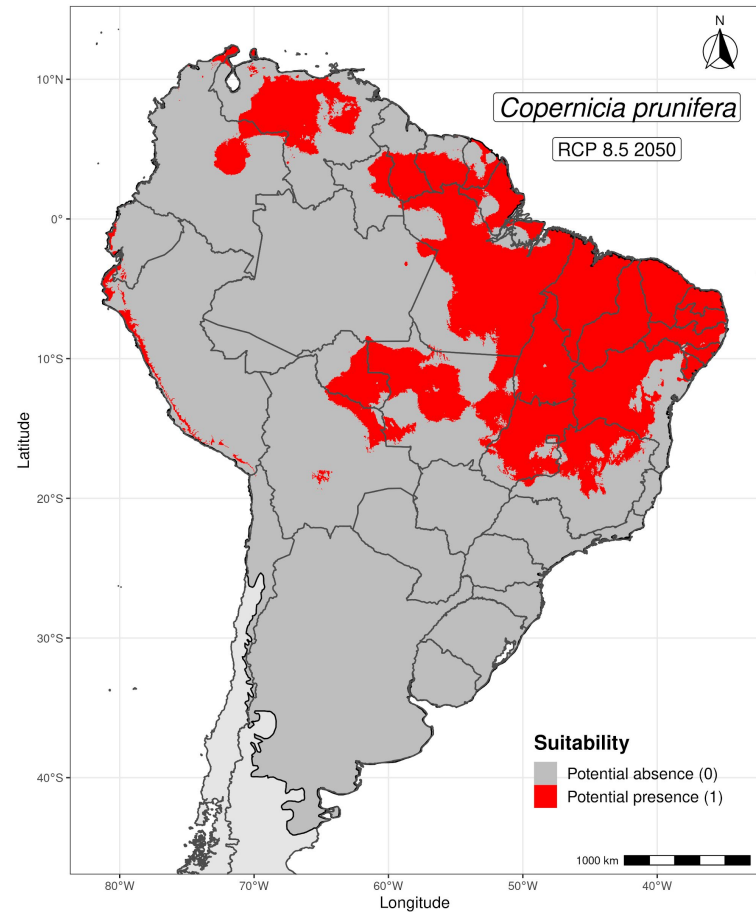
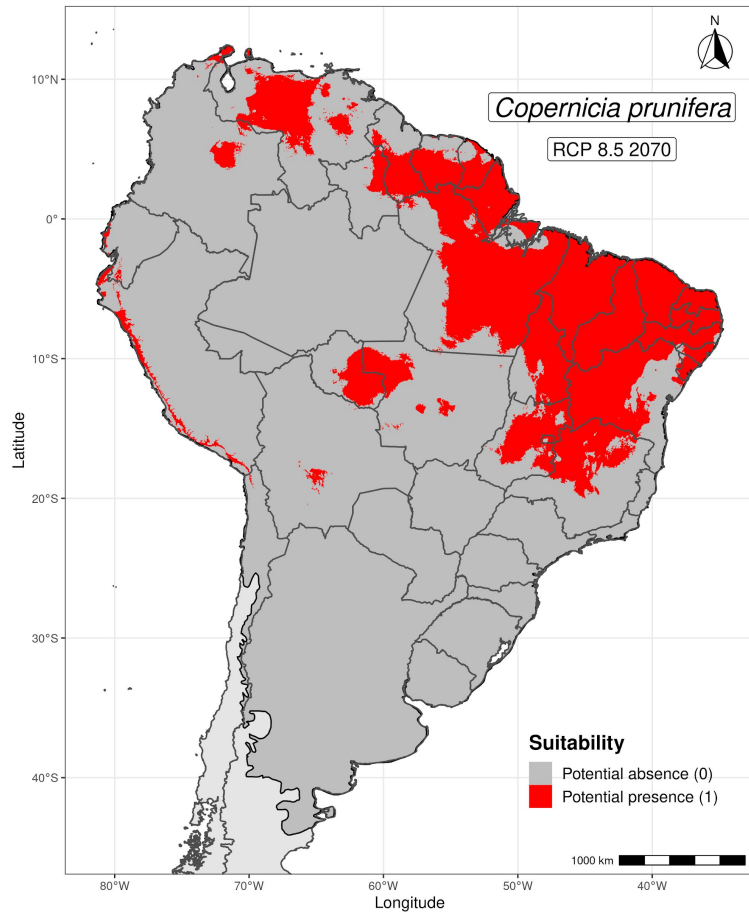
ENMs - Cenário otimista (RCP 4.5)

Copernicia prunifera



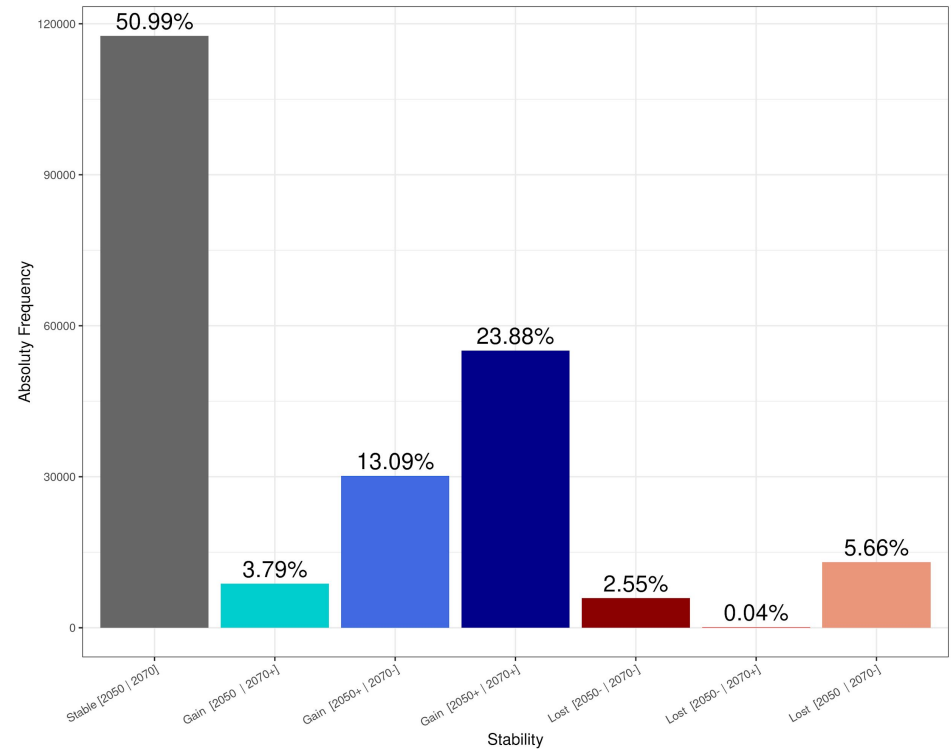
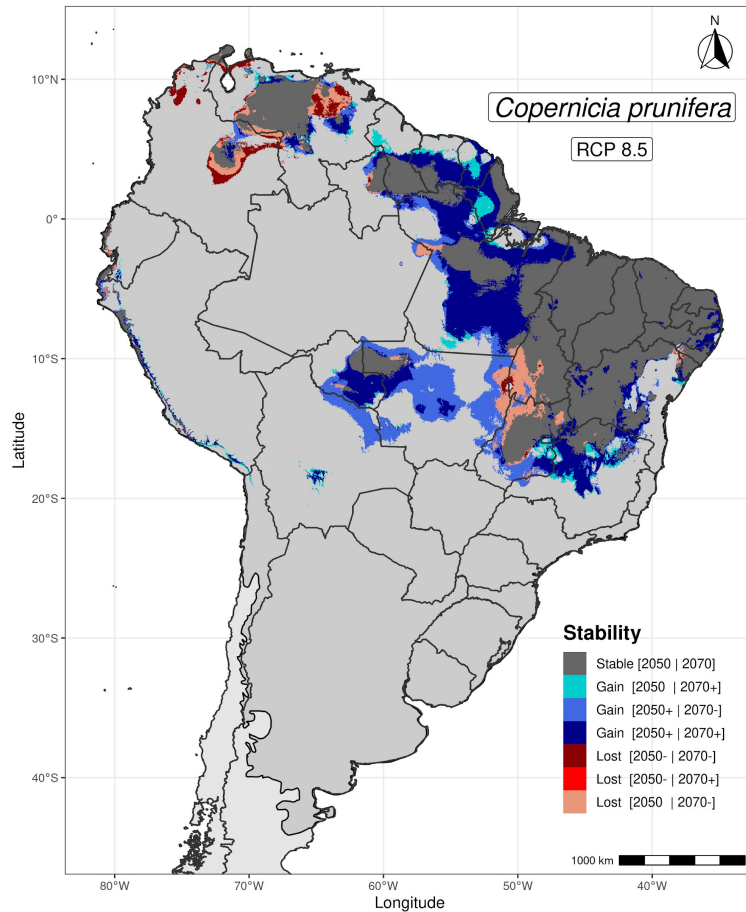
ENMs - Cenário pessimista (RCP 8.5)

Copernicia prunifera



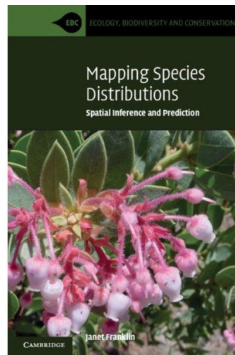
ENMs - Estabilidade (RCP 8.5)

Copernicia prunifera

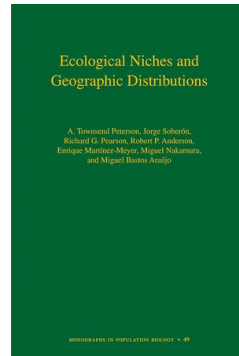


Mais informações

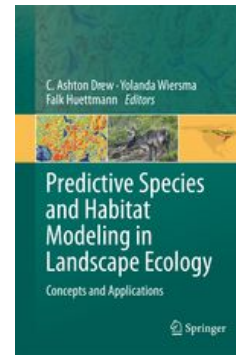
Livros



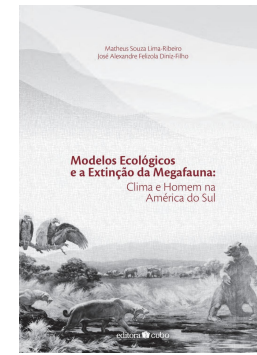
Franklin (2009)



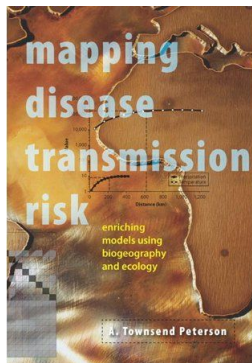
Peterson et al. (2011)



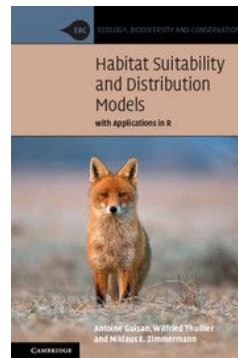
Drew et al. (2011)



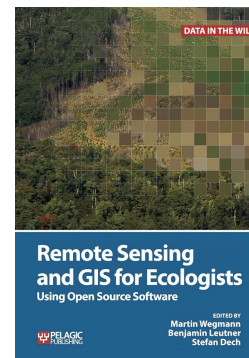
Lima-Ribeiro &
Diniz-Filho (2013)



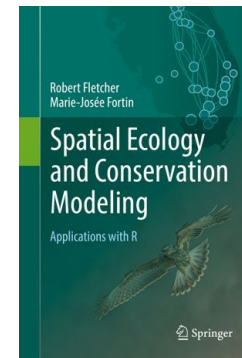
Peterson (2014)



Guisan et al. (2017)



Wegmann et al. (2016)
Cap. 13

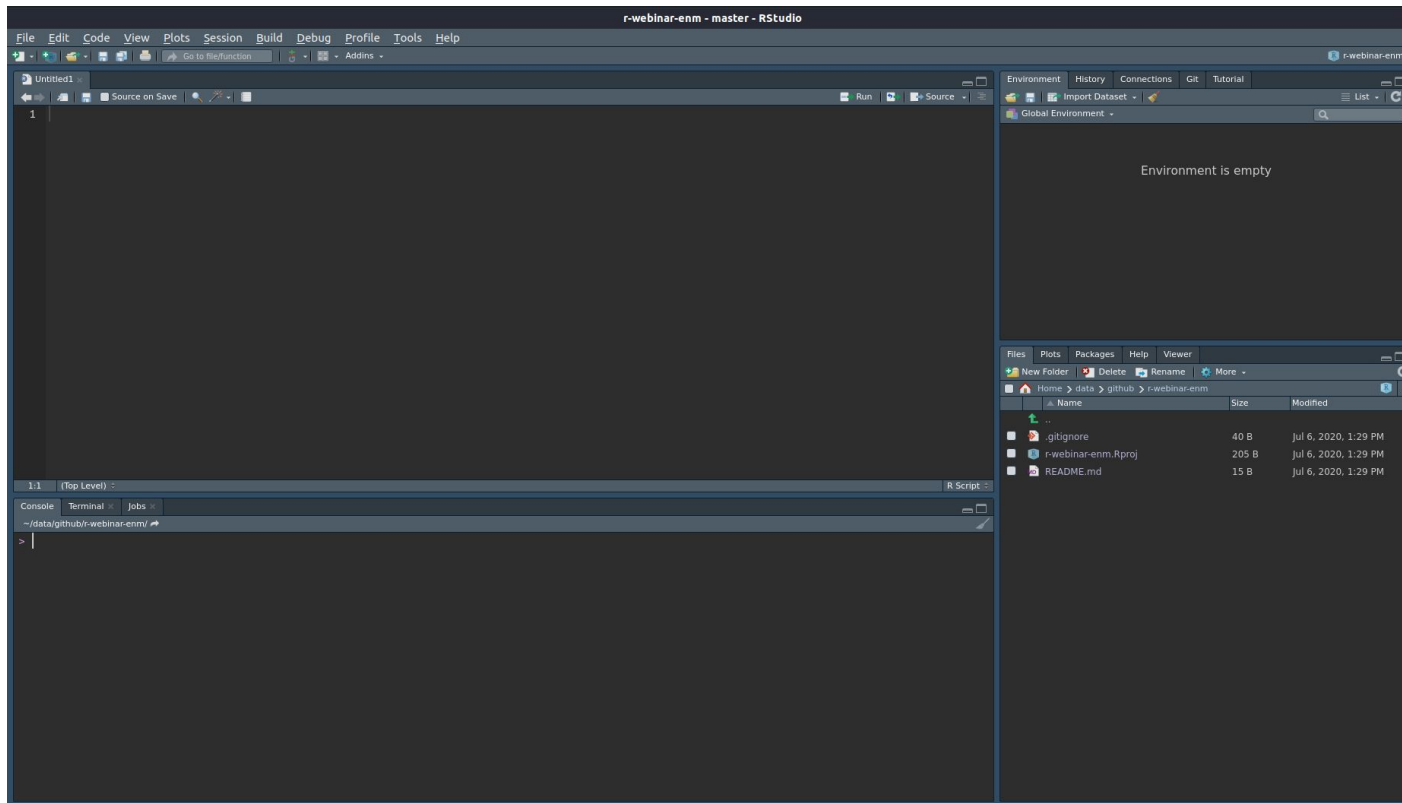


Fletcher and Fortin (2018)
Cap. 07

Prática

R

<https://github.com/mauriciovancine/r-webinar-enm>



Muito obrigado!



Contato e informações

✉ mauricio.vancine@gmail.com

🐦 [@mauriciovancine](https://twitter.com/mauriciovancine)

🔗 mauriciovancine.netlify.com